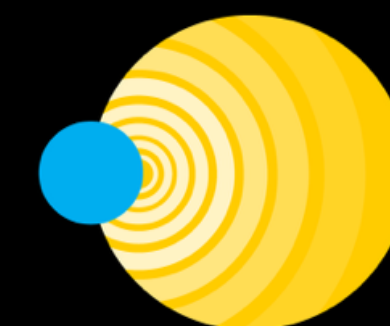


BREAKTHROUGH LISTEN

Detecting ISM Scintillation in Narrowband Signals: A New Filter for Radio SETI

BRYAN BRZYCKI

**UNIVERSITY OF CALIFORNIA BERKELEY
PSETI SEMINAR, SEPTEMBER 21, 2023**



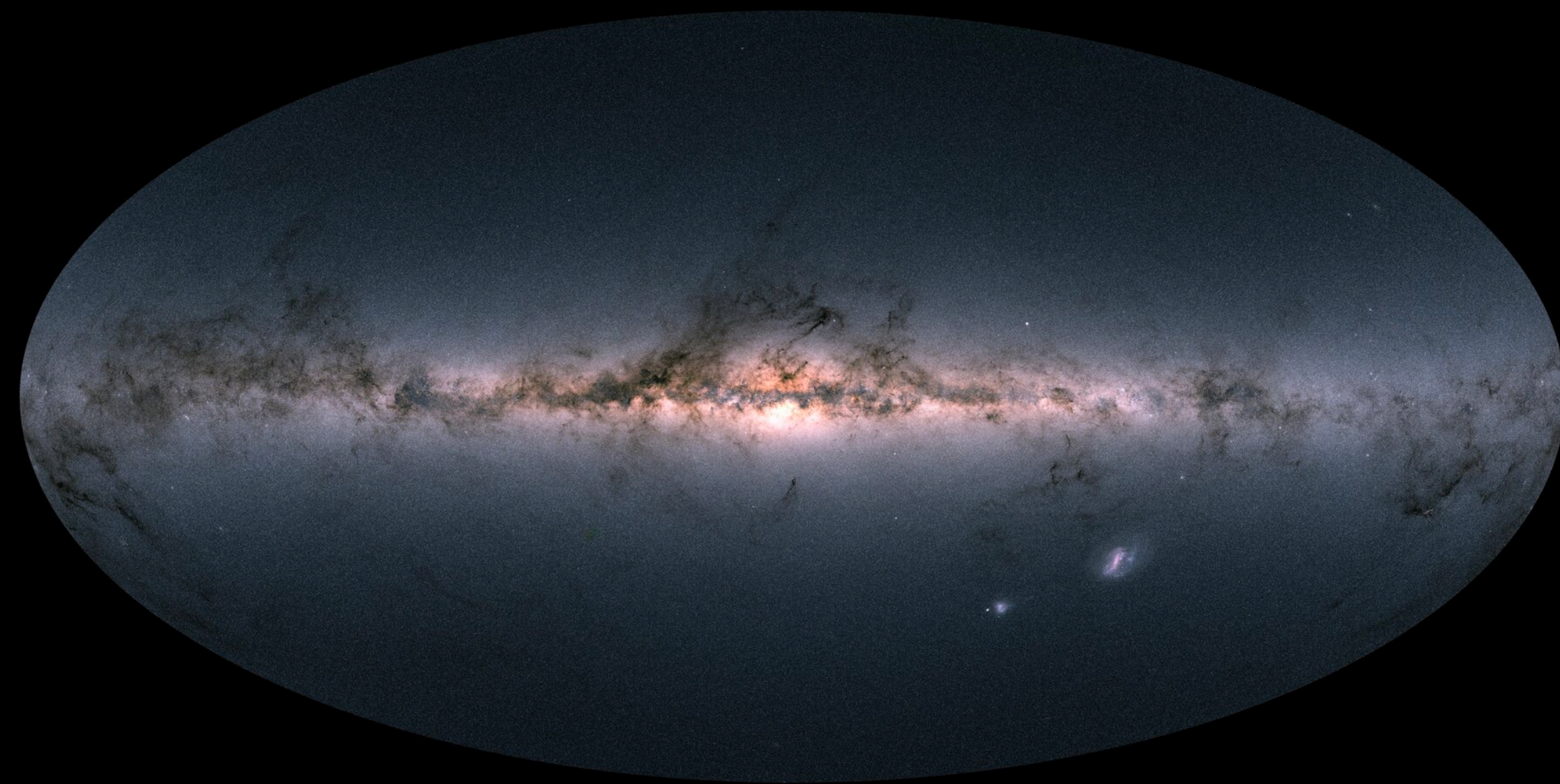
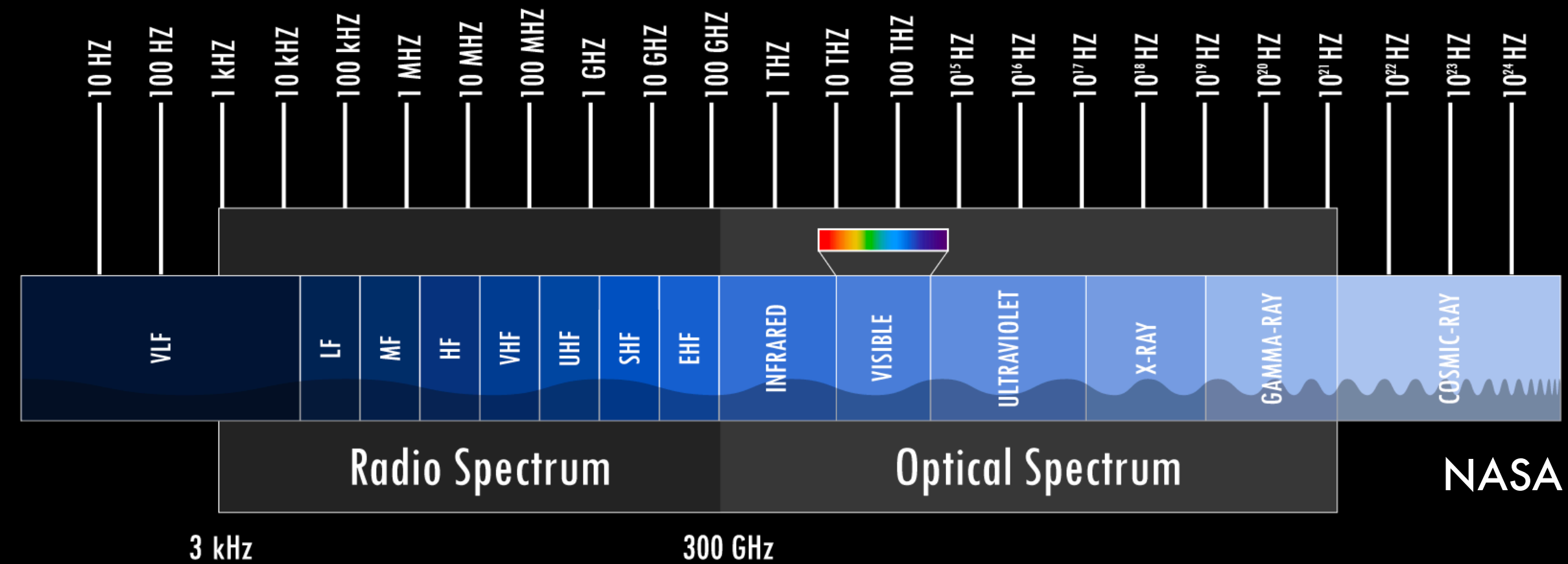
BERKELEY SETI
RESEARCH CENTER

The Search for Extraterrestrial Intelligence (SETI)

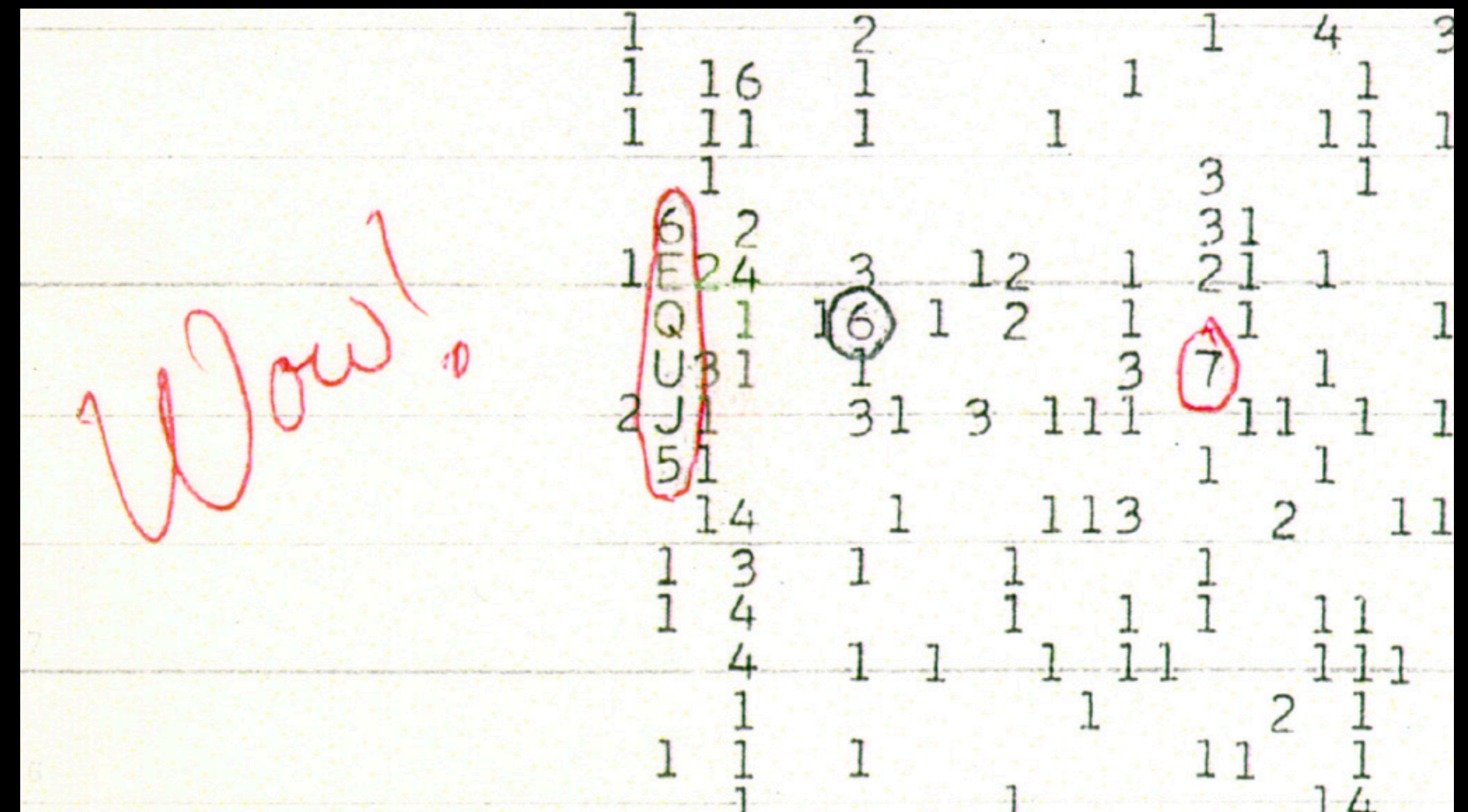
- Modern radio SETI began in the 1960s
- Vast improvements and expansion in:
 - Instantaneous bandwidth
 - Sensitivity
 - Survey size
 - Search strategies



Where should we look?



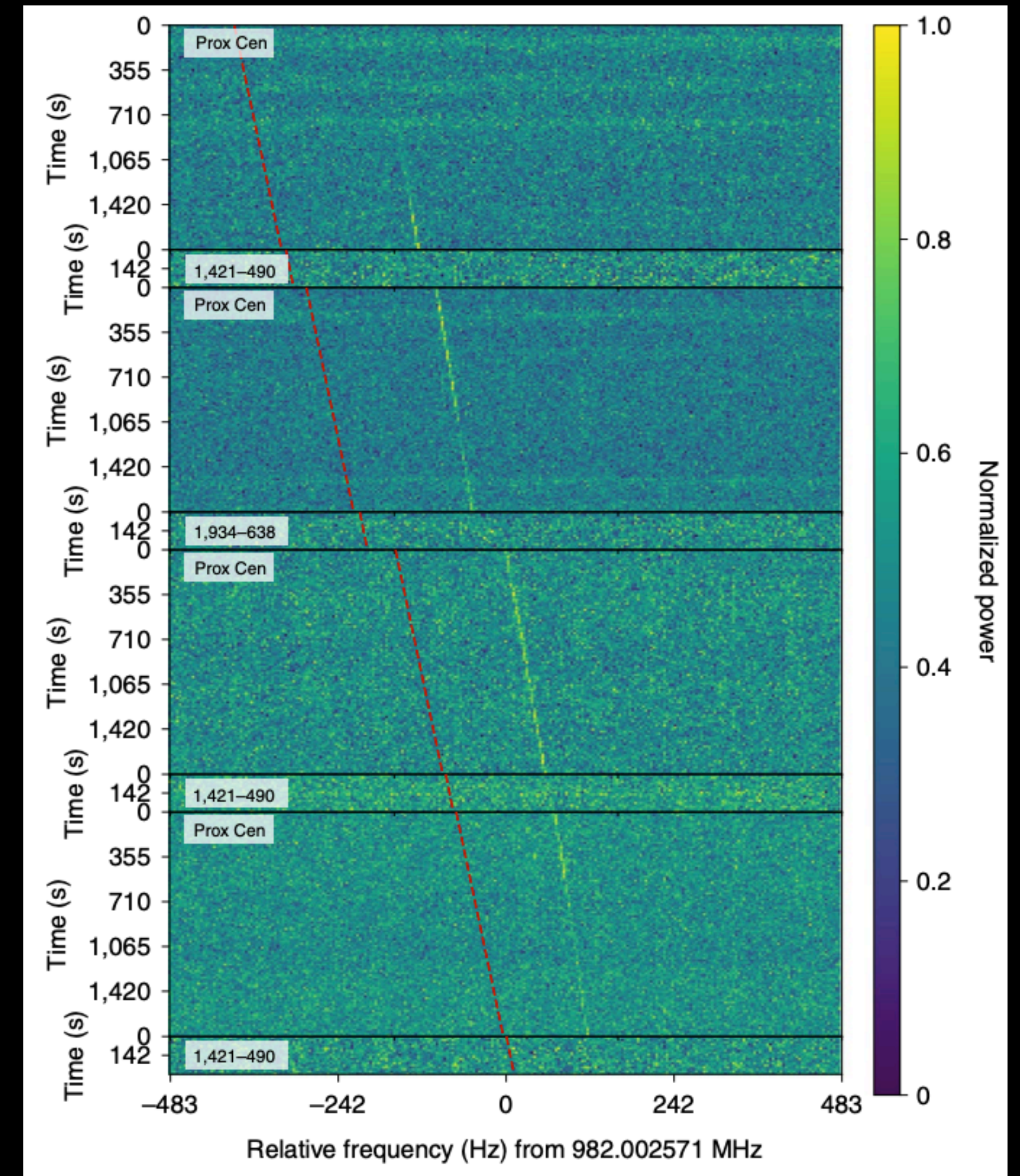
ESA



Big Ear Radio Observatory

How should we look? What makes for a convincing candidate?

- **Narrowband vs. astrophysical sources**
- **Non-zero drift rate vs. RFI**
- **Sky localization vs. RFI**



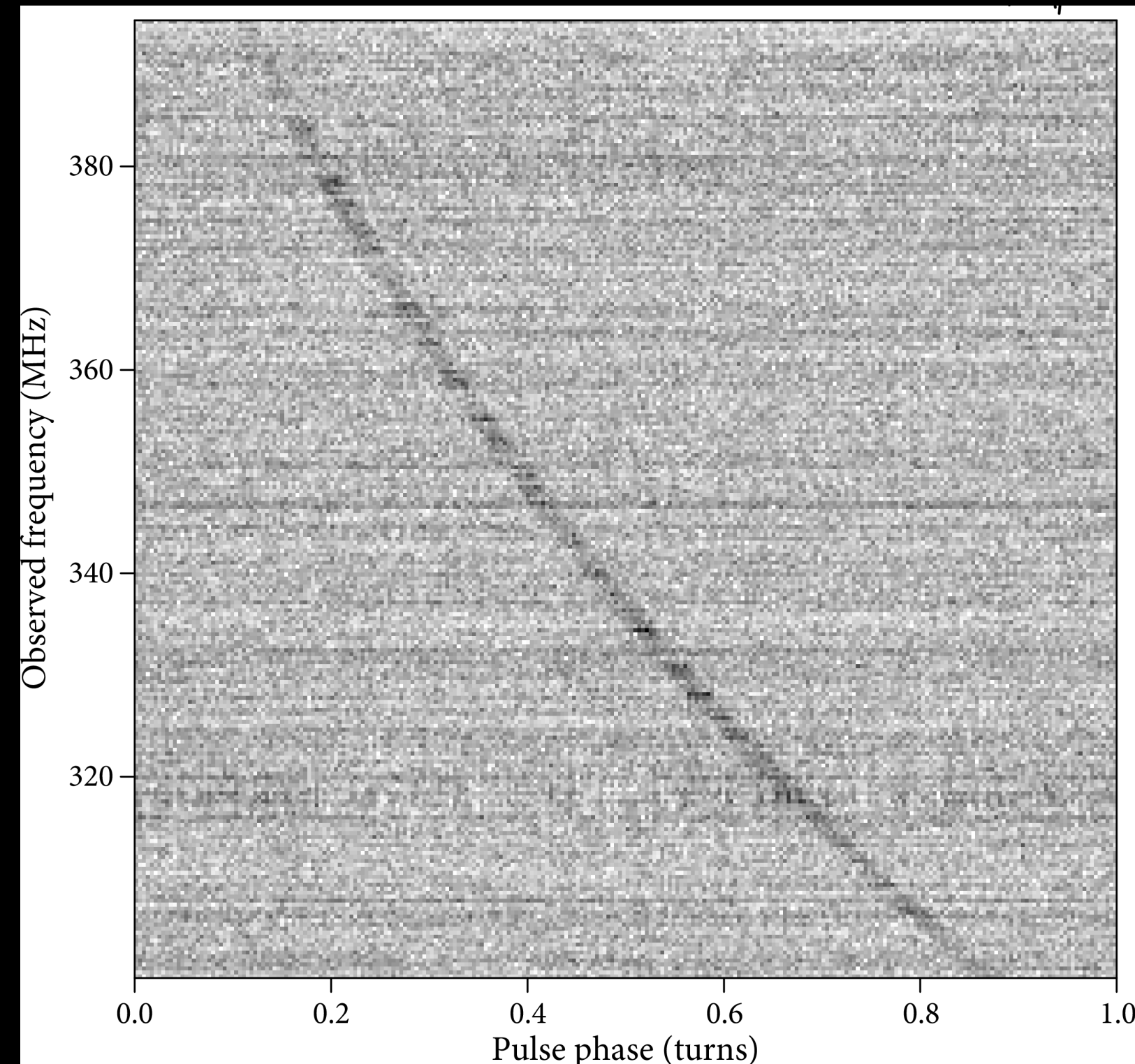
Can we use astrophysical phenomena as a filter to distinguish technosignatures from RFI?



ESA

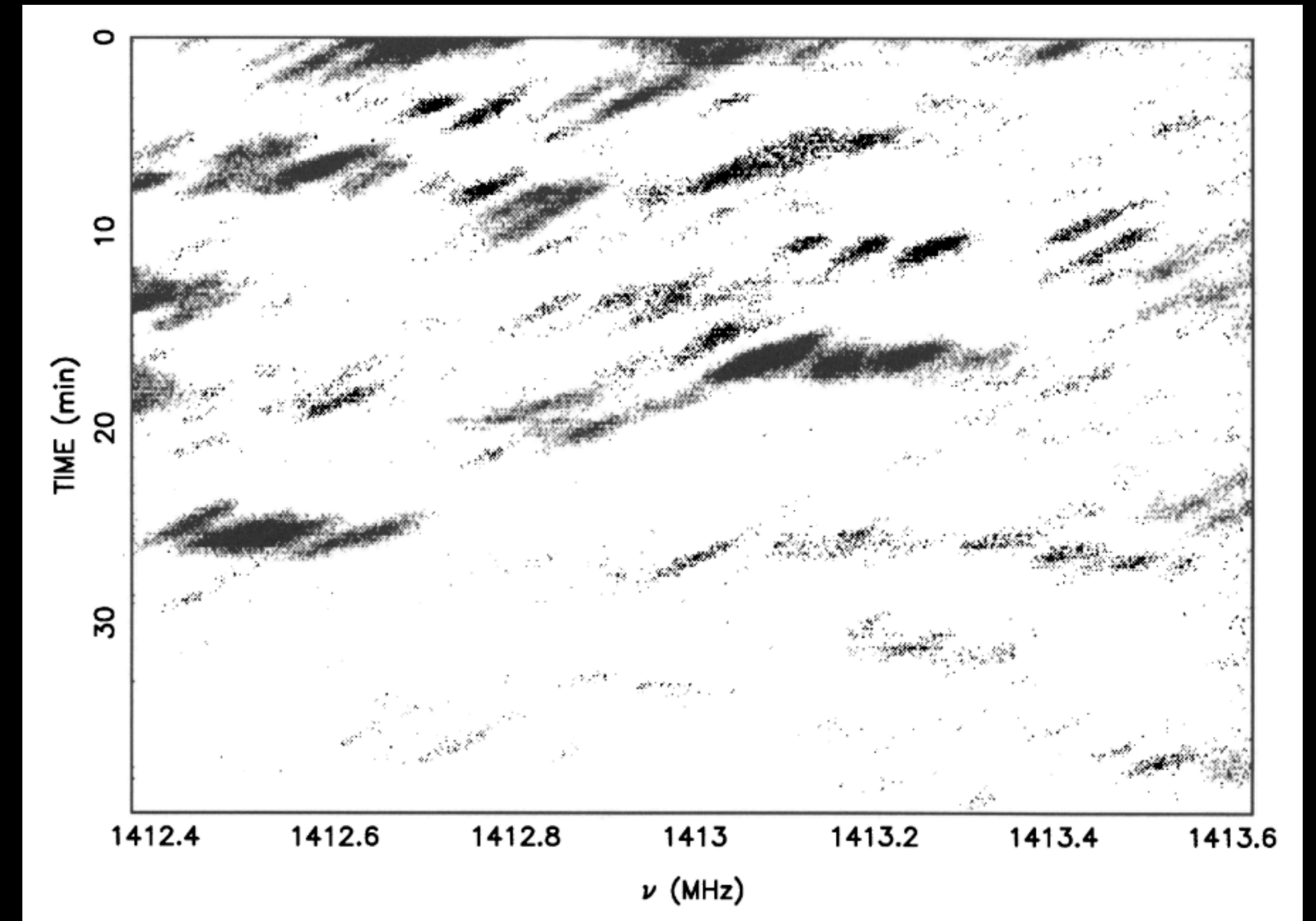
Pulsar observations probe radio plasma effects

Dispersion



Condon & Ransom 2016

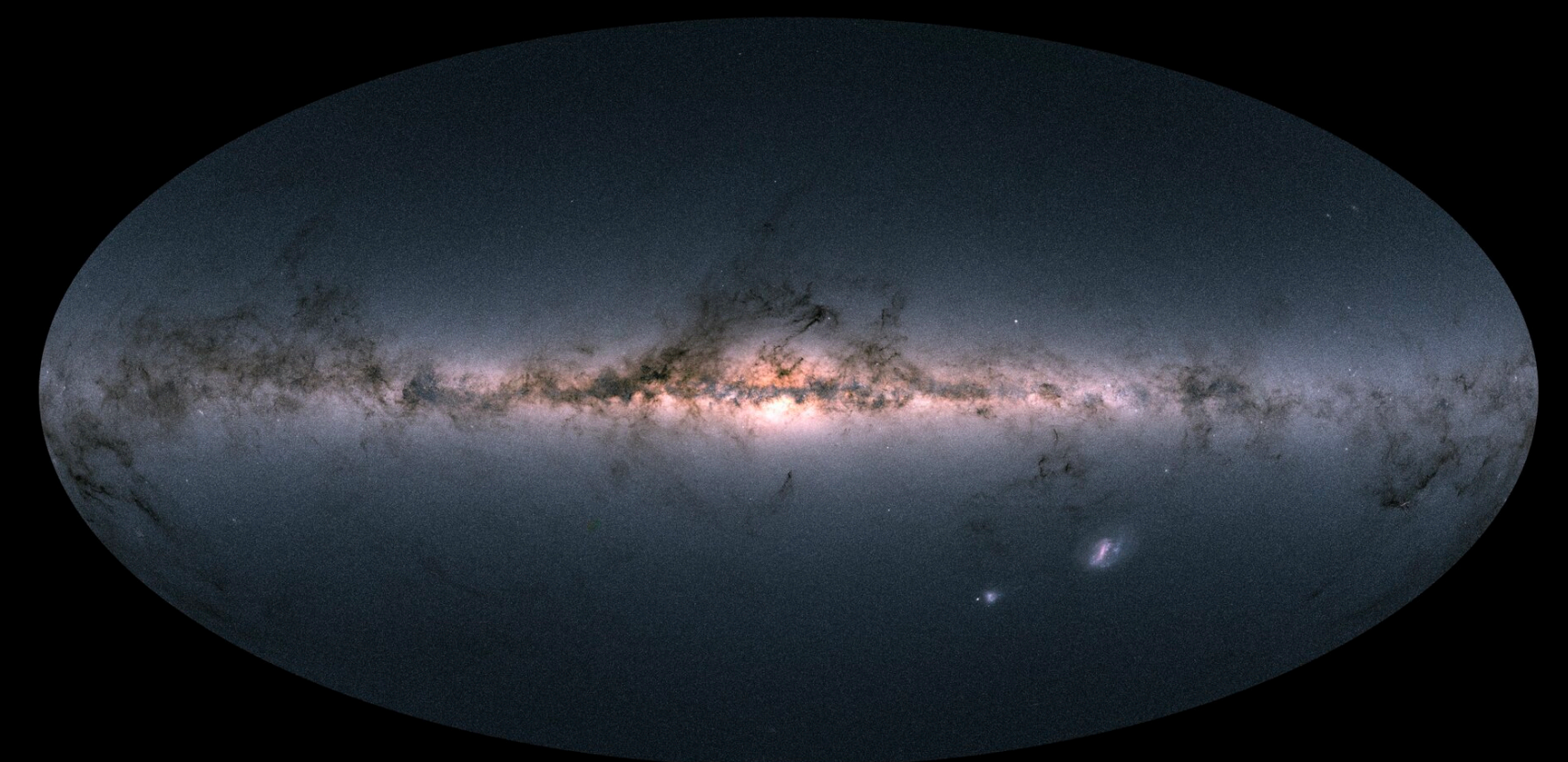
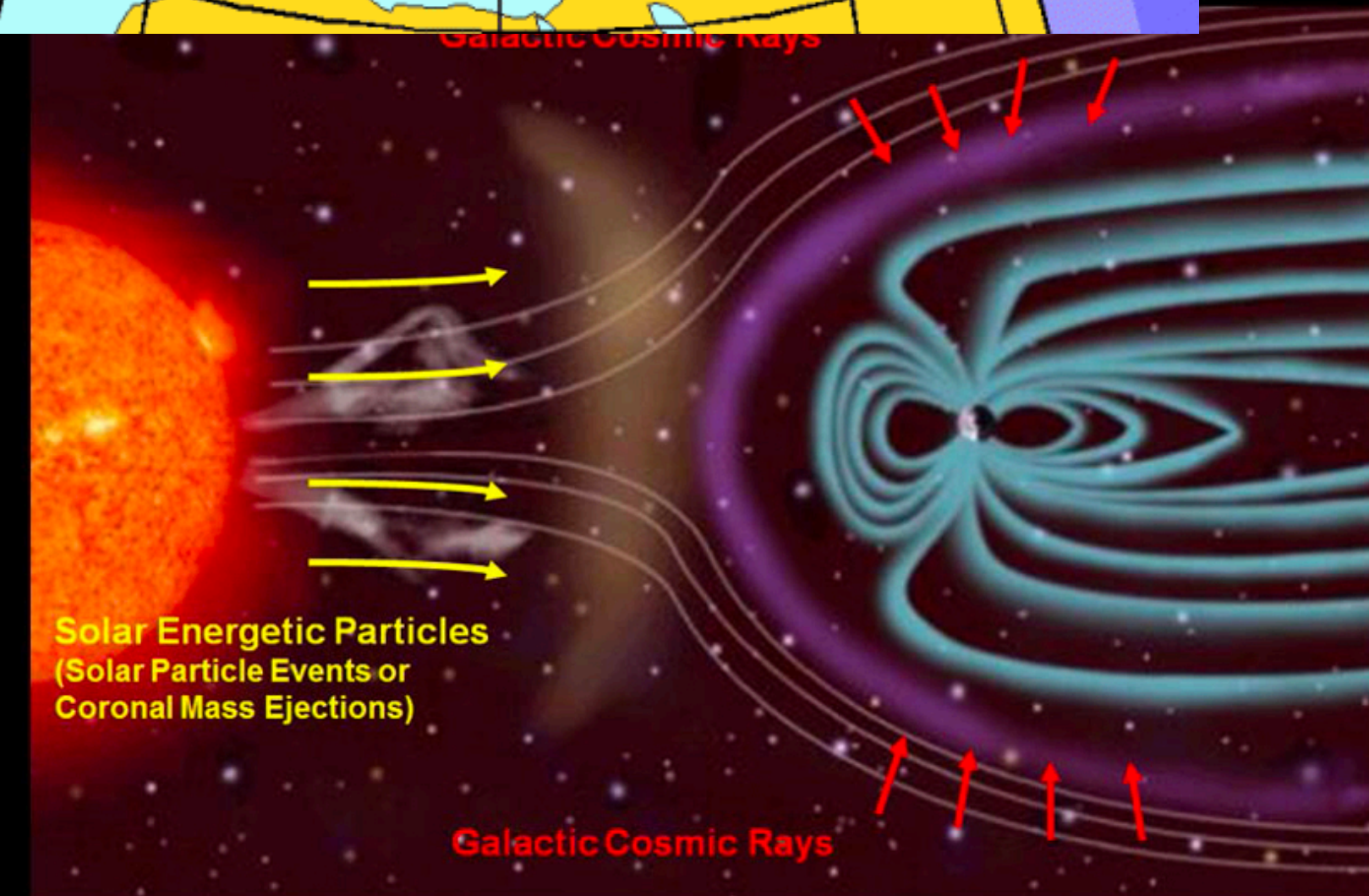
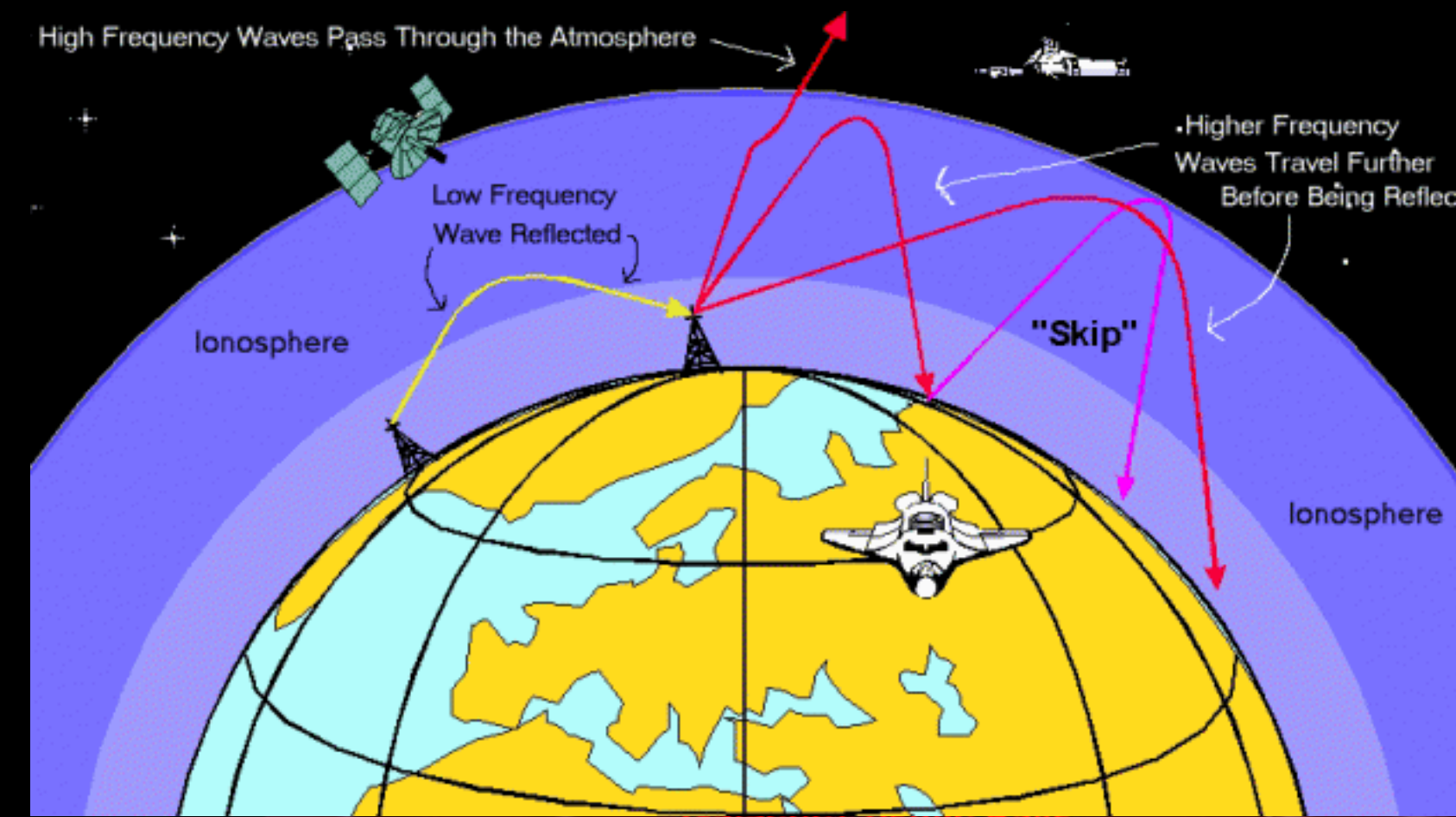
Scattering



Cordes & Lazio 1991

Regions of ionized plasma

- Ionosphere
- Interplanetary Medium (IPM)
- Interstellar Medium (ISM)

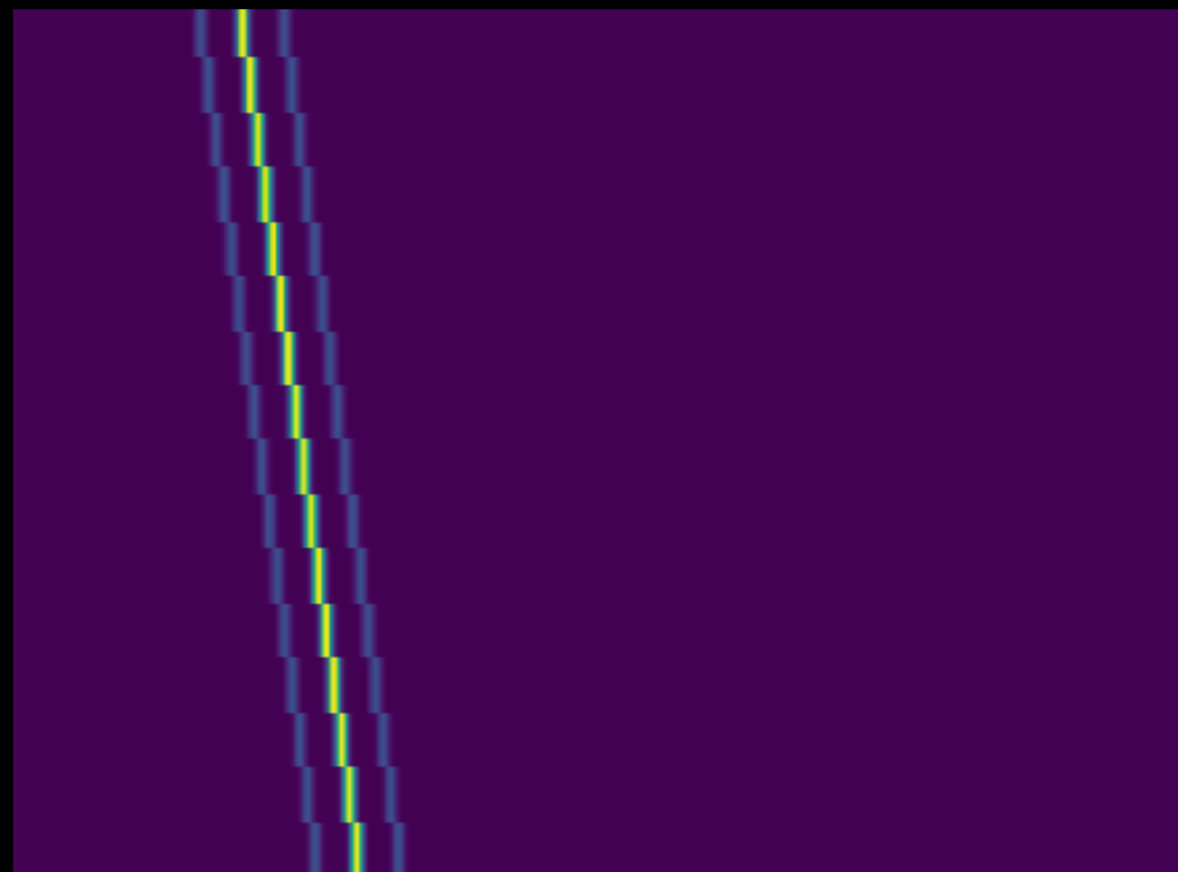


What would strongly scattered signals look like?

- Temporal scintillation
- Spectral broadening
- Pulse broadening
- Spectral de-correlation

What would strongly scattered signals look like?

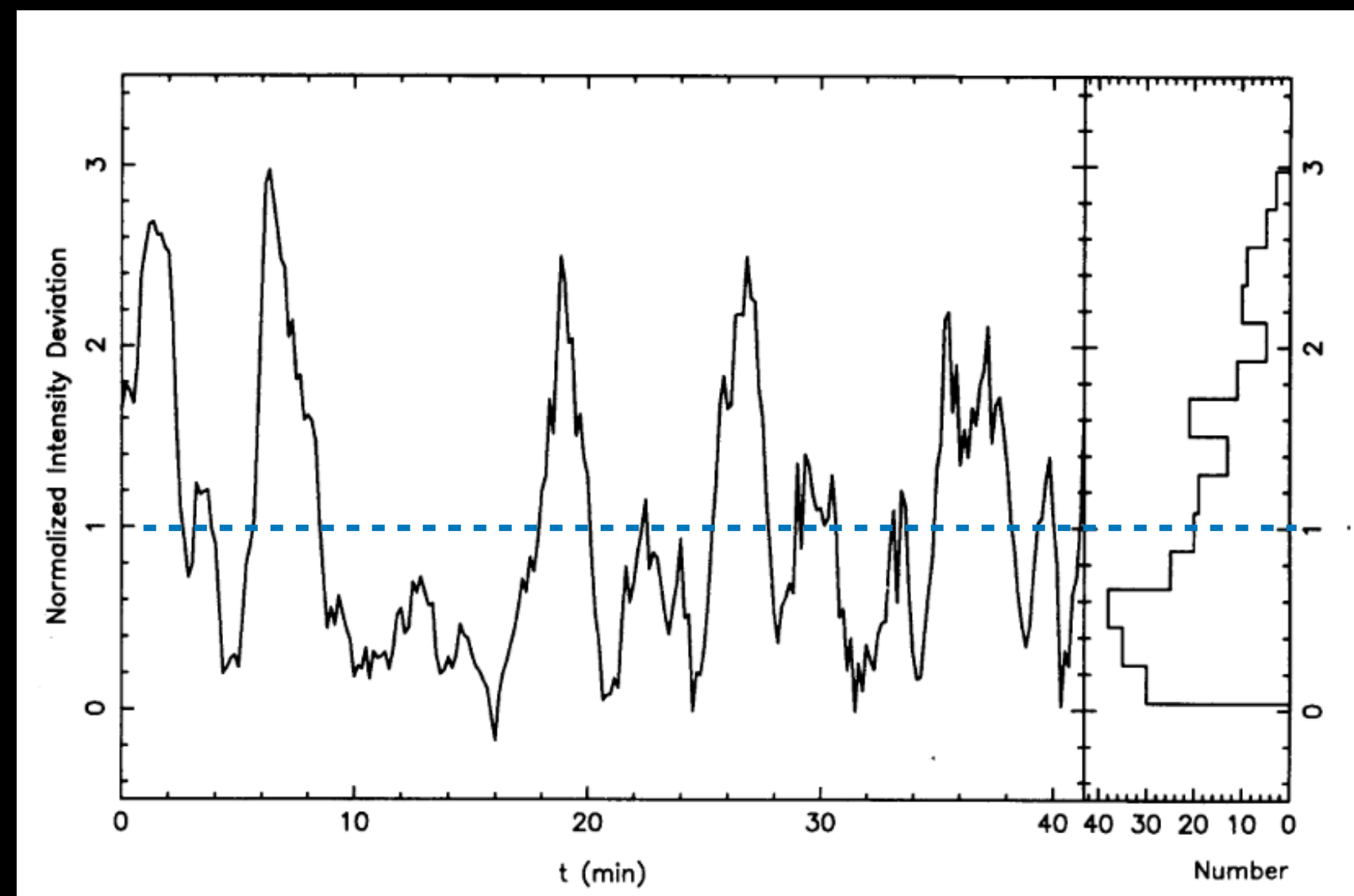
- Assuming a 100% duty-cycle narrowband transmitter



- Temporal scintillation
- Spectral broadening
- ~~Pulse broadening~~
- ~~Spectral de-correlation~~

What would strongly scattered signals look like?

- Assuming a 100% duty-cycle narrowband transmitter



Cordes & Lazio 1991

- **Temporal scintillation**
- ~~Spectral broadening~~
- ~~Pulse broadening~~
- ~~Spectral de-correlation~~

Prior SETI research on scattering

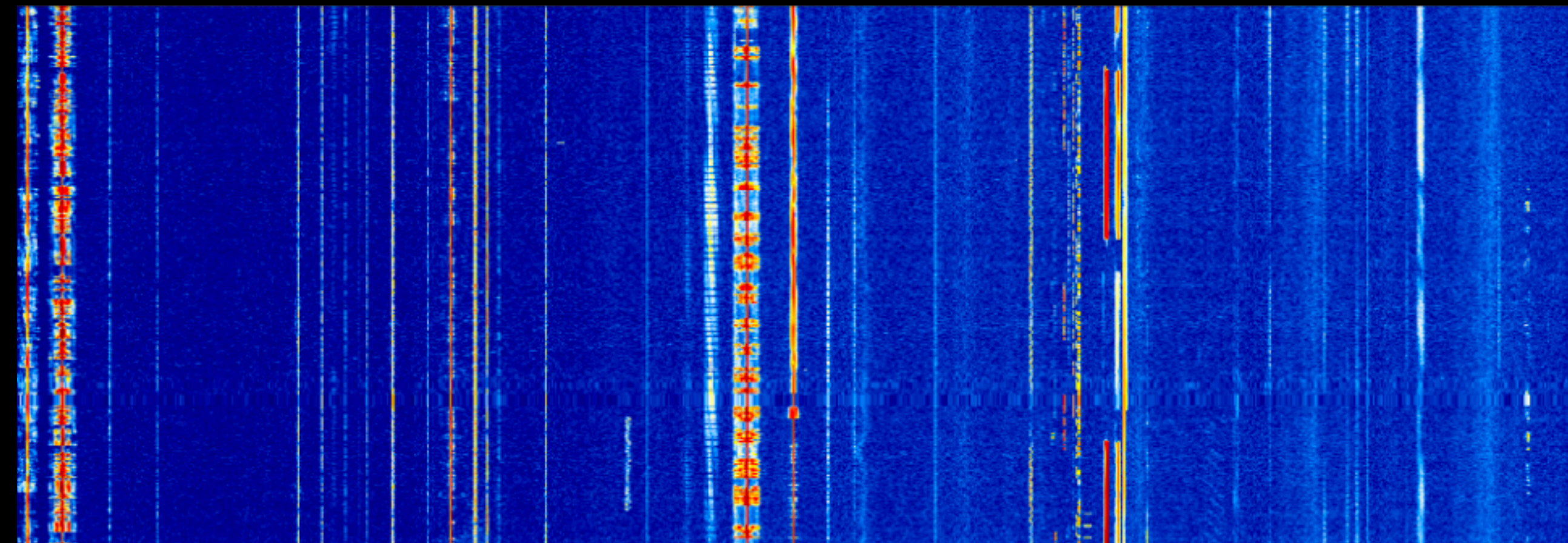
INTERSTELLAR SCATTERING EFFECTS ON THE DETECTION OF NARROW-BAND SIGNALS

JAMES M. CORDES AND T. JOSEPH LAZIO

National Astronomy and Ionosphere Center and Department of Astronomy, Cornell University, Ithaca, NY 14853

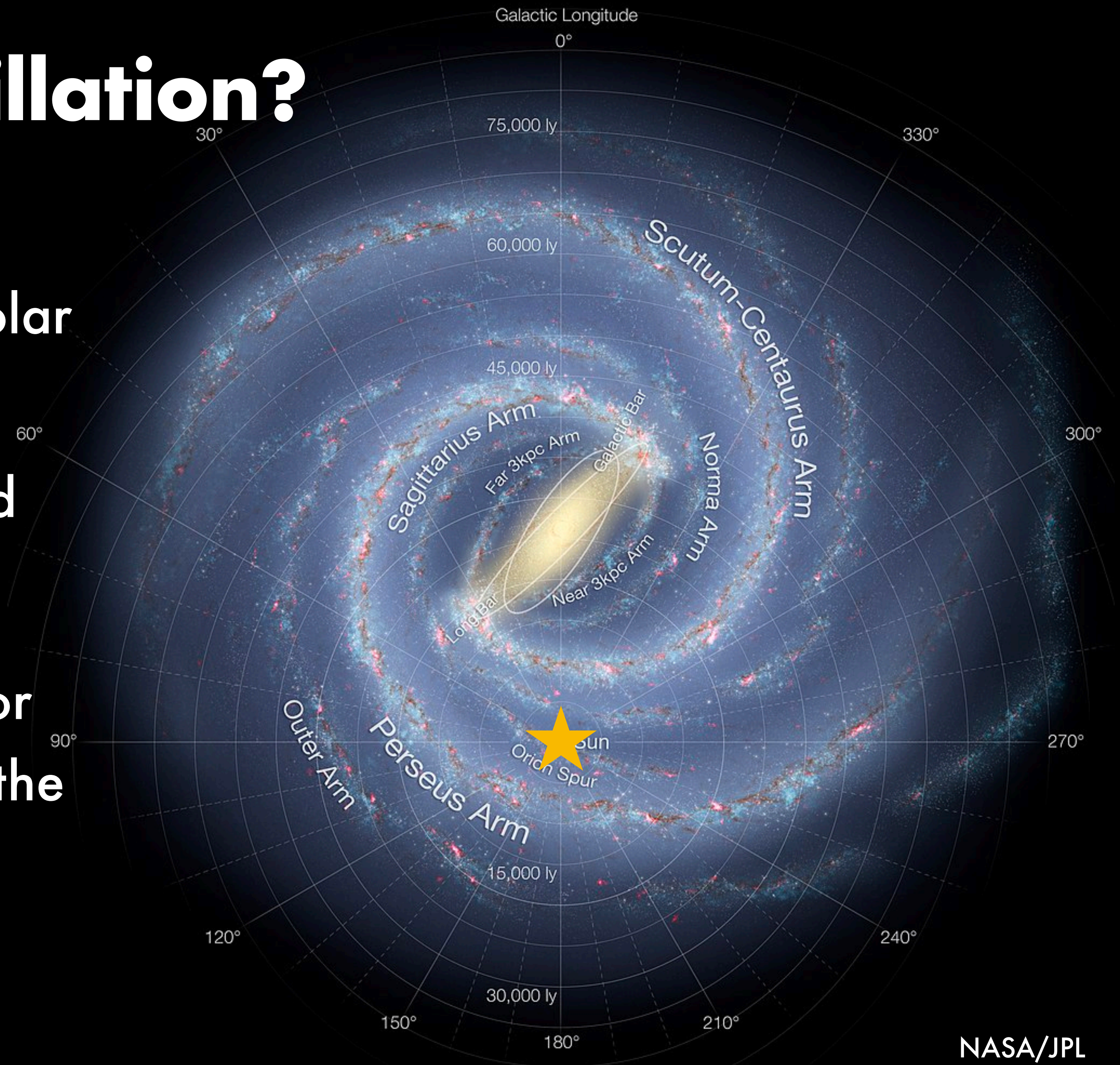
Received 1990 October 4; accepted 1991 January 15

- Many studies acknowledge scattering but attempt to avoid it
- Generally, SETI techniques aren't sensitive to detailed morphology
 - Noise, modulation, S/N
- Stochastic effects are hard to describe



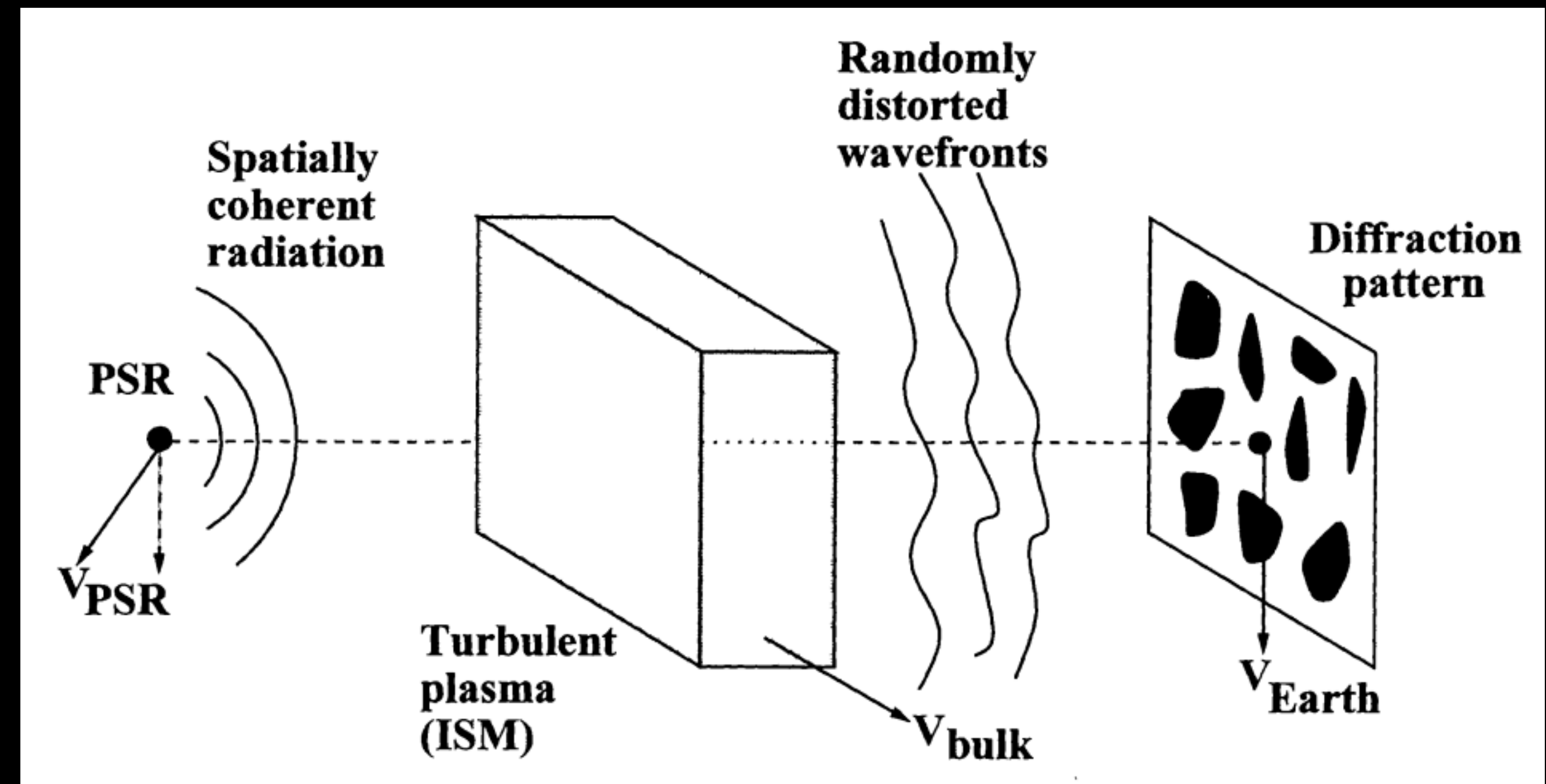
Why search for scintillation?

- A filter that directly implies extra-solar origin
- Well-suited for continuous or pulsed *narrowband* signals
- One of the best places to search for scintillation corresponds to one of the best places to look for ETI
— the Galactic Center



Diffraction scintillation in the ISM

- Electron density fluctuations in ionized plasma creates interference pattern
- Can lead to 100% intensity modulation, especially towards the Galactic center, with characteristic temporal scales Δt_d



Cordes 2002

INTERSTELLAR SCATTERING EFFECTS ON THE DETECTION OF NARROW-BAND SIGNALS

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SCINTILLATION-INDUCED INTERMITTENCY IN SETI

JAMES M. CORDES,^{1,2,3} T. JOSEPH W. LAZIO,^{1,2} AND CARL SAGAN^{1,3,4,5}

Received 1996 May 15; accepted 1997 May 9

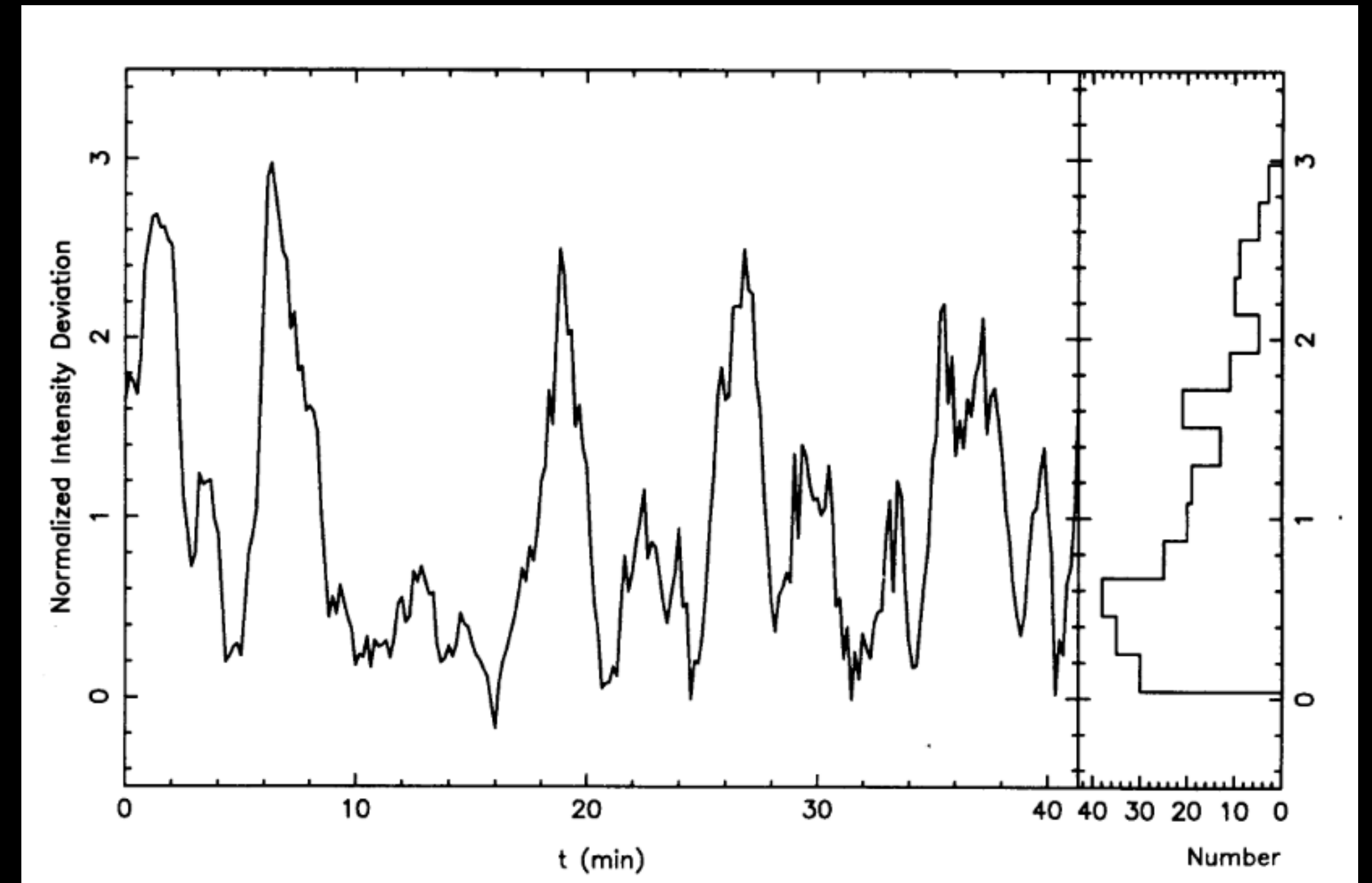
- Showed that scattering can both help and hinder SETI
- Developed asymptotic expressions for detectability

What would strongly scintillated signals look like?

- Assuming a 100% duty-cycle narrowband transmitter
- Exponential intensity distribution

$$p(I) \propto e^{-I/\langle I \rangle}$$

- Near Gaussian auto-correlation (ACF)

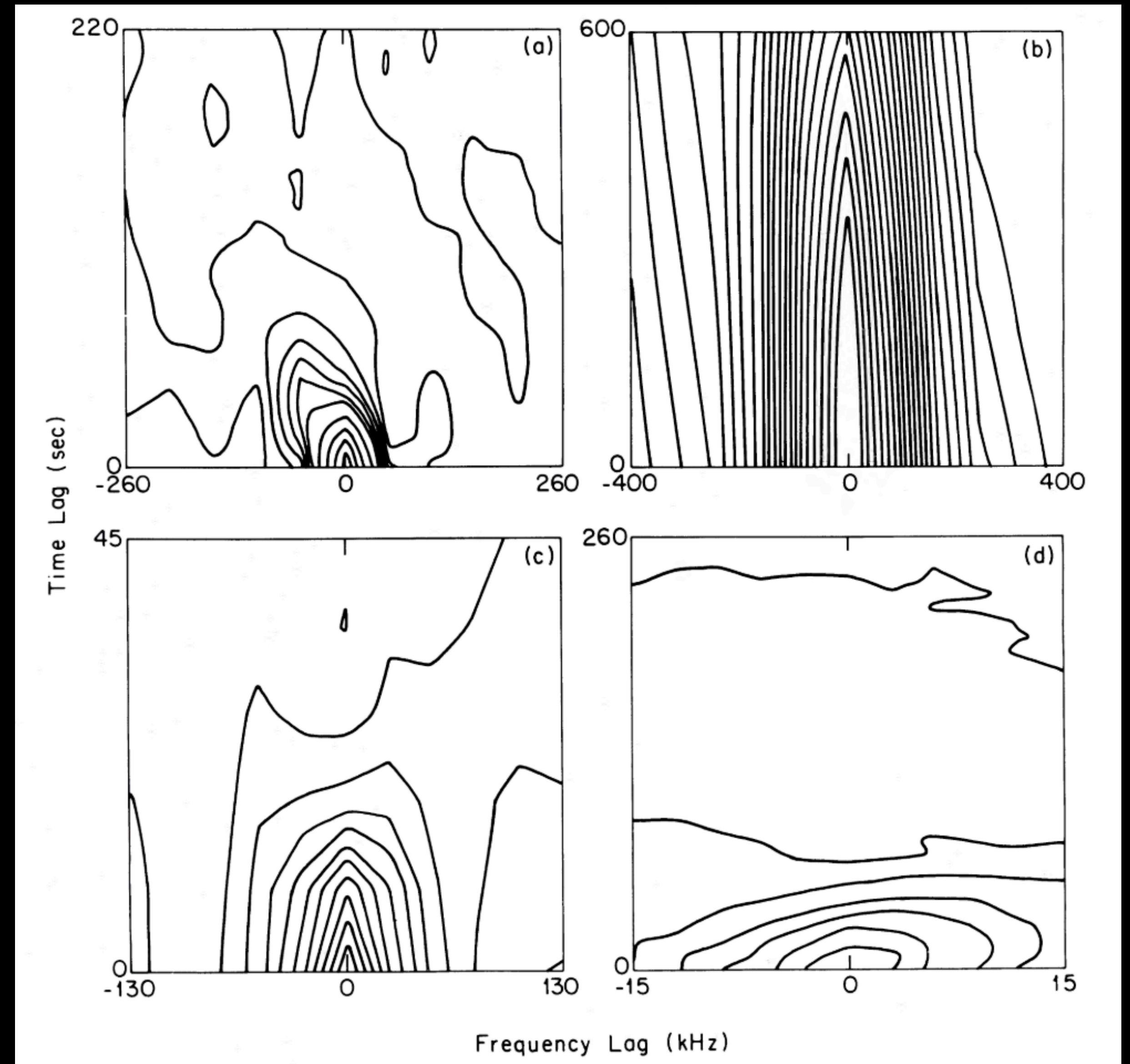


Cordes & Lazio 1991

What would strongly scintillated signals look like?

- Assuming a 100% duty-cycle narrowband transmitter
- Exponential intensity distribution
- Near Gaussian auto-correlation (ACF)

$$\rho(\tau) \sim e^{-(\tau/\Delta t_d)^{5/3}}$$



Cordes 1986

Can we detect scintillated narrowband technosignatures?

1. What scintillation timescales should we expect?
2. How can we probe asymptotic statistics?
3. Can we differentiate scintillated signals from existing RFI?
4. How can we design a survey to search for scintillated technosignatures?

Can we detect scintillated narrowband technosignatures?

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Estimating scattering strength

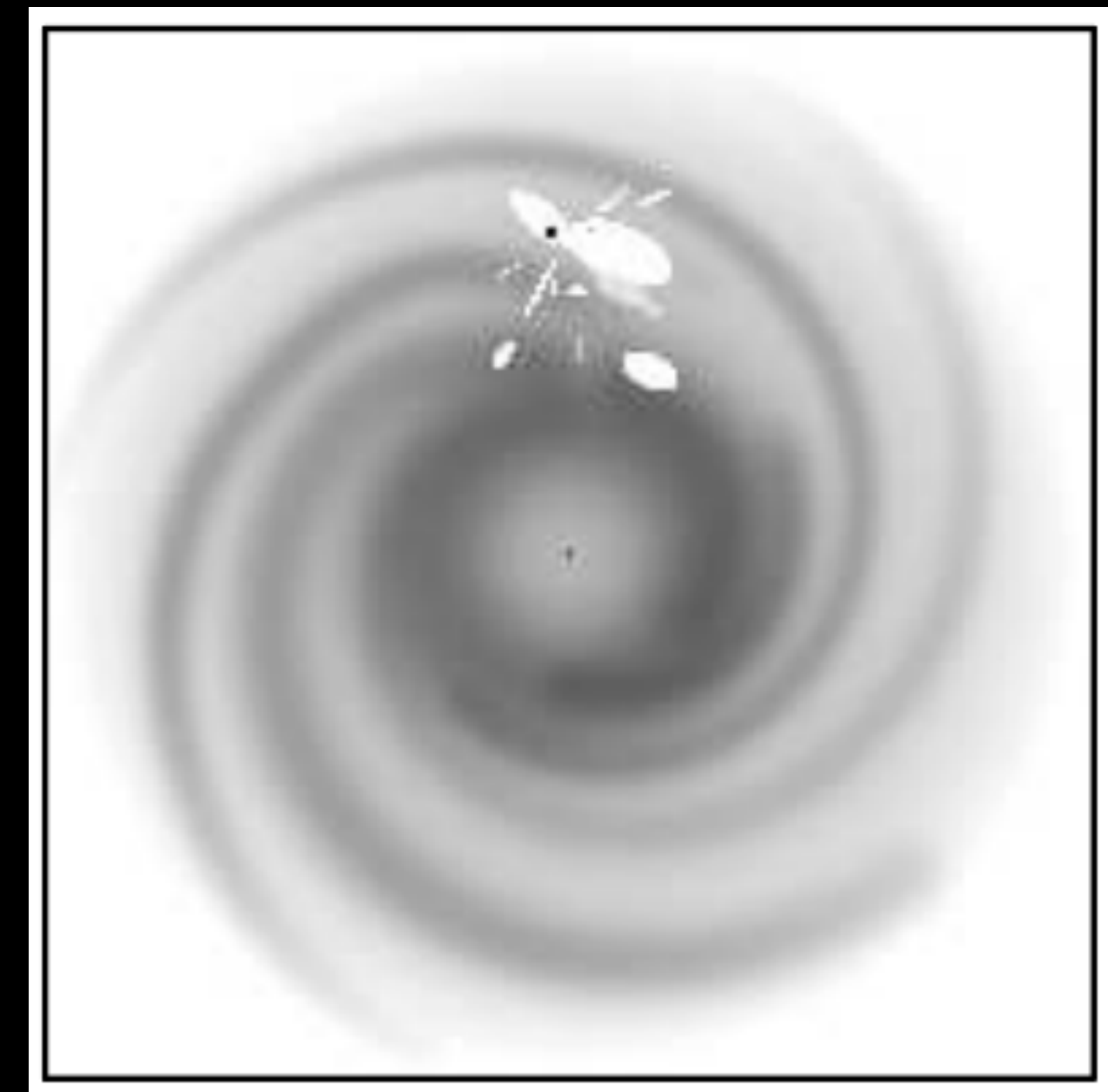
- NE2001 model: the standard for estimating pulsar distances for a while
- Estimates scattering parameters
- Computes values assuming defaults of 1 GHz and 100 km/s – requires scaling!

$$\Delta t_d \propto \nu^{6/5} \nu_T^{-1}$$

NE2001. I. A NEW MODEL FOR THE GALACTIC DISTRIBUTION
OF FREE ELECTRONS AND ITS FLUCTUATIONS

J. M. CORDES
Astronomy Department and NAIC, Cornell University, Ithaca, NY 14853
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Naval Research Lab, Code 7213, Washington, D.C. 20375-5351
Joseph.Lazio@nrl.navy.mil



Parameter space exploration of scattering parameters

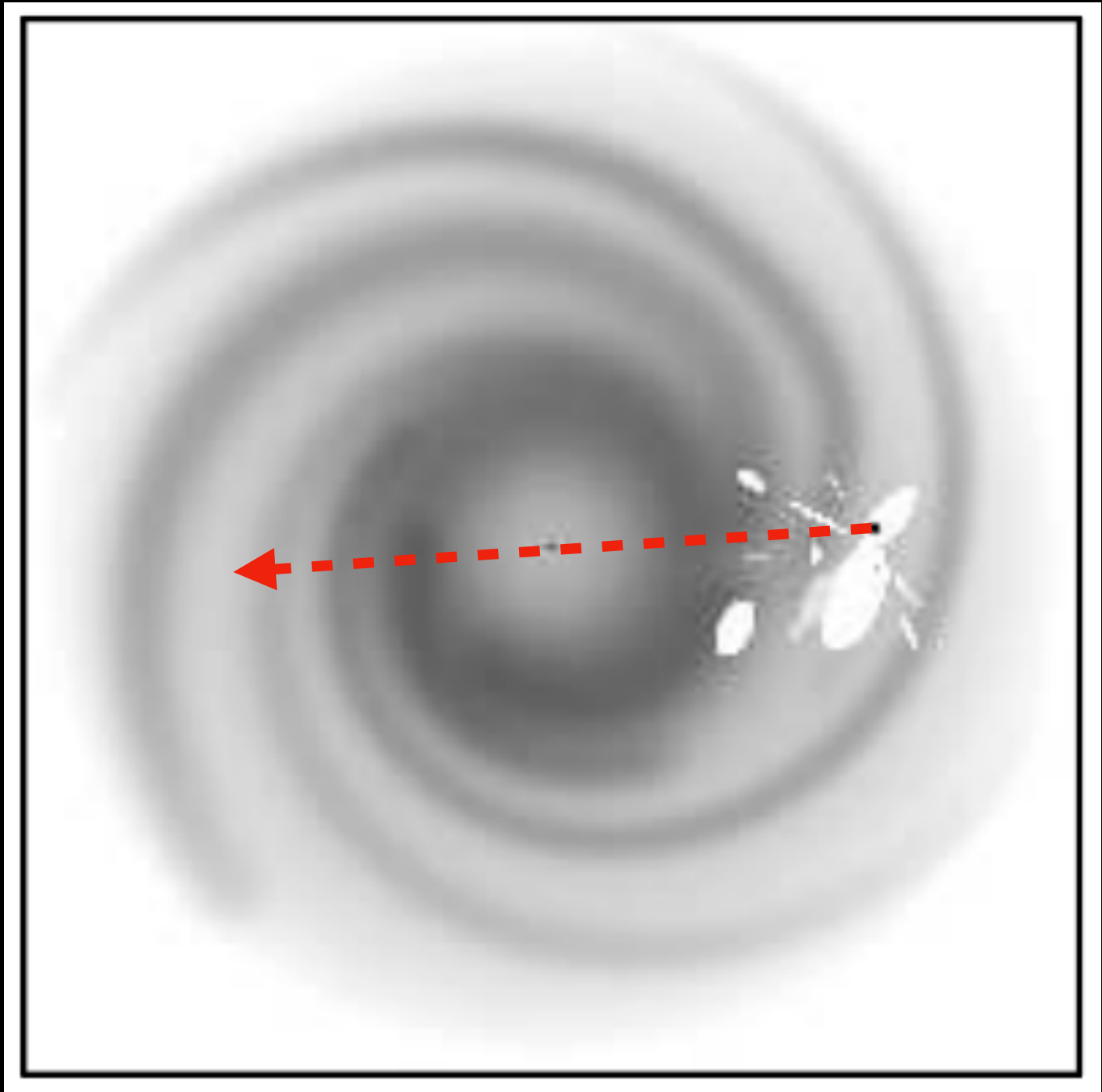
- A priori, we do not know:
 - Sky direction
 - Frequency
 - Distance
 - Transverse velocity

Monte Carlo sampling!

$$\Delta t_d \propto \nu^{6/5} \nu_T^{-1}$$

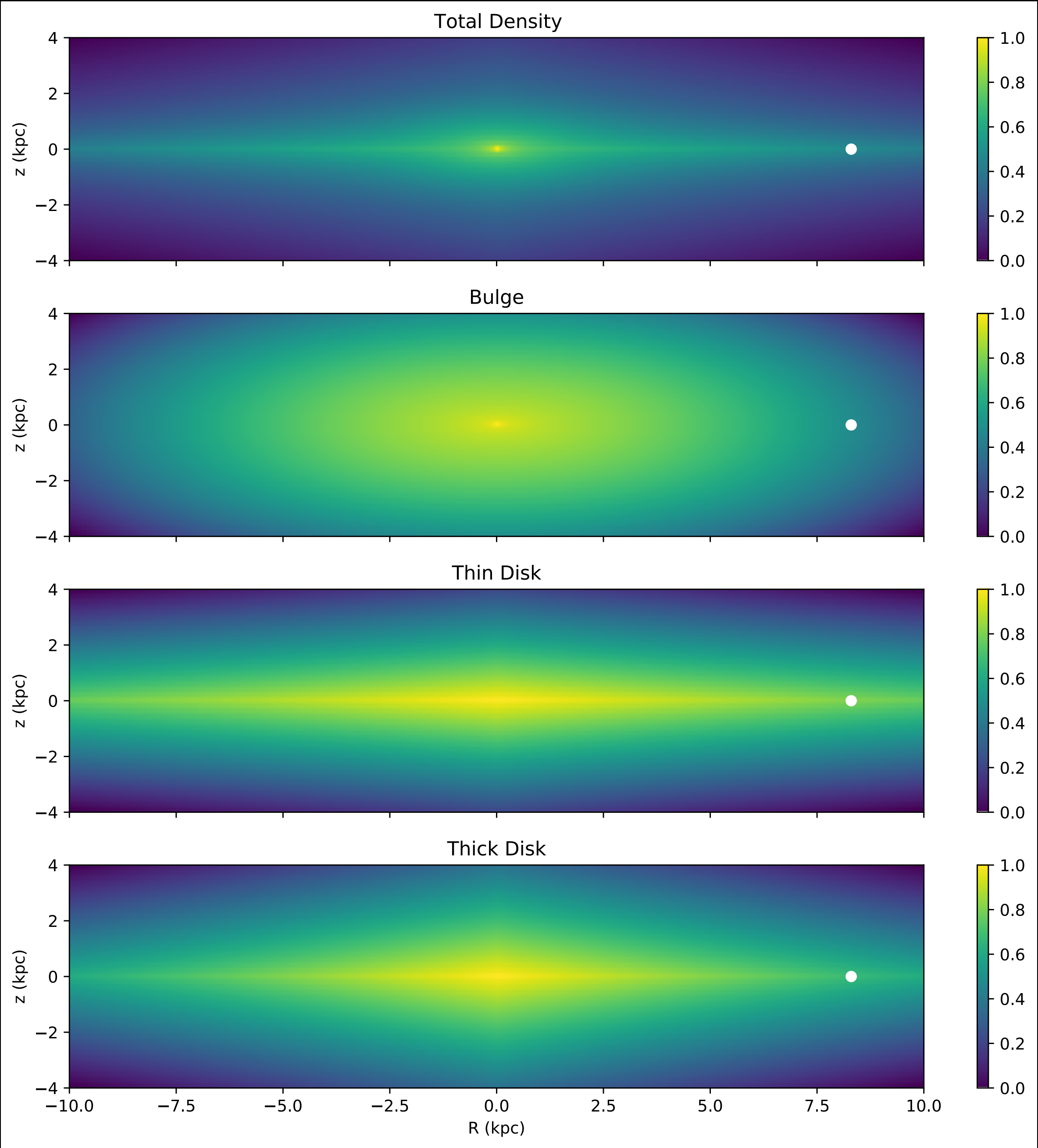
- | | | |
|-----------------------|---|-------------------------------------|
| • Sky direction | → | Chosen parameter |
| • Frequency | → | Uniform sampling within chosen band |
| • Distance | → | Uniform or density based sampling |
| • Transverse velocity | → | Uniform sampling |

Density-based sampling

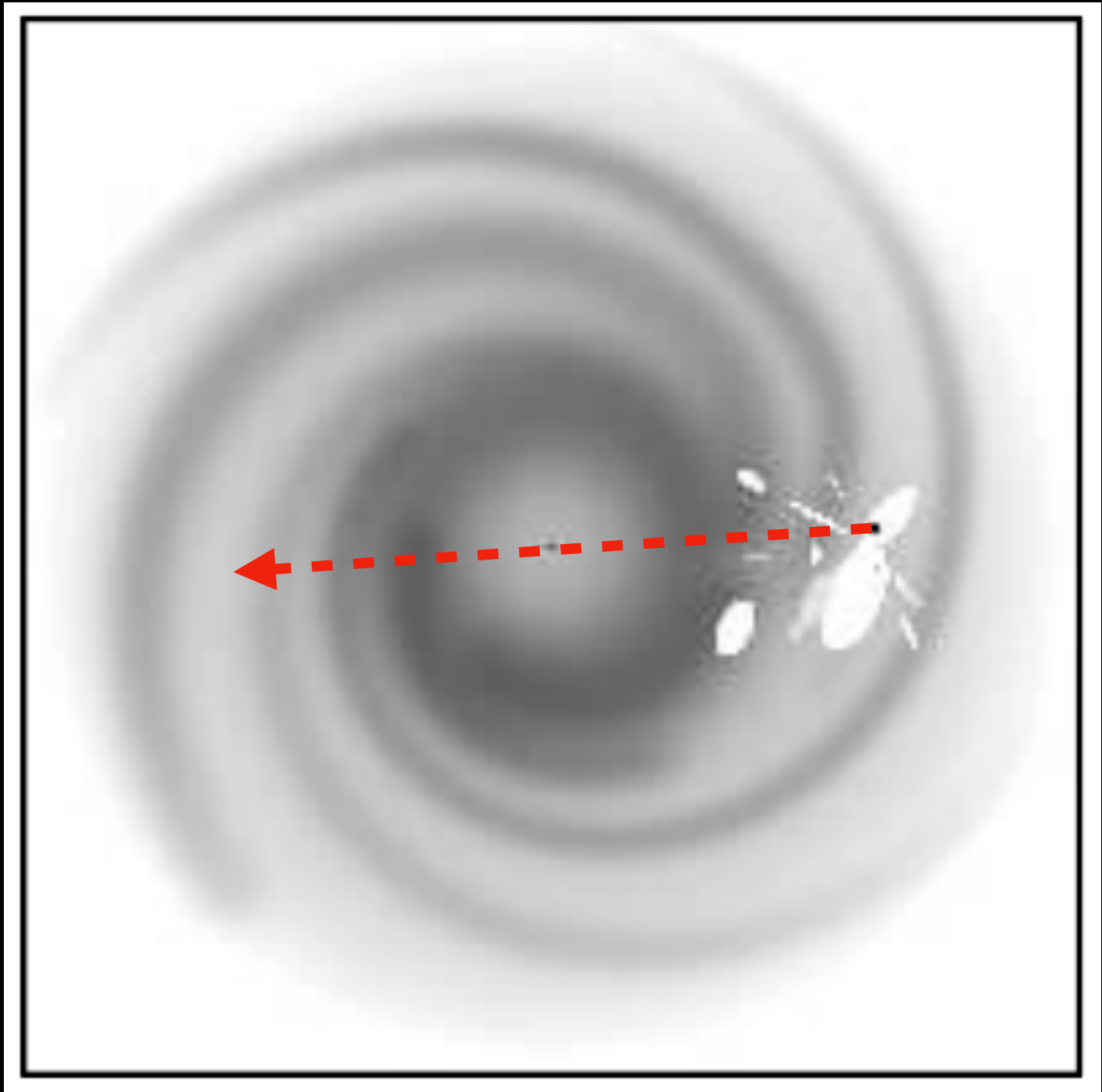


Cordes & Lazio 2002

McMillan 2017, Gowanlock et al. 2011, Carroll & Ostlie 2007

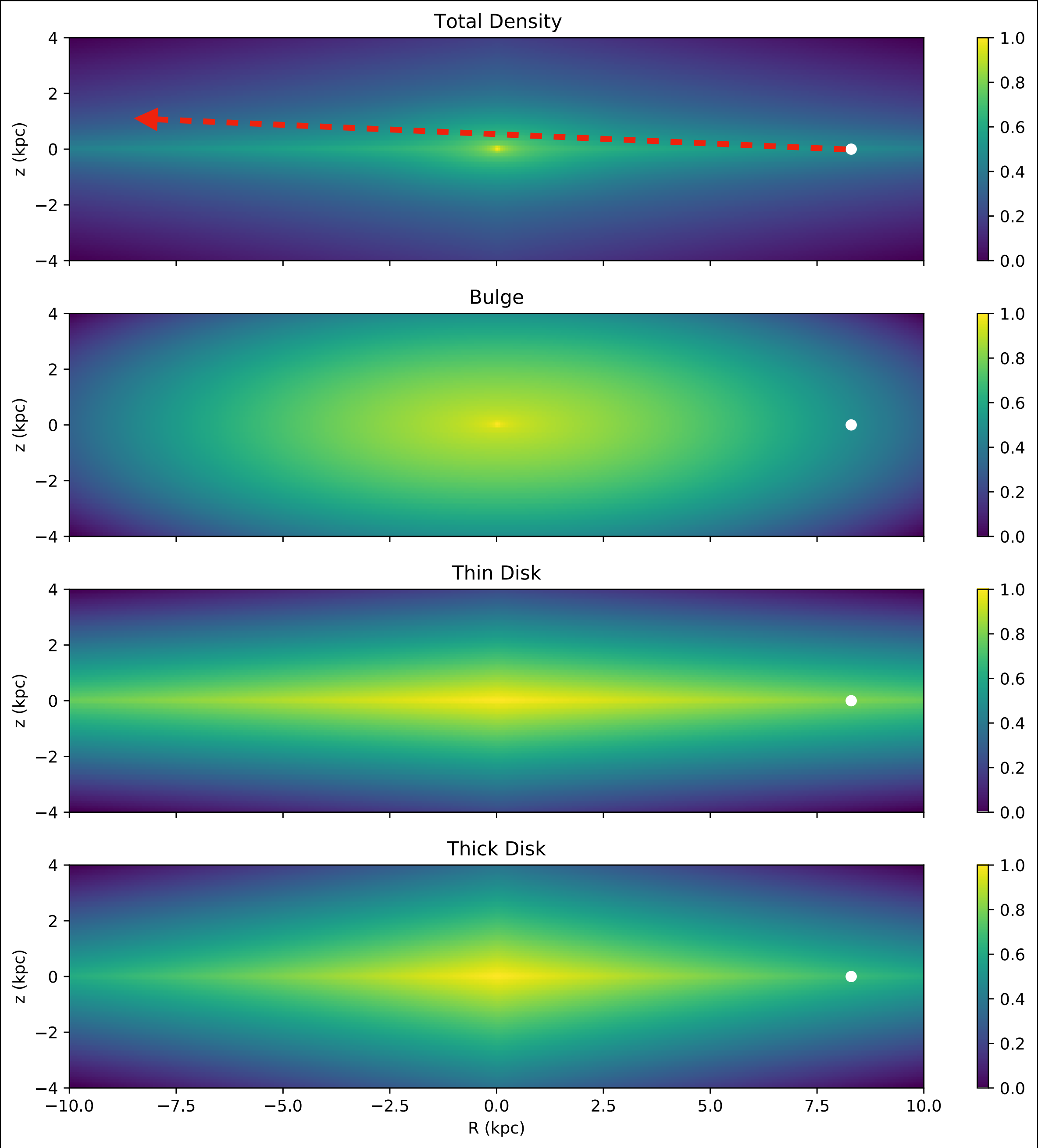


Density-based sampling

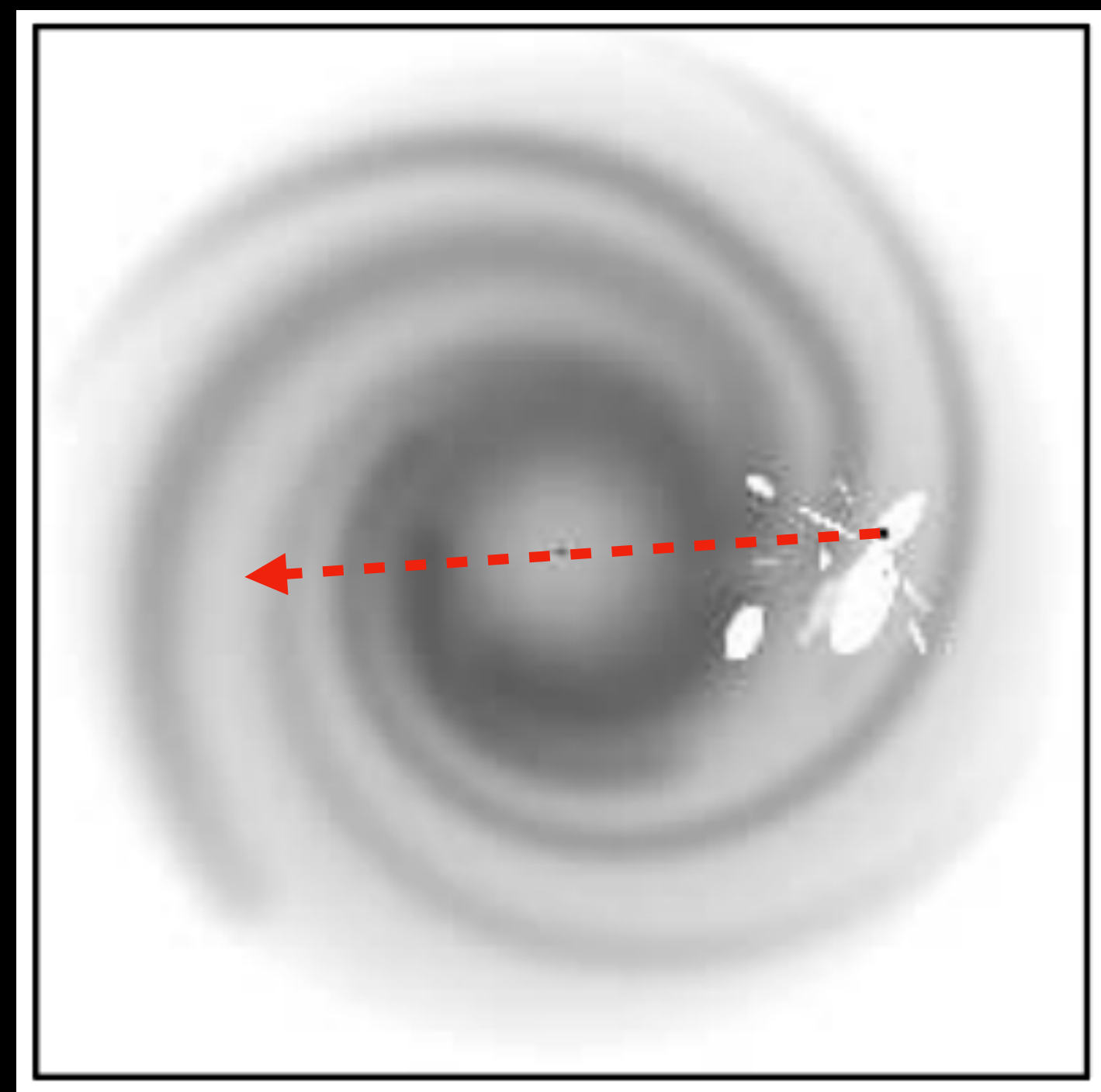


Cordes & Lazio 2002

McMillan 2017, Gowanlock et al. 2011, Carroll & Ostlie 2007



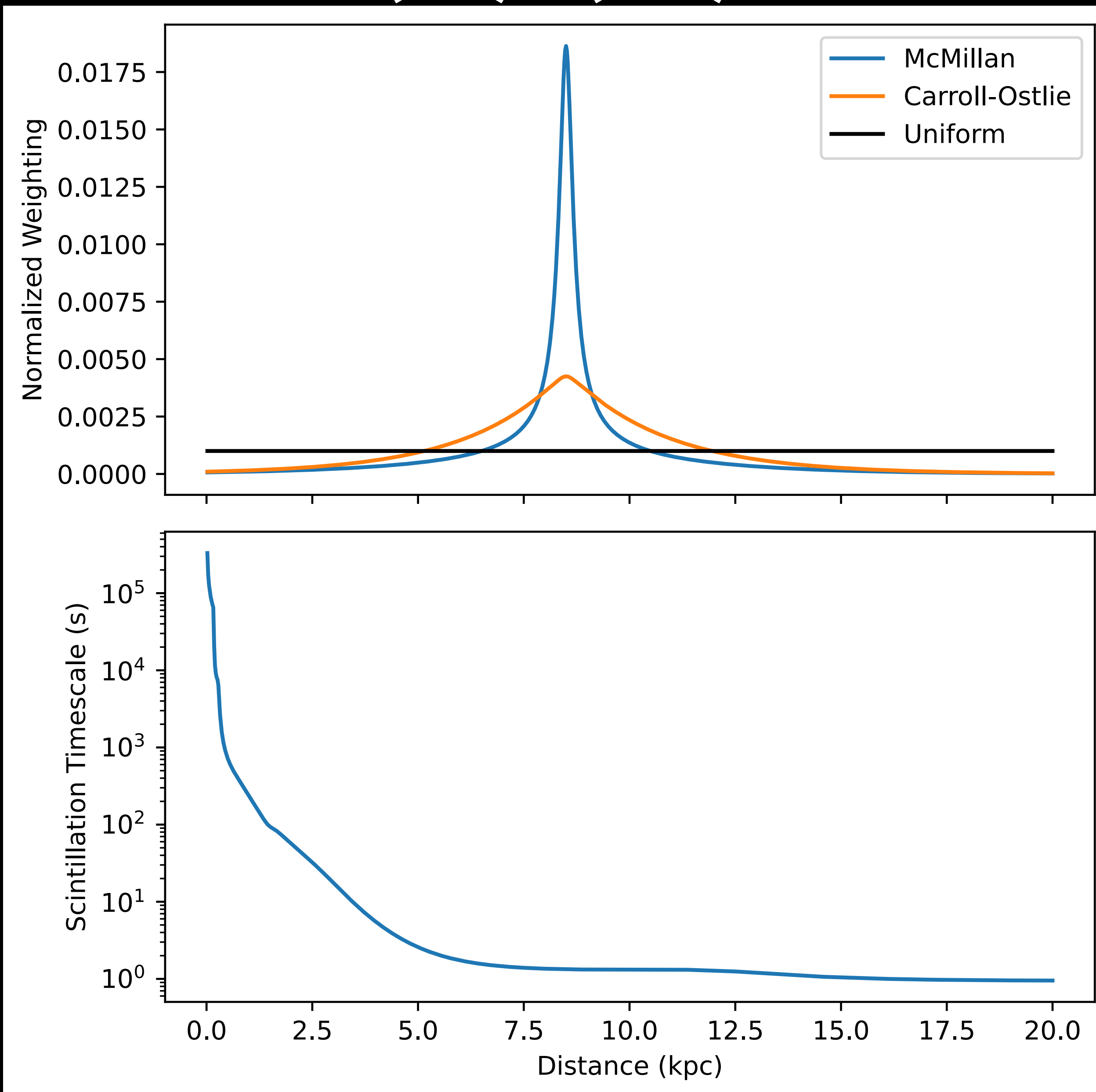
Density-based sampling



Cordes & Lazio 2002

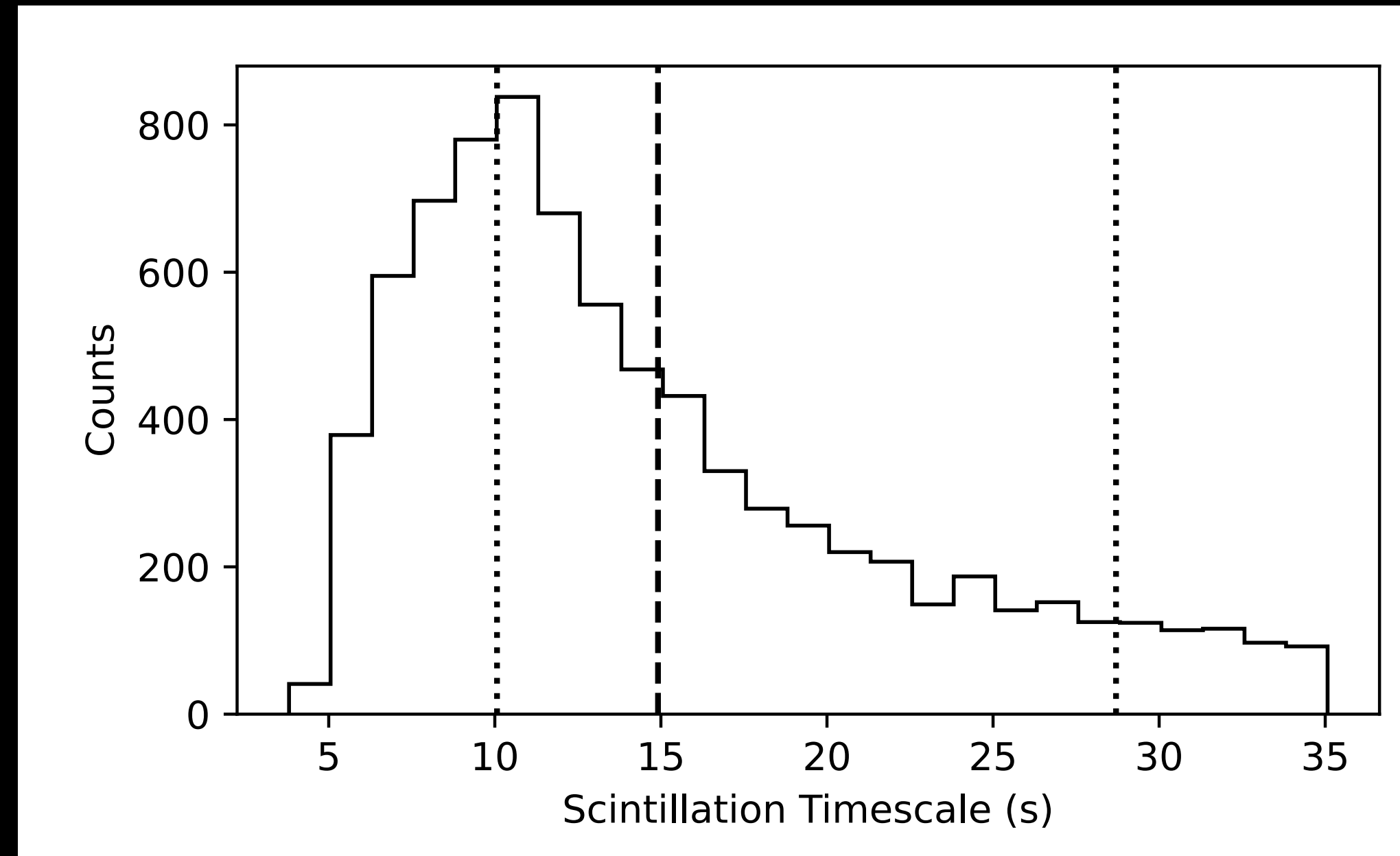
McMillan 2017, Gowanlock et al. 2011, Carroll & Ostlie 2007

$$(l, b) = (1, 0)$$



Inter-quartile range

Median



C-band

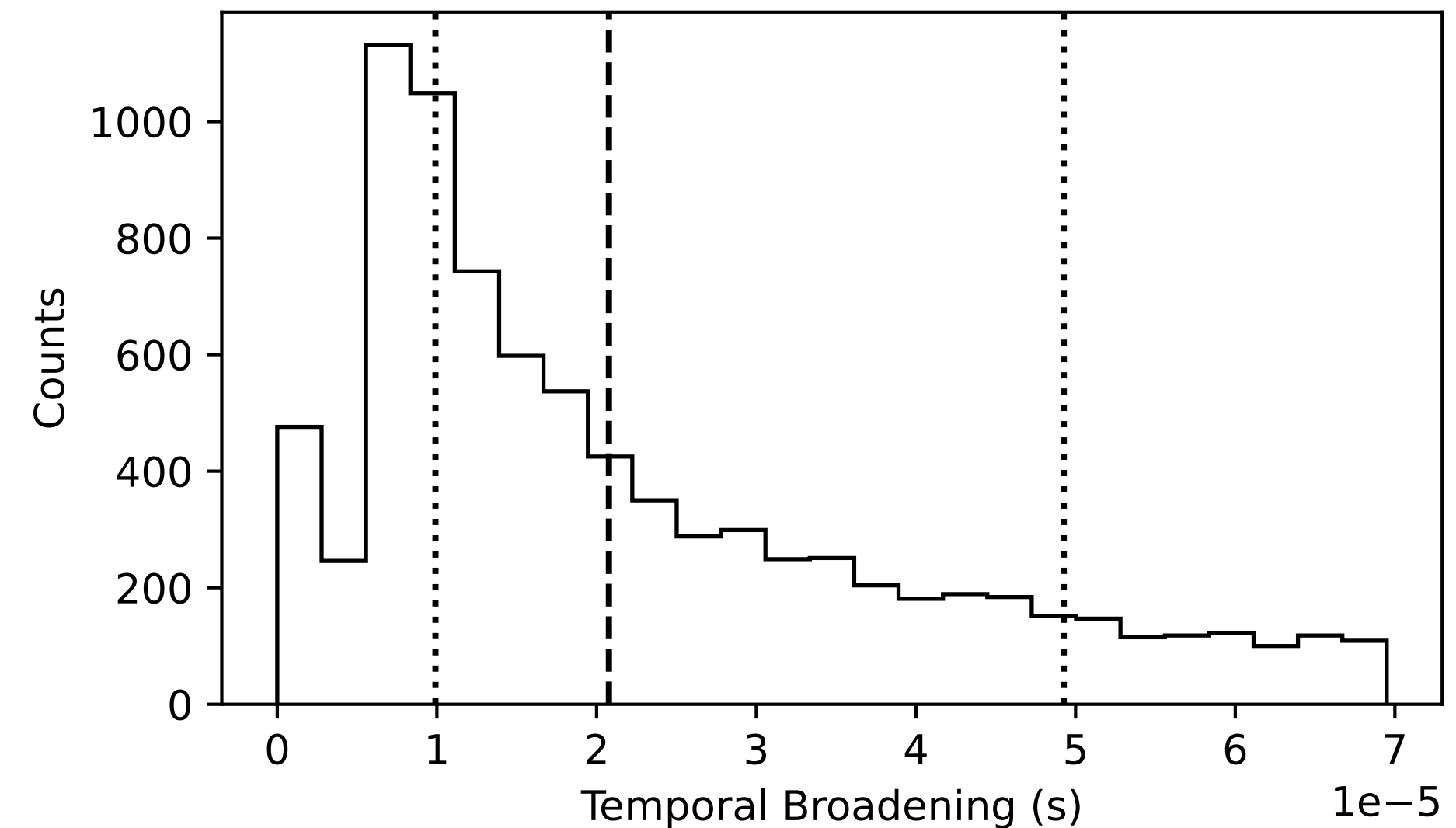
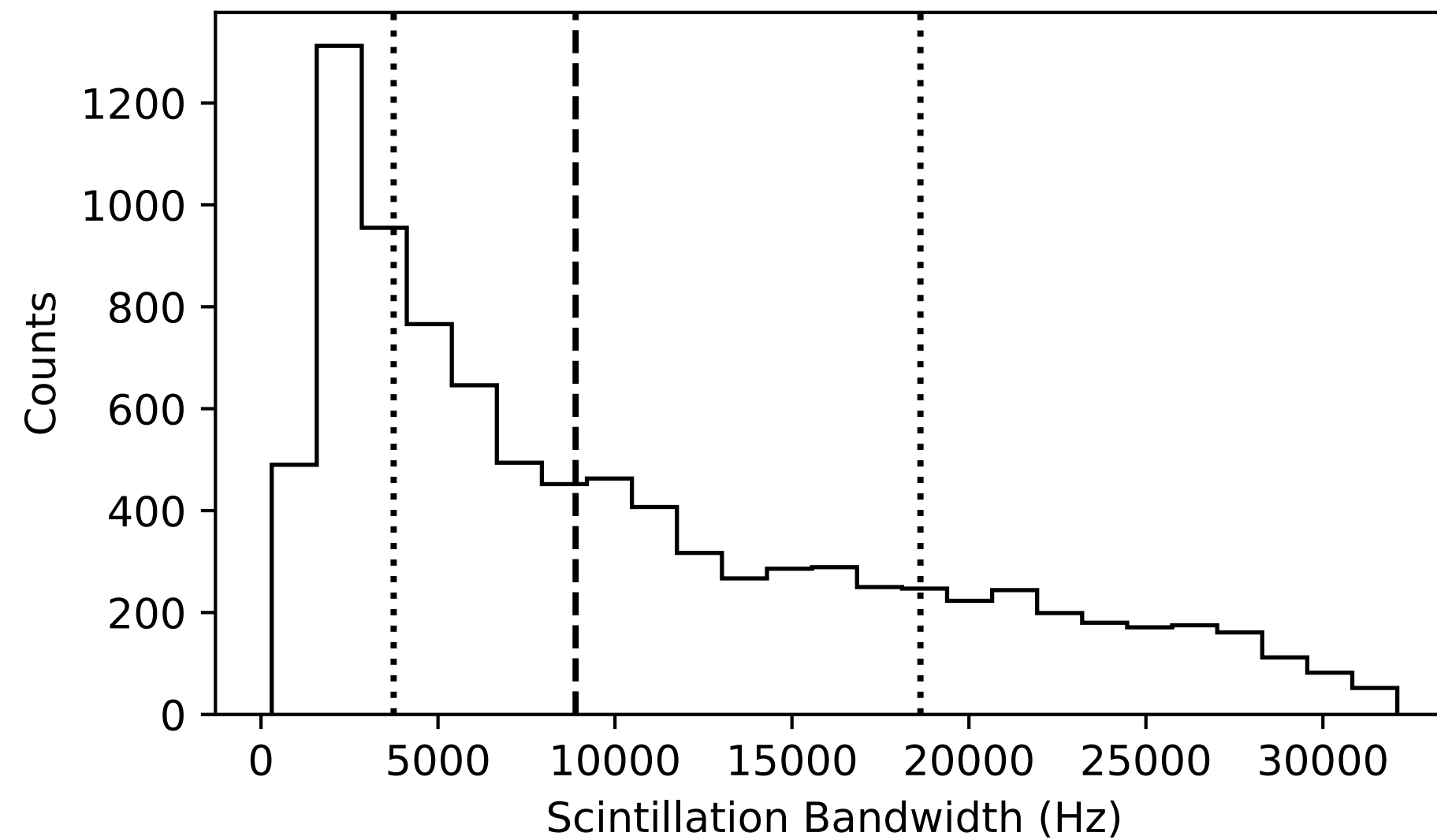
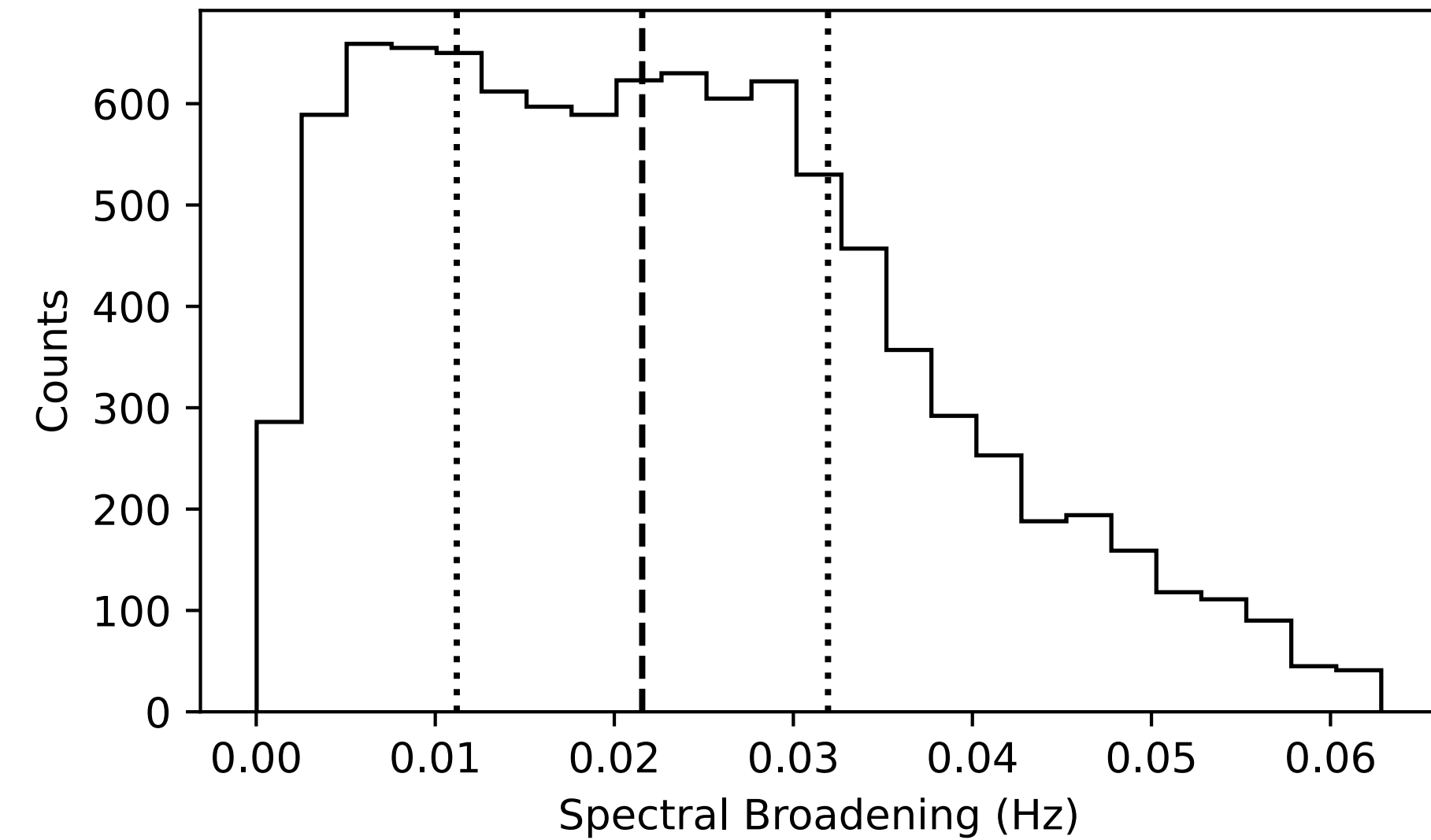
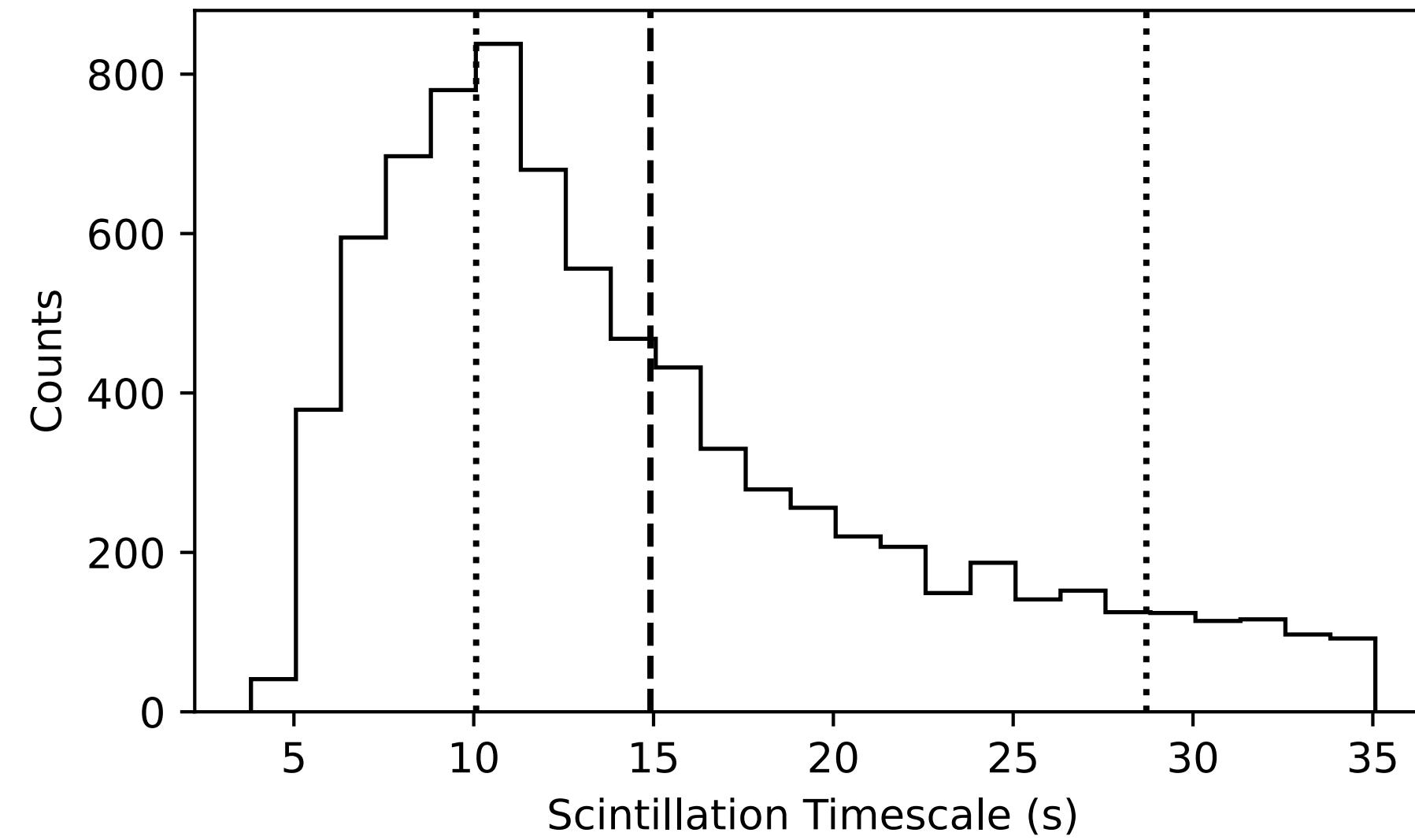
(l, b) = (1, 0)

Inter-quartile range

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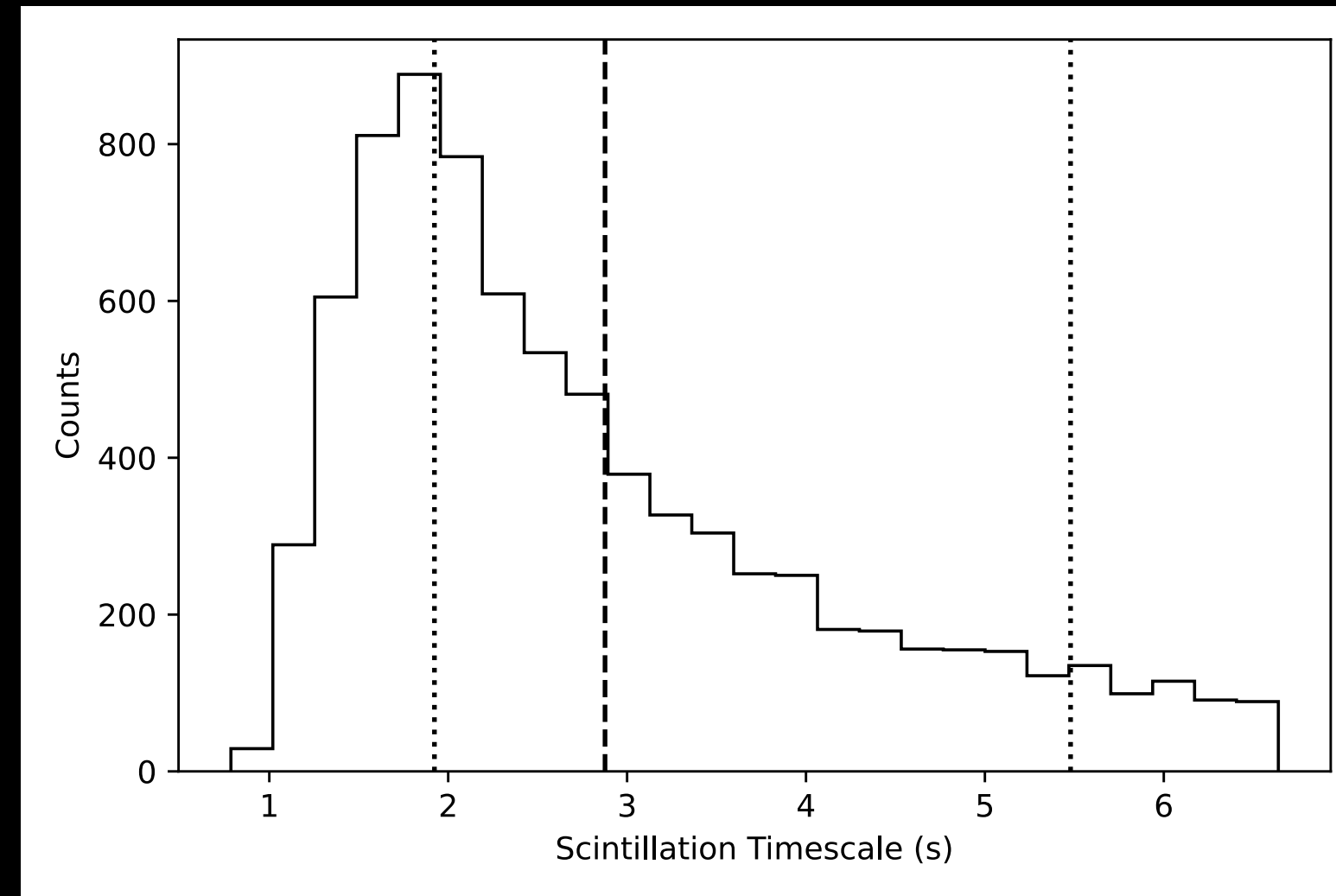
Monte Carlo-sampled timescales

Distance sampling

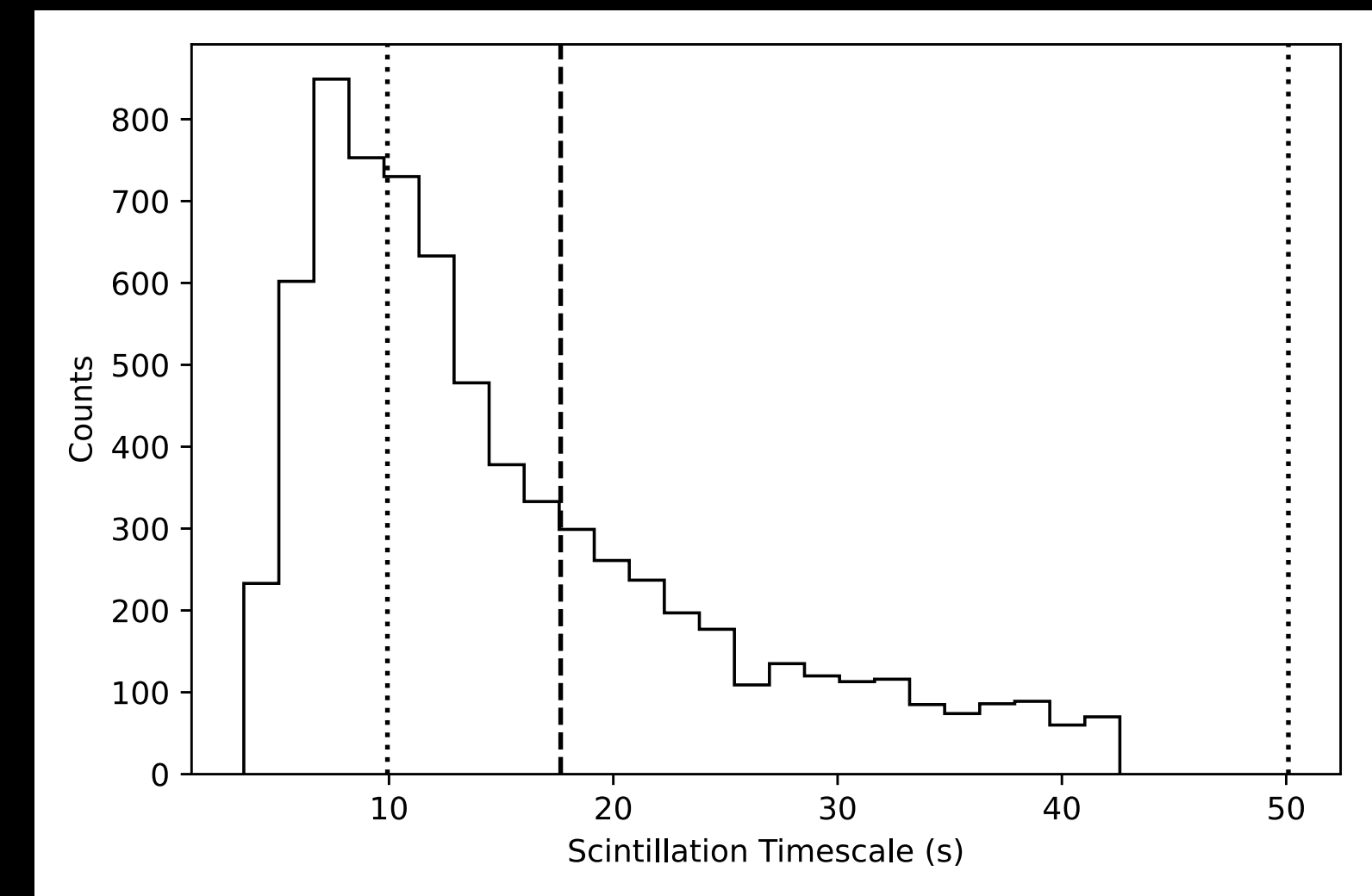
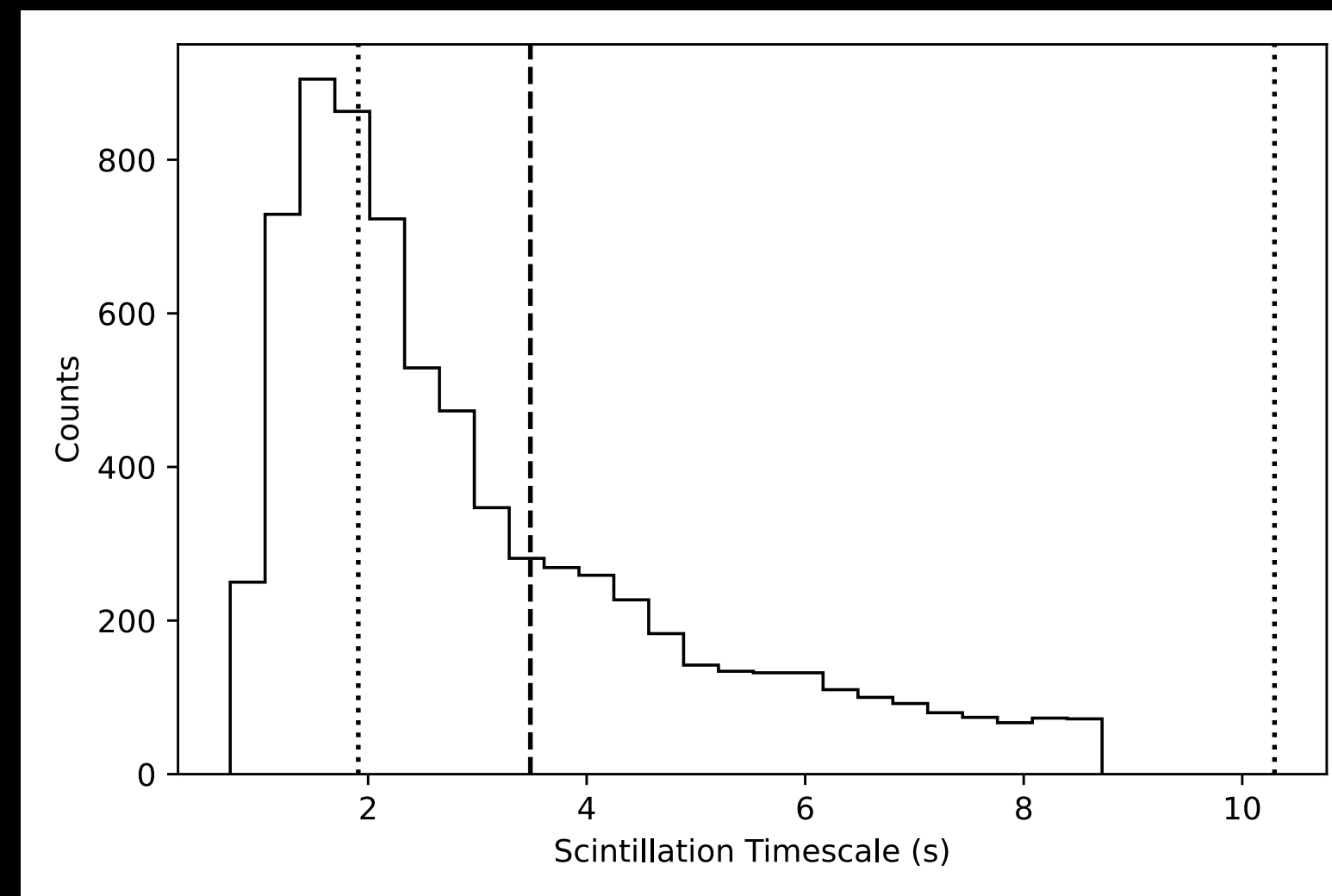
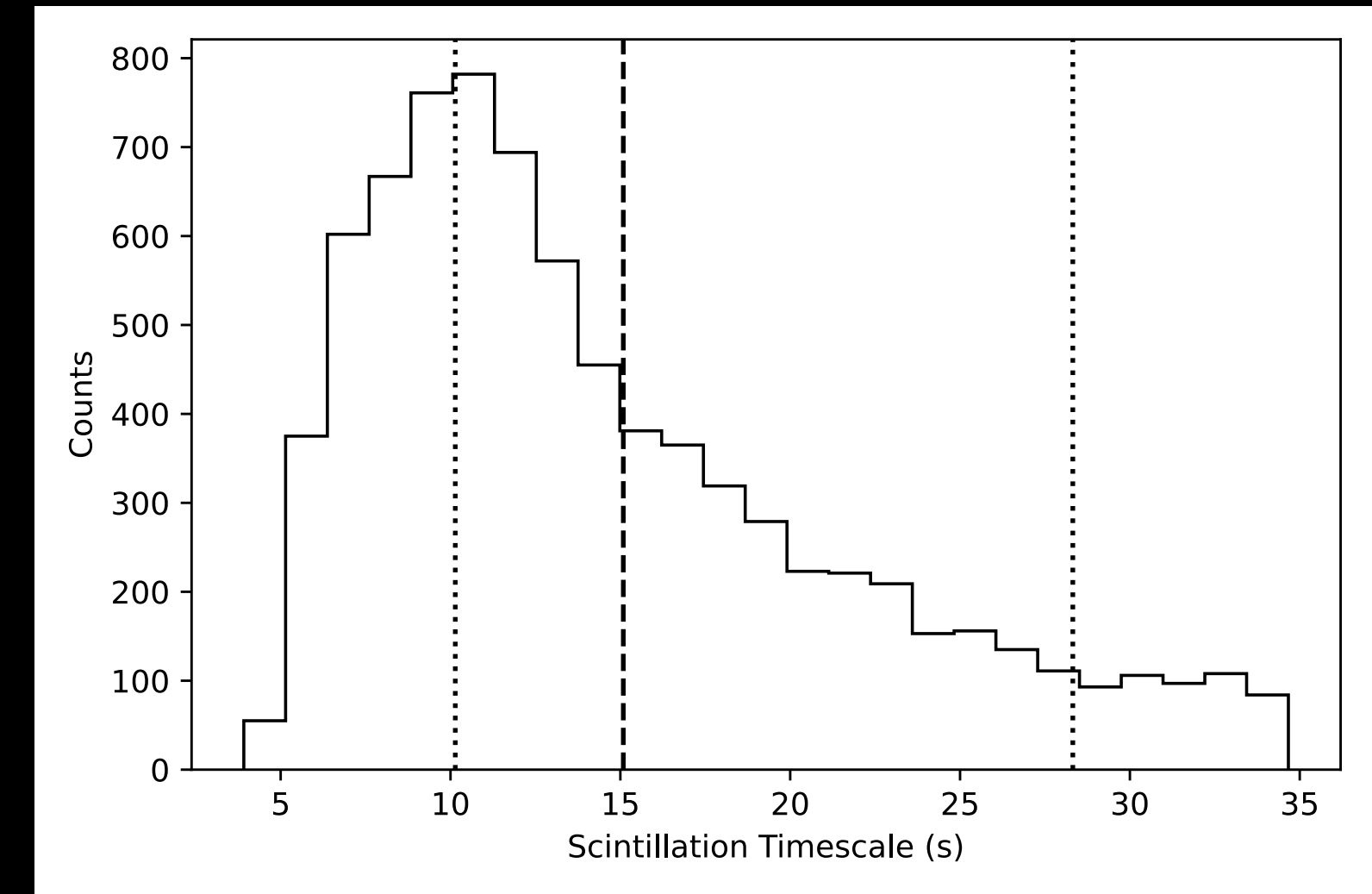
Weighted

Uniform

L band



C band



Example: Statistics at different bands

$(l, b) = (1, 0)$

Band	Frequency (GHz)	Median (s)	IQR (s)	Mode (s)
LOFAR	0.110 – 0.240	0.22	0.14 – 0.41	0.14
L	1.1 – 1.9	2.9	1.9 – 5.6	1.9
S	1.8 – 2.8	4.8	3.3 – 9.0	3.1
C	3.95 – 8	15	10 – 28	11
X	8 – 11.6	28	19 – 52	16

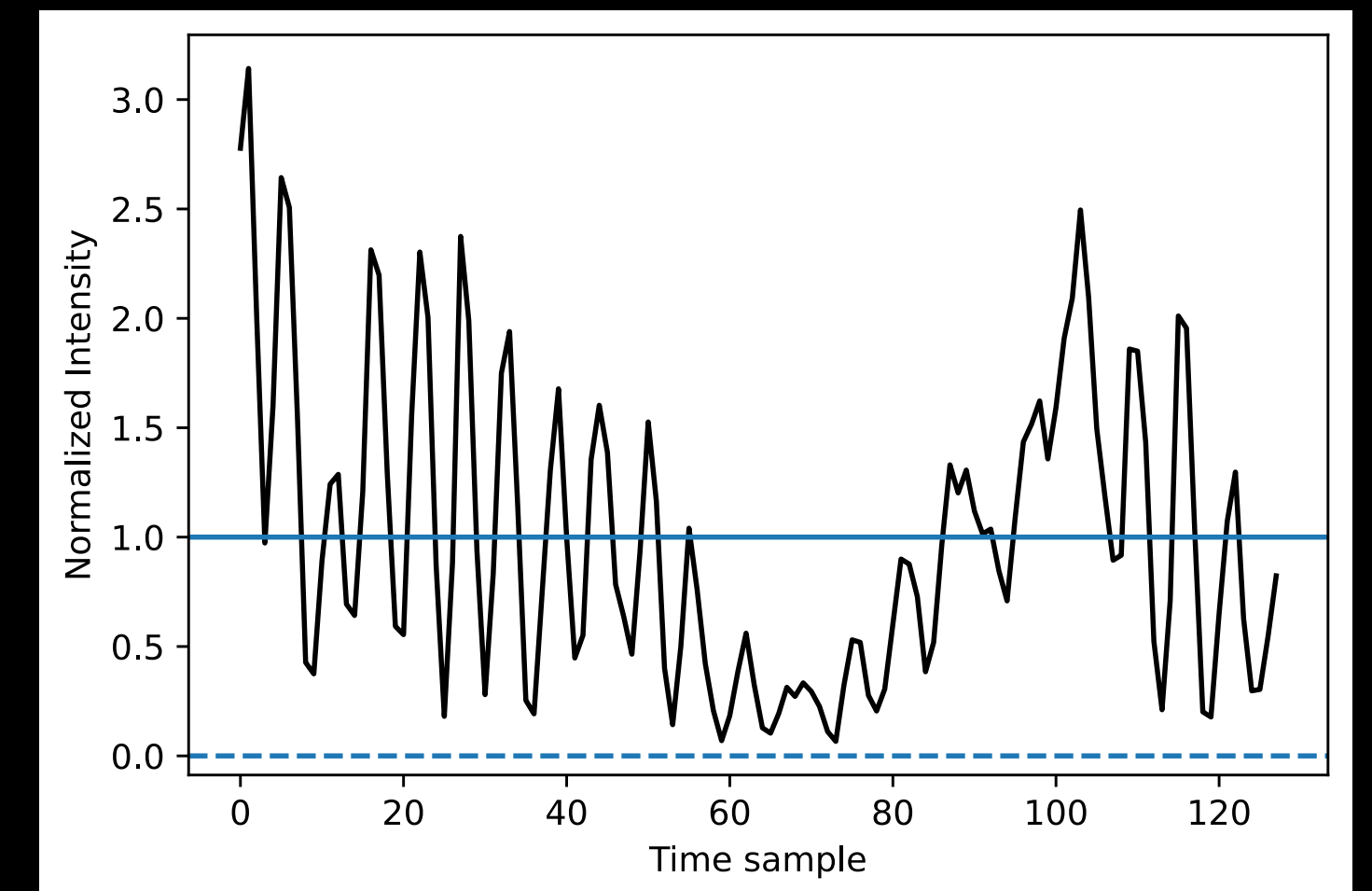
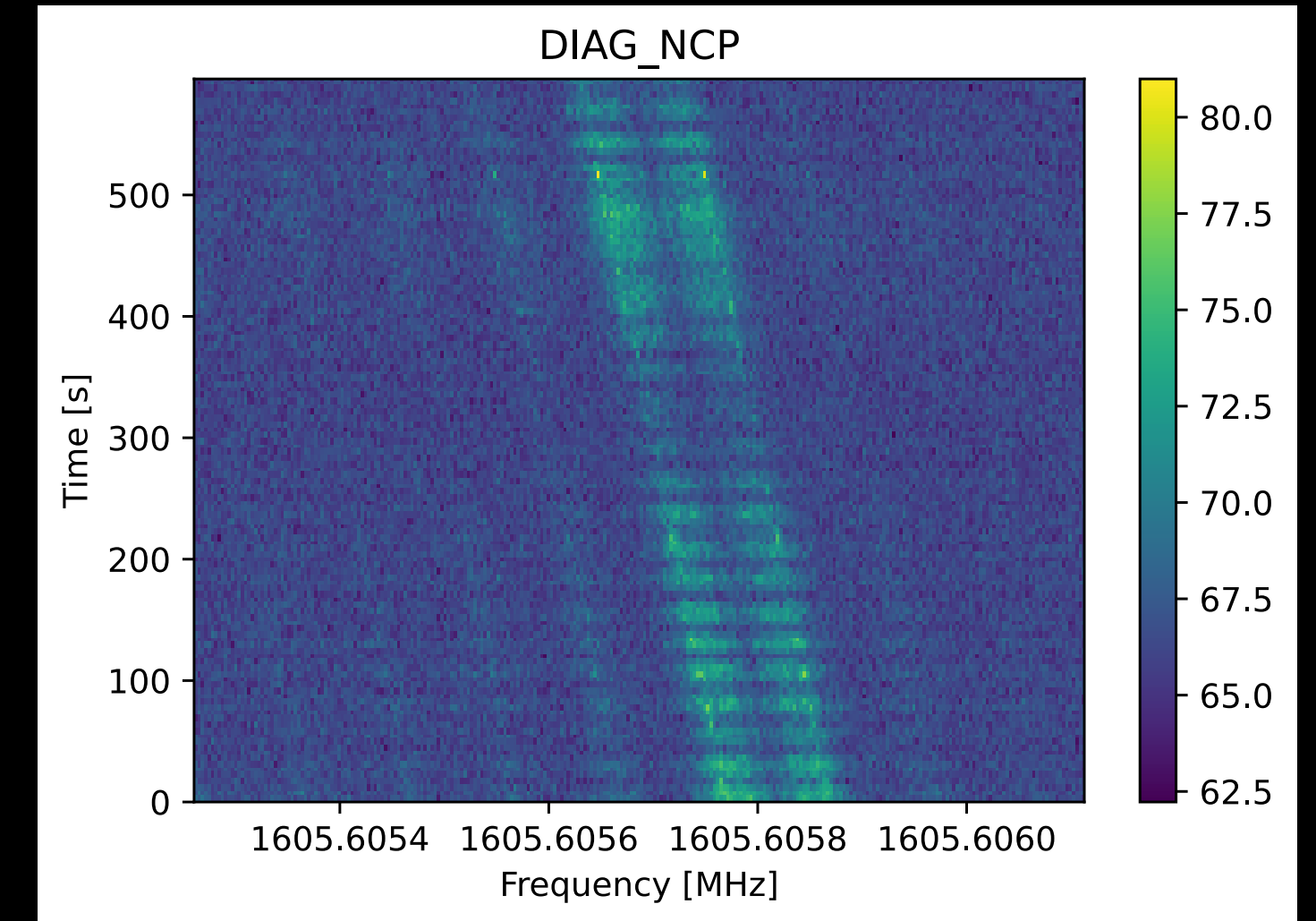
$$\Delta t_d \propto \nu^{6/5} \nu_T^{-1}$$

Can we detect scintillated narrowband technosignatures?

1. What scintillation timescales should we expect?
- 2. How can we probe asymptotic statistics?**
3. Can we differentiate scintillated signals from existing RFI?
4. How can we design a survey to search for scintillated technosignatures?

How might we detect scintillation?

- Estimate intensity time series from signals detected with deDoppler methods
- Since scintillation is stochastic, identify **measurable statistics** for asymptotic behavior
- Would existing RFI modulation confound real scintillation?
 - Methods for creating synthetic scintillated intensities
 - Compare statistics of detected signals with those of synthetic scintillated signals



Set of diagnostic statistics

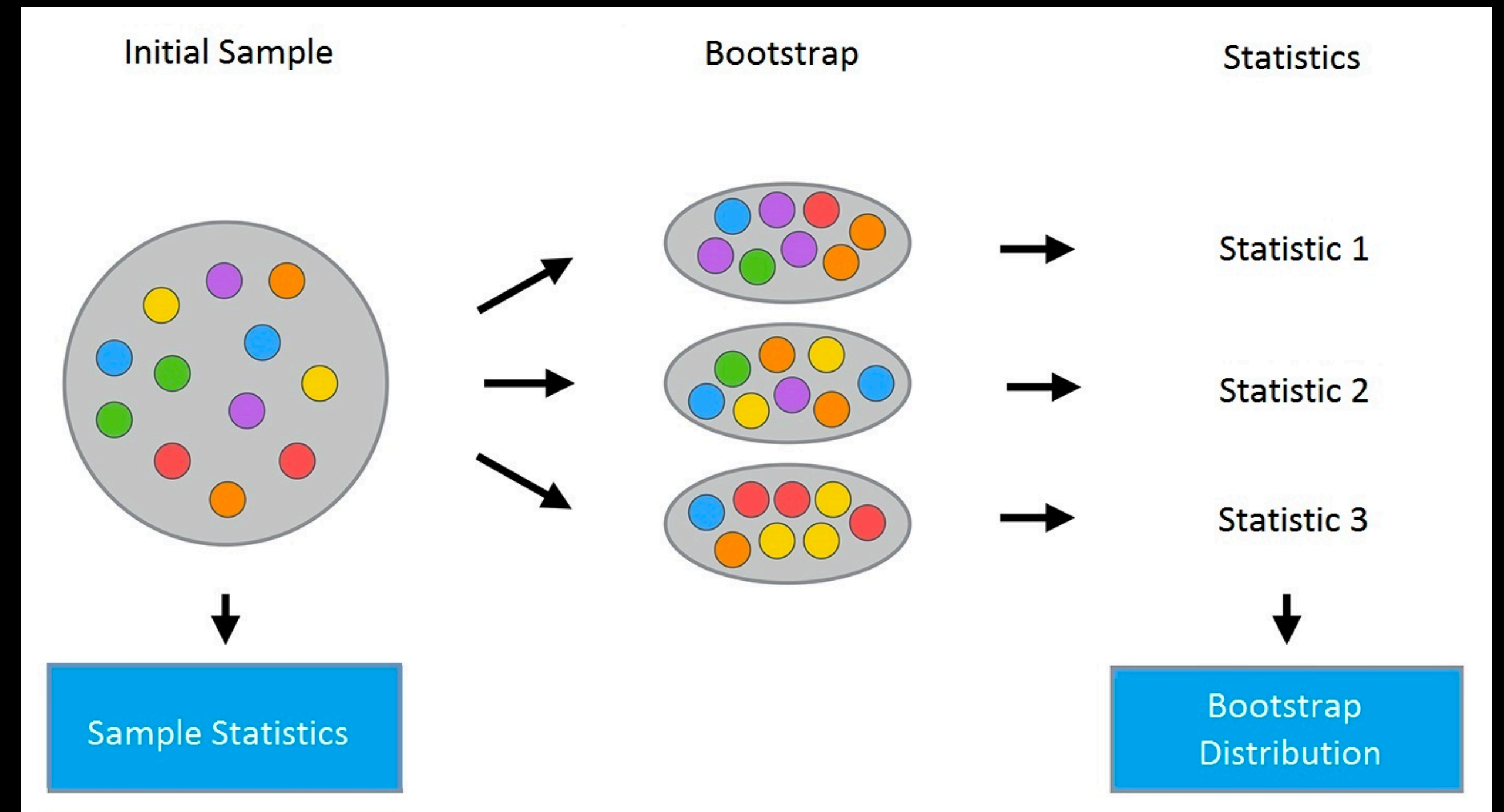
Statistic	Asymptotic Value (with no noise)	Data Type	Theoretical Behavior
Standard Deviation (RMS)	1	Intensity	Exponential
Minimum	0	Intensity	Exponential
Kolmogorov-Smirnoff statistic	0	Intensity	Exponential
Autocorrelation lag	Variable	Autocorrelation	Near-Gaussian
Least squares fit to autocorrelation	Variable	Autocorrelation	Near-Gaussian

There are a number of constraints...

- Time resolution
 - Sufficiently resolve scintles
- Observation time
 - Collect enough scintles, gain stability
- Signal brightness
 - Compute accurate statistics embedded in noise
- RFI environment
 - Bad normalization, false narrowband detections, confounding modulation

Low sample regime

- Spread of values around the asymptotic “truth”
- Both correlated and uncorrelated samples within the same observation



Kaggle

- How can we evaluate this?

Synthetic random processes: Autoregressive-to-anything (ARTA)

- (Cario & Nelson 1996) The ARTA process:
 - Matches a target intensity distribution
 - Matches a target autocorrelation structure (with custom asymptotic precision)

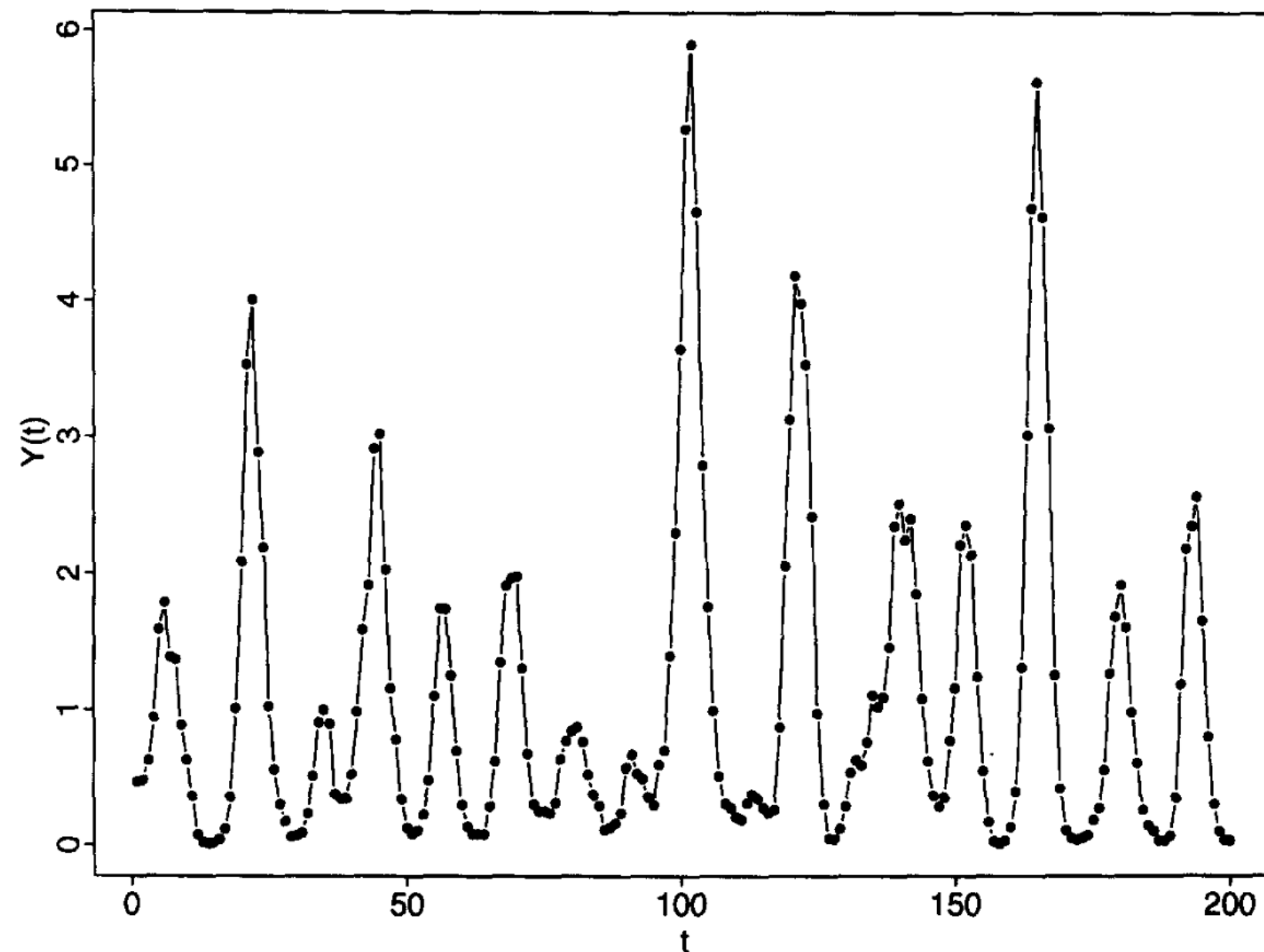
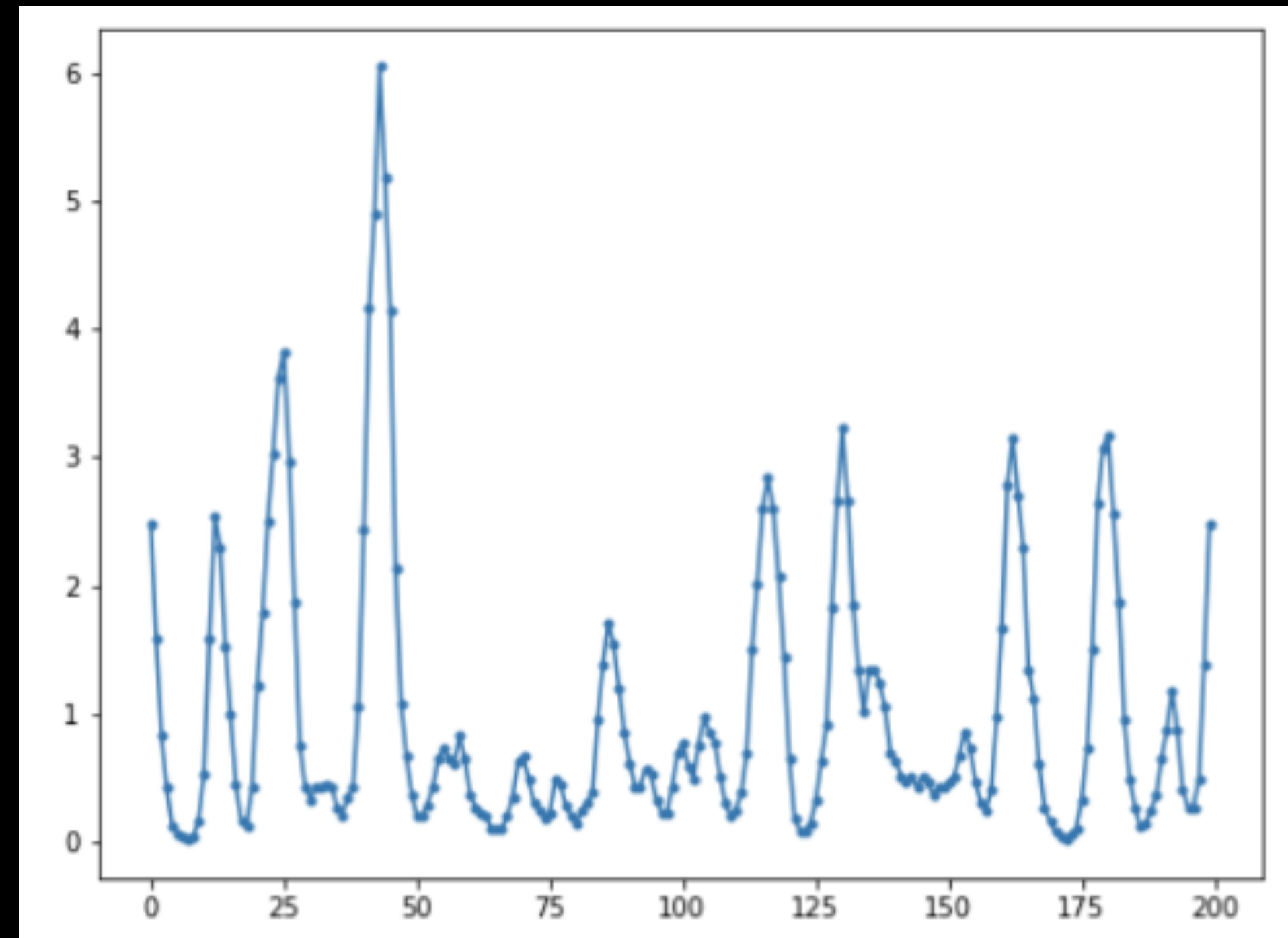


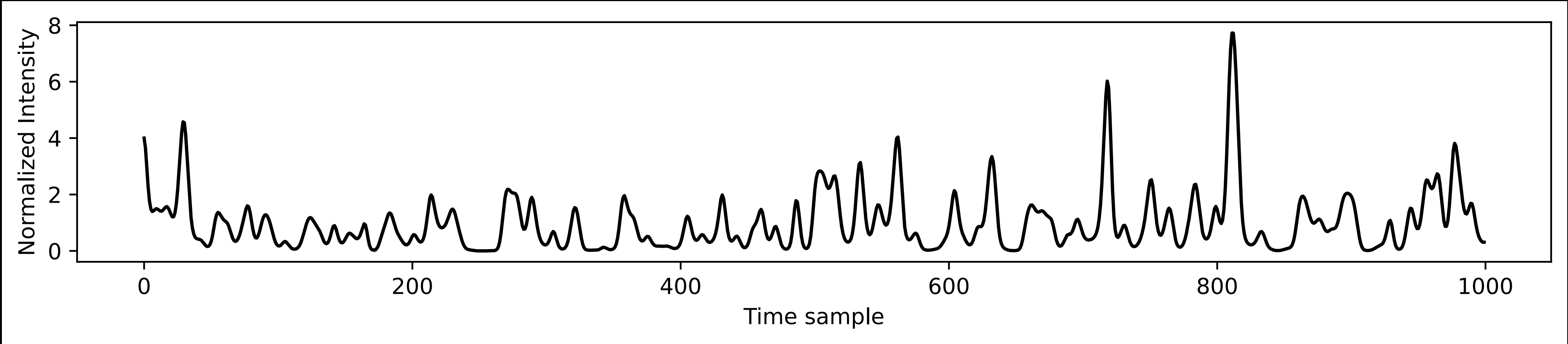
Fig. 1. Sample path of an ARTA process with exponential marginals and autocorrelations $\rho_1 = 0.9$ and $\rho_2 = 0.6$.



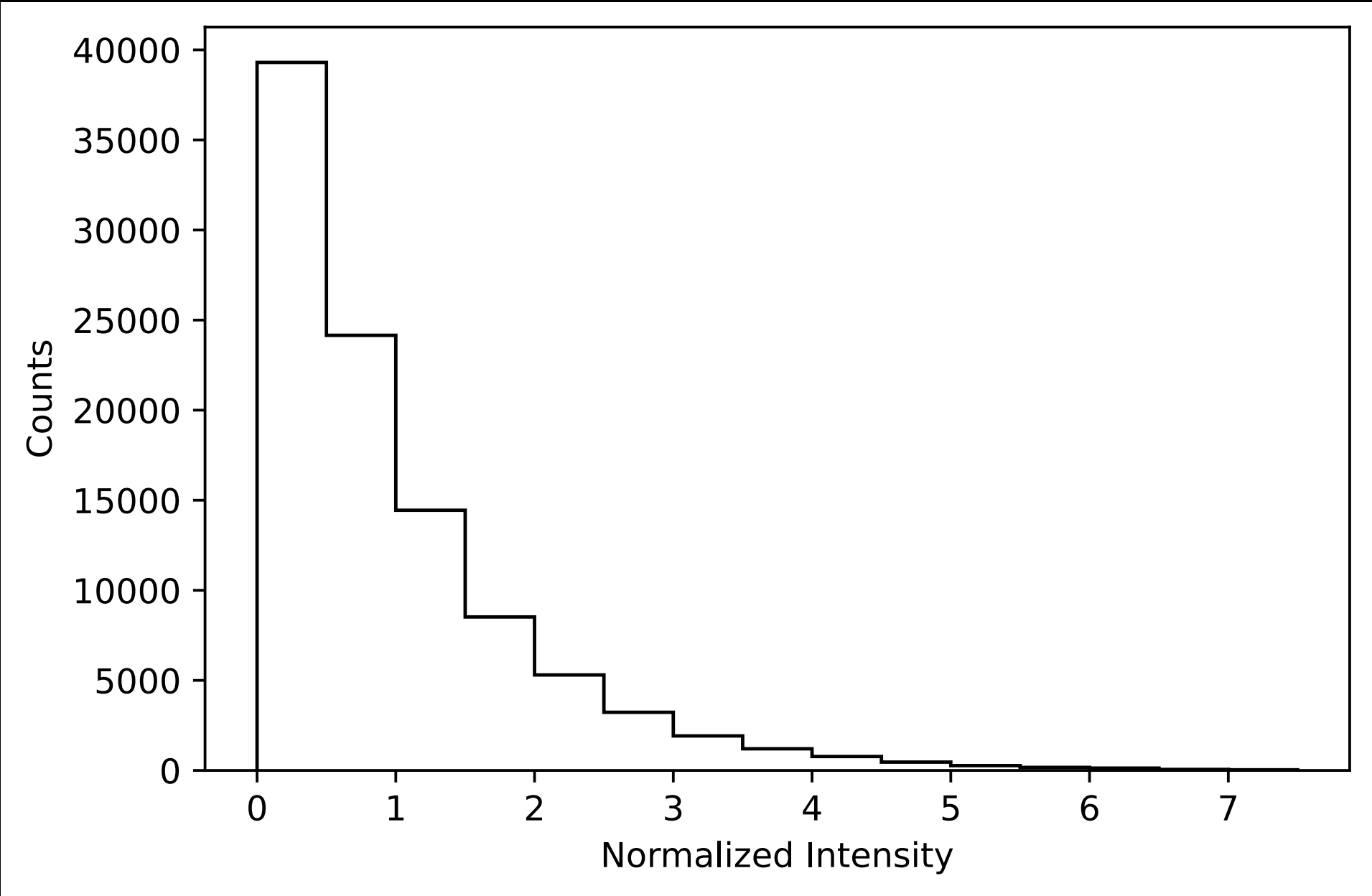
Cario & Nelson 1996

Our implementation

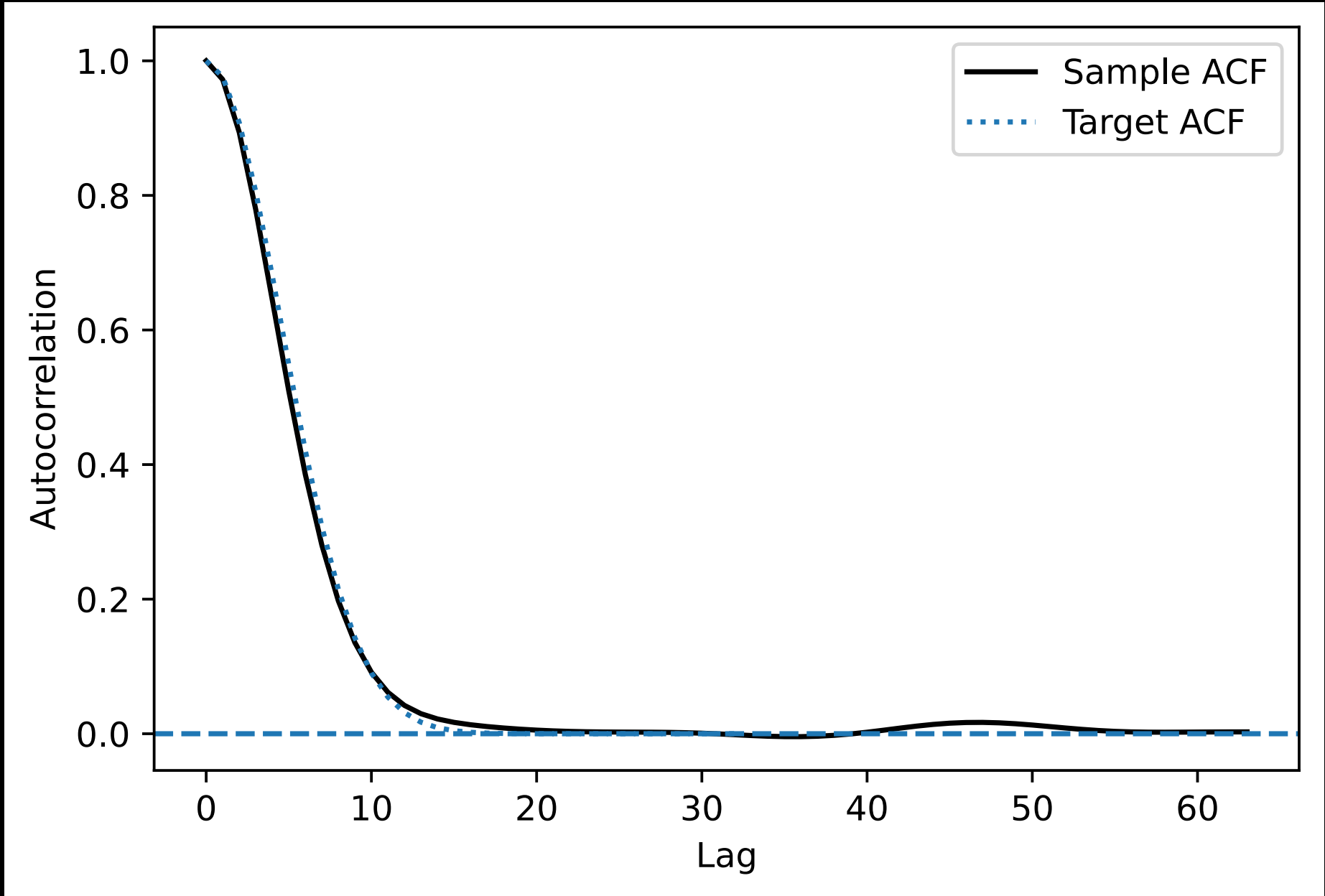
Large sample limit ($n = 1\text{e}5$) for $\Delta t_d = 30\text{ s}$



Intensity histogram

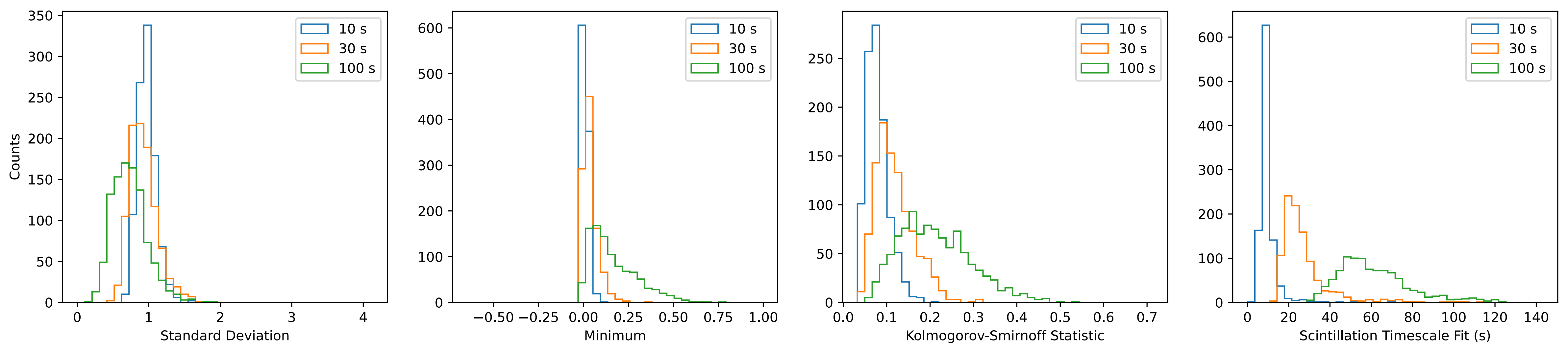


Autocorrelation



Statistics using low number of synthetic samples

10 min “observation”, 4.65 s resolution



Standard Deviation

Minimum

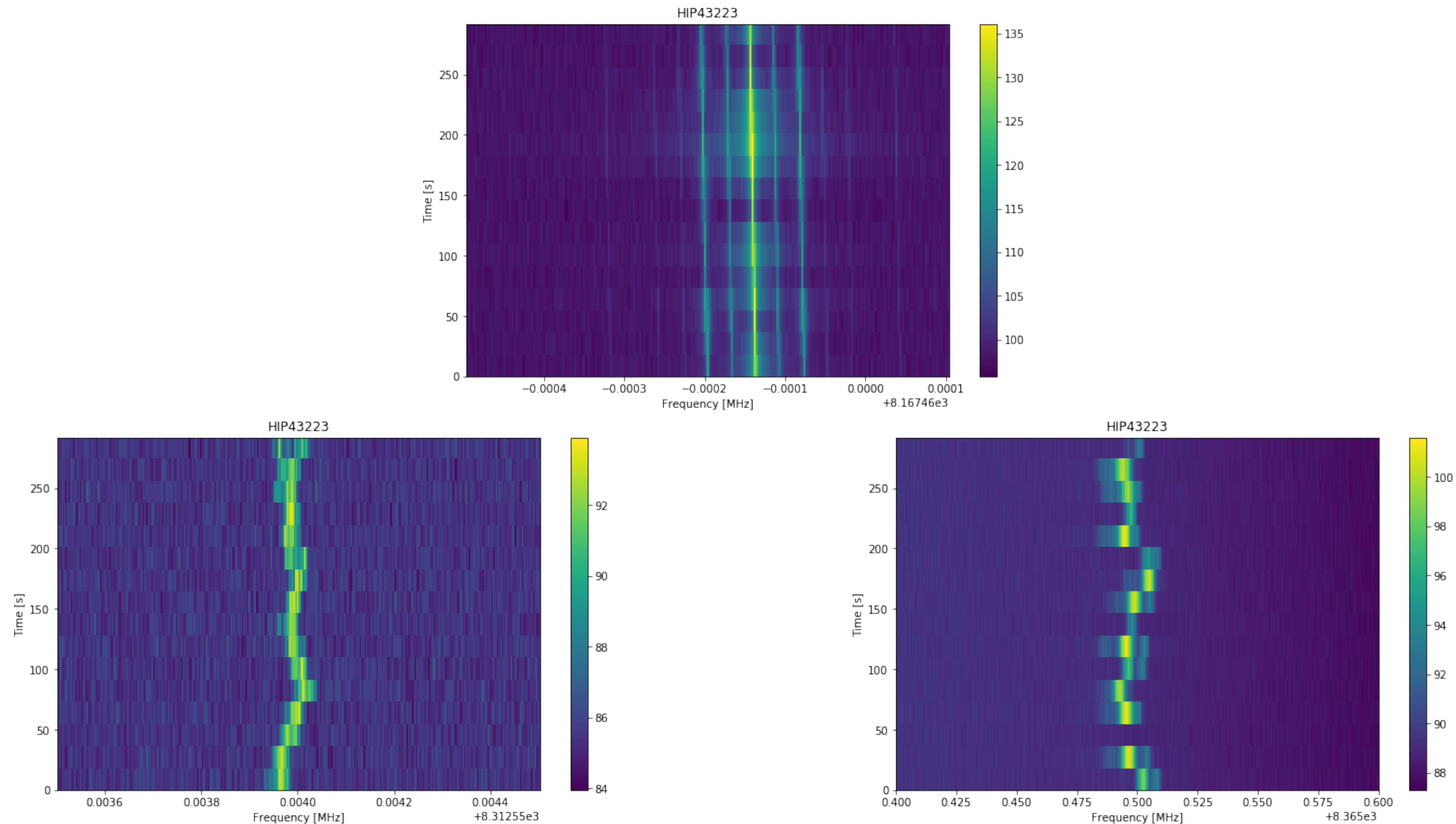
Kolmogorov-Smirnoff Statistic

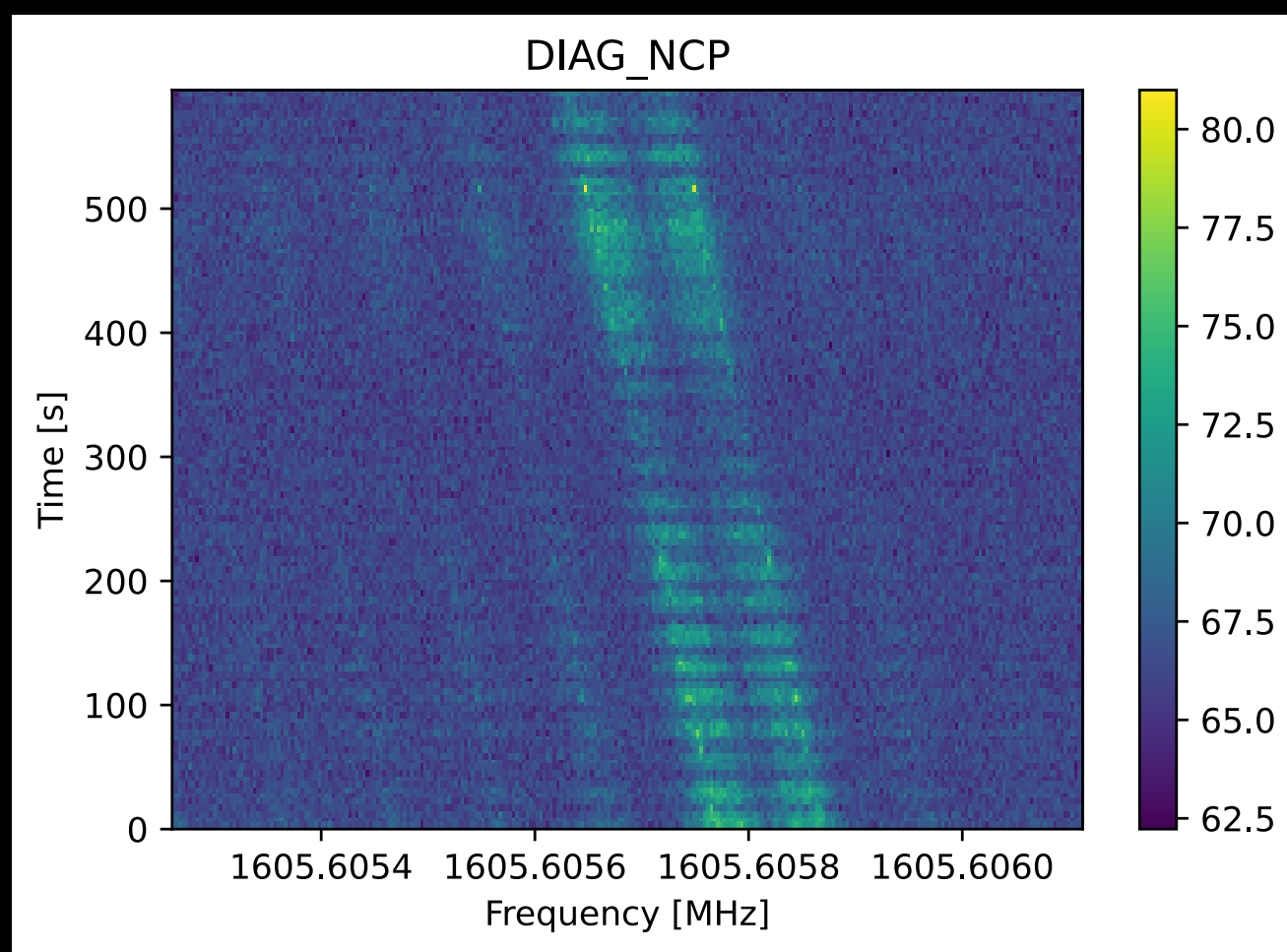
Scintillation Timescale Fit

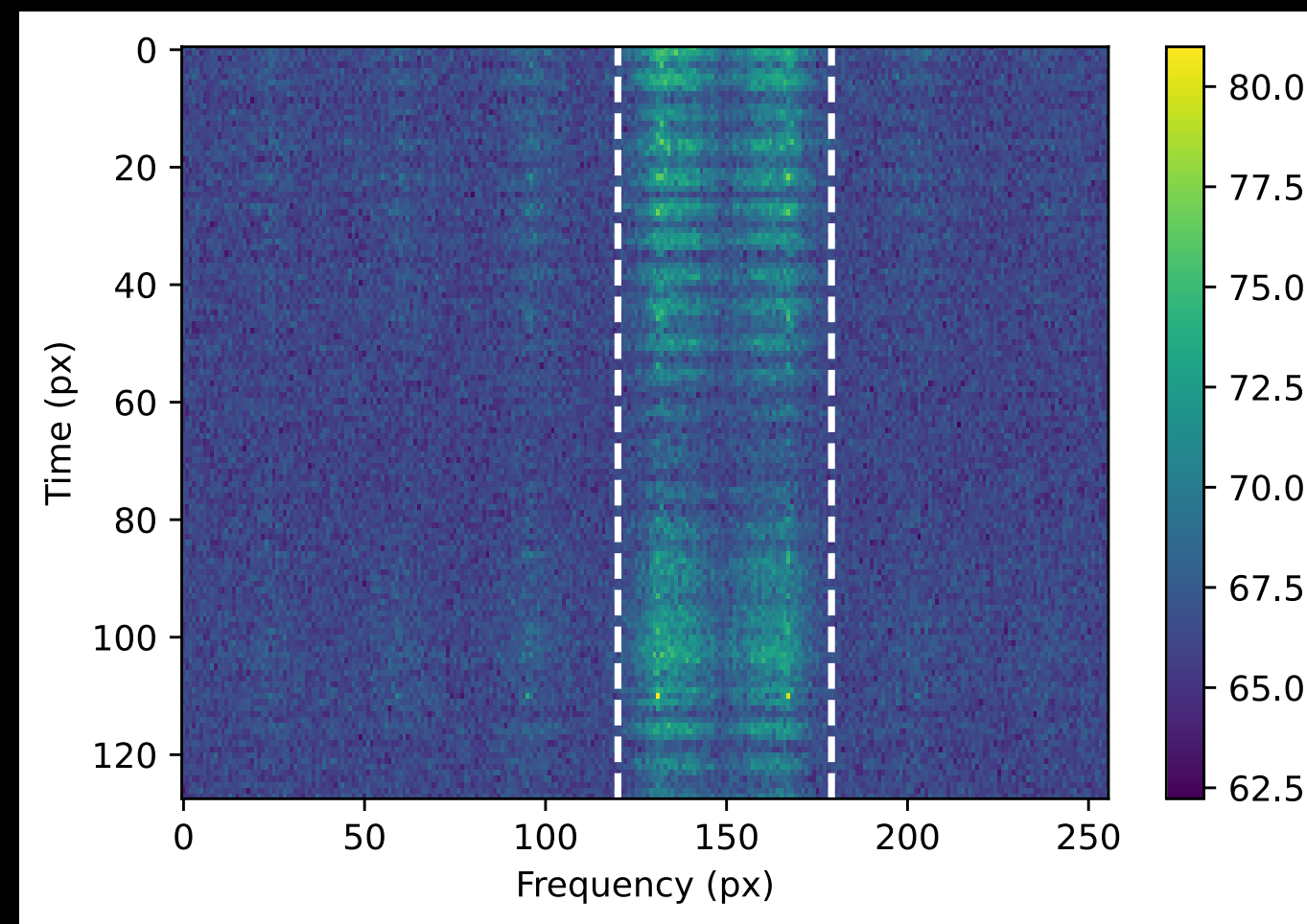
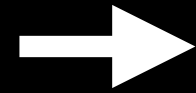
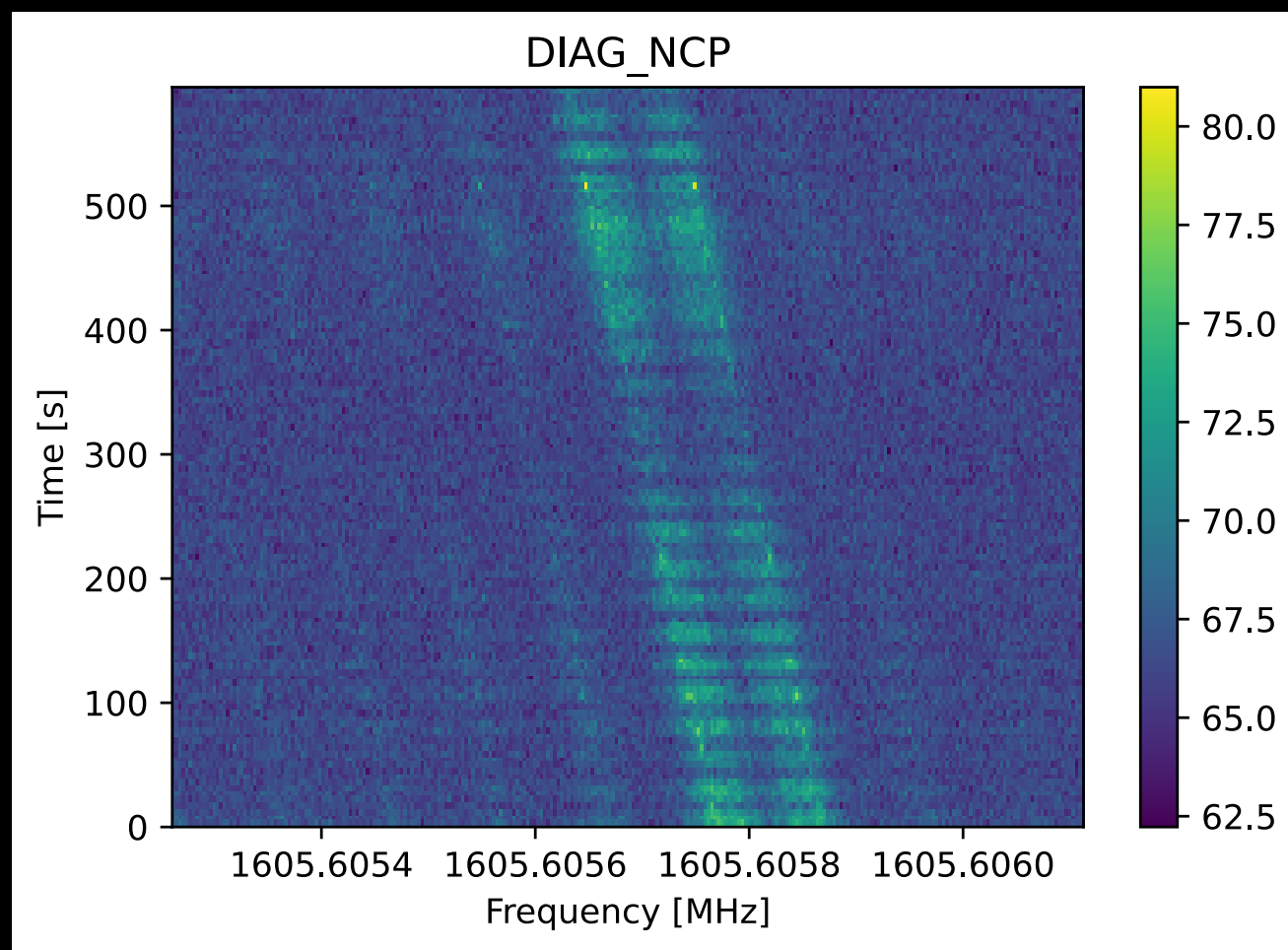
Can we detect scintillated narrowband technosignatures?

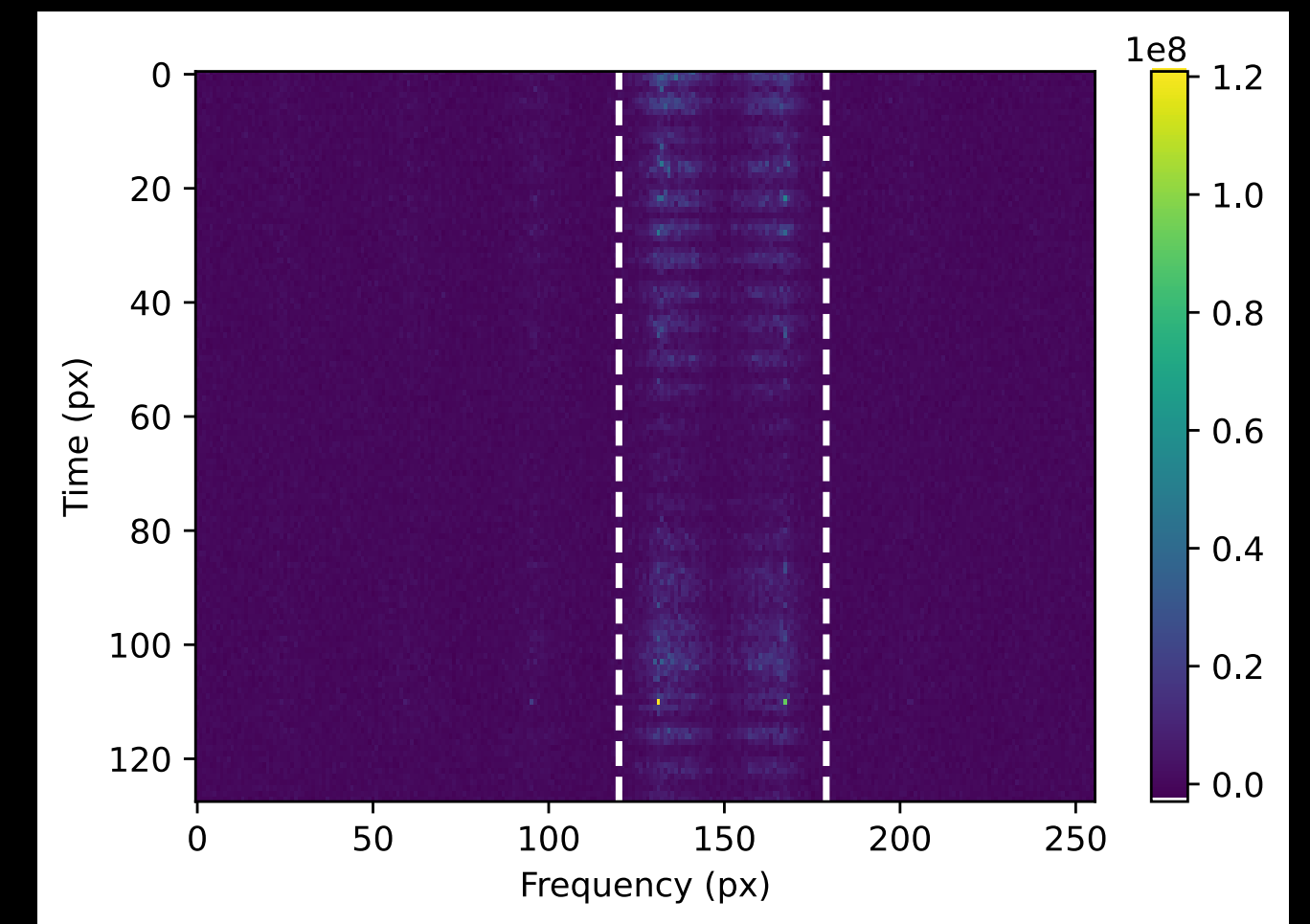
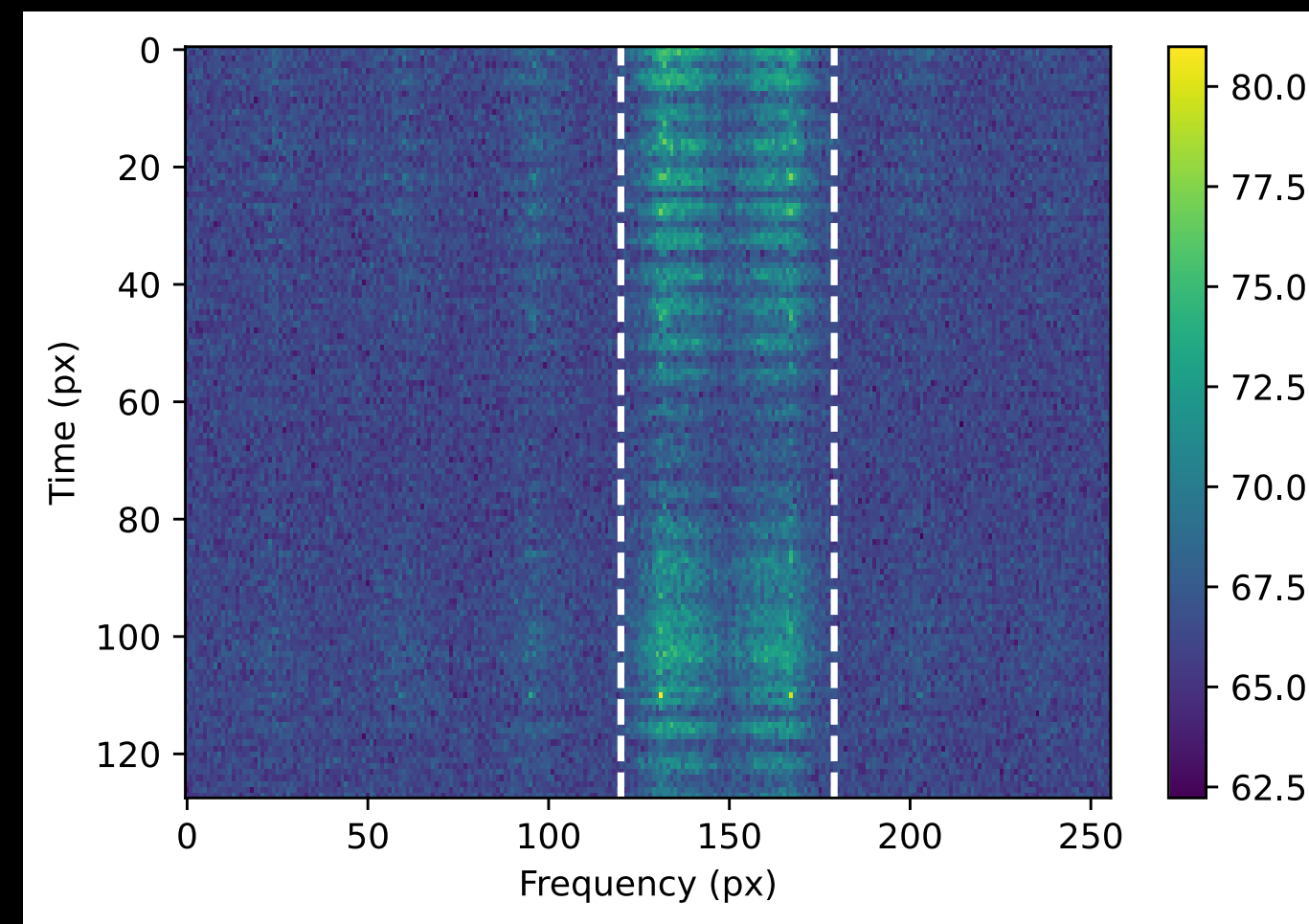
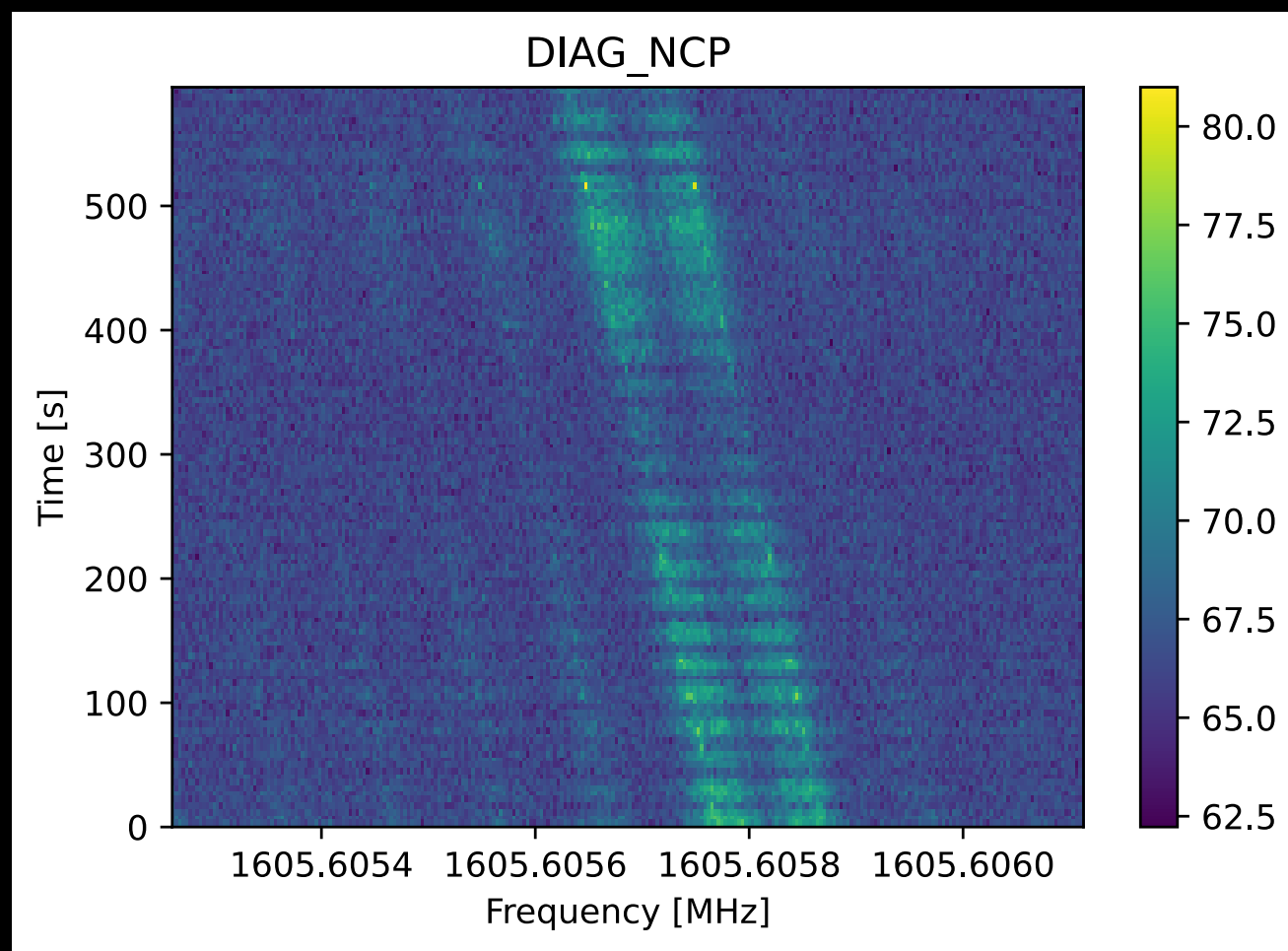
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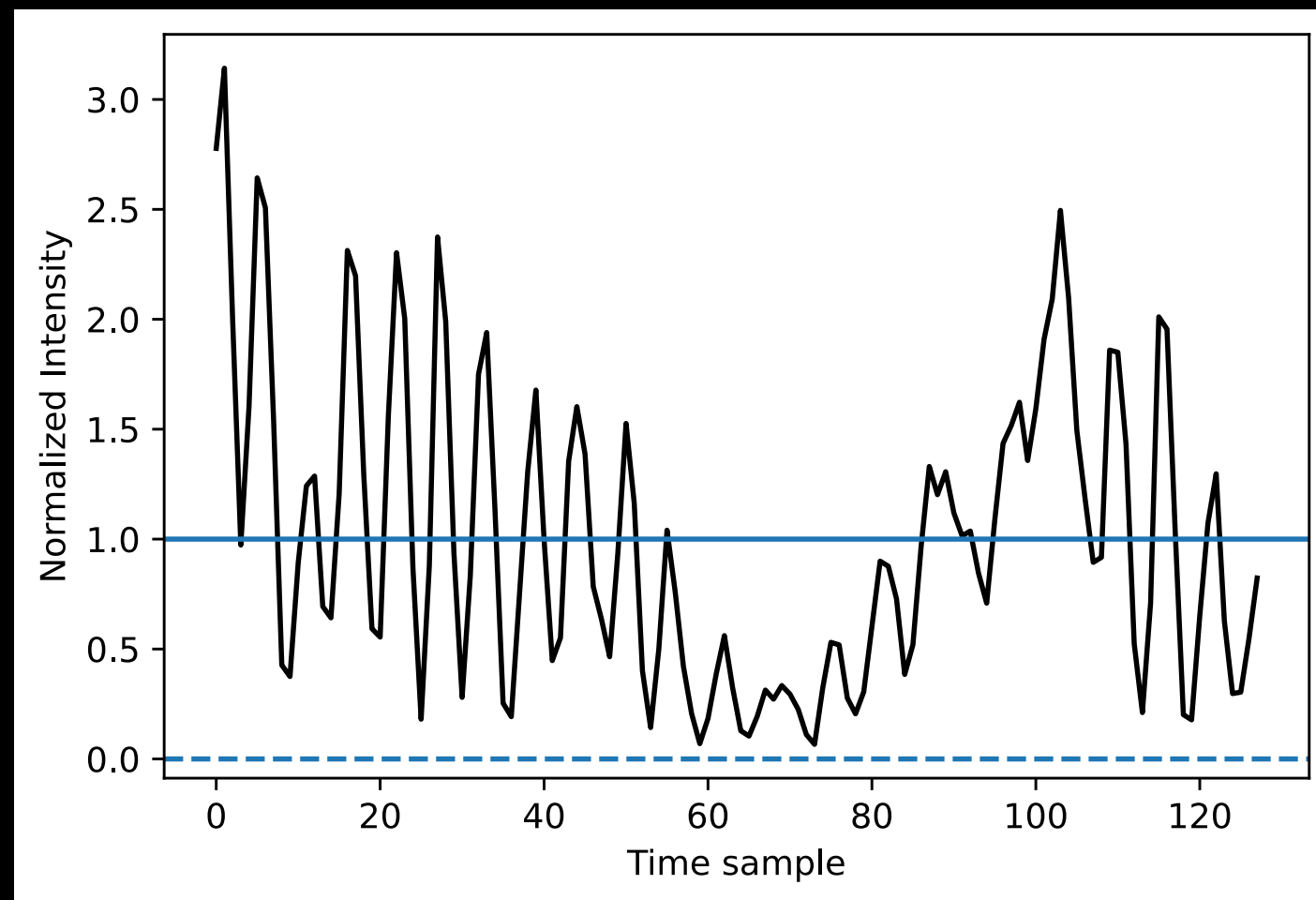
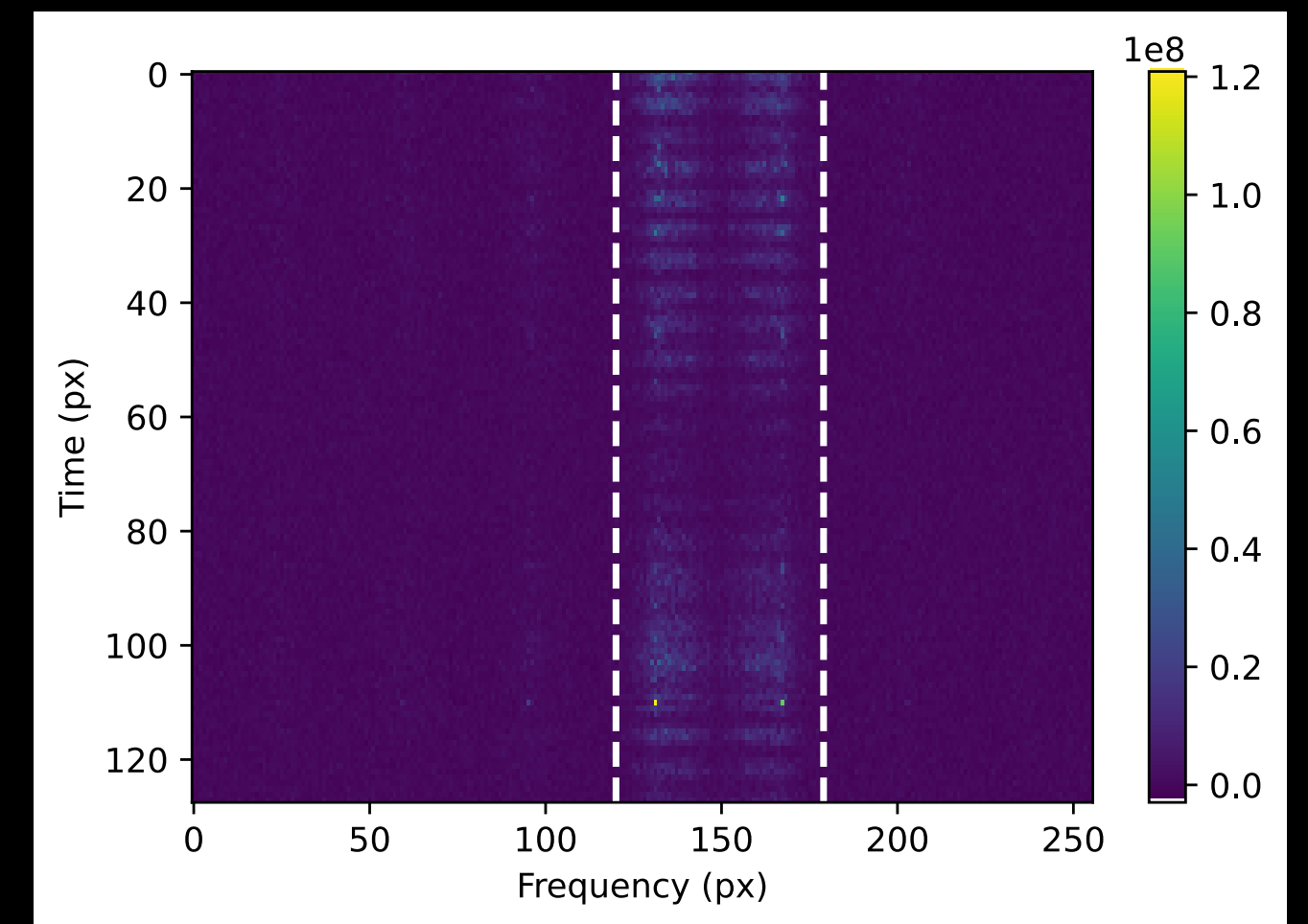
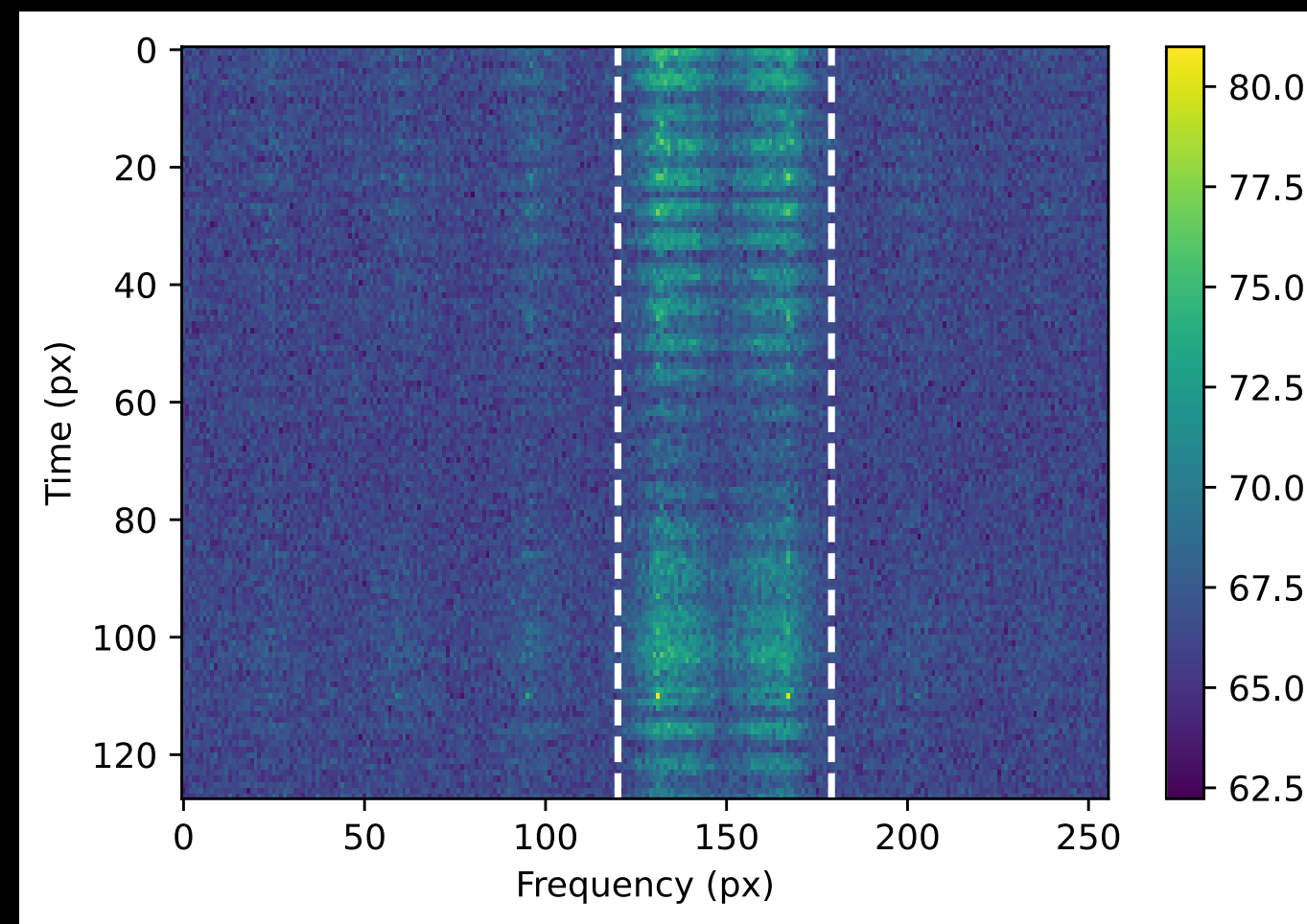
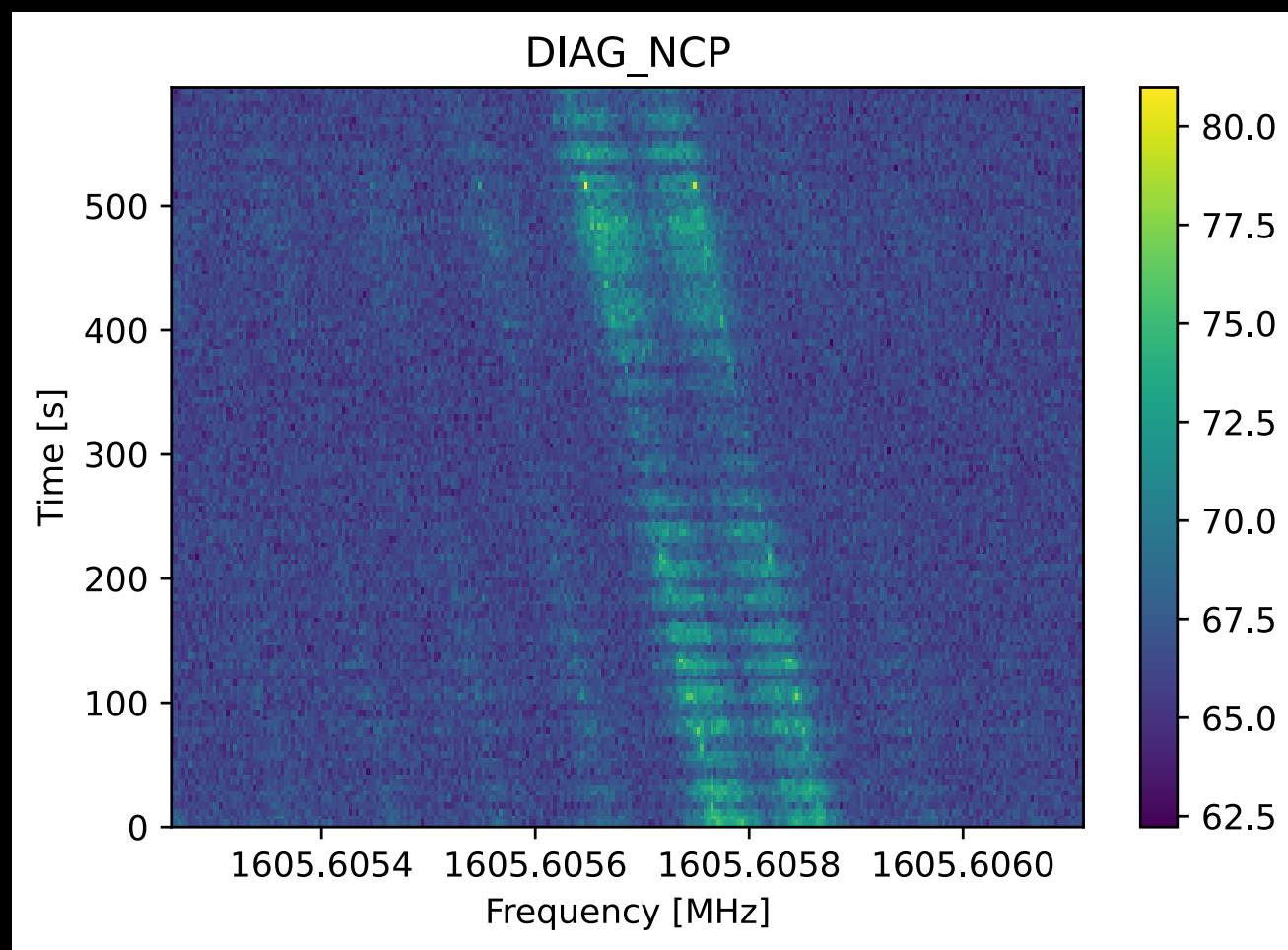
What does the RFI environment look like?

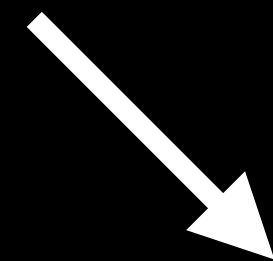
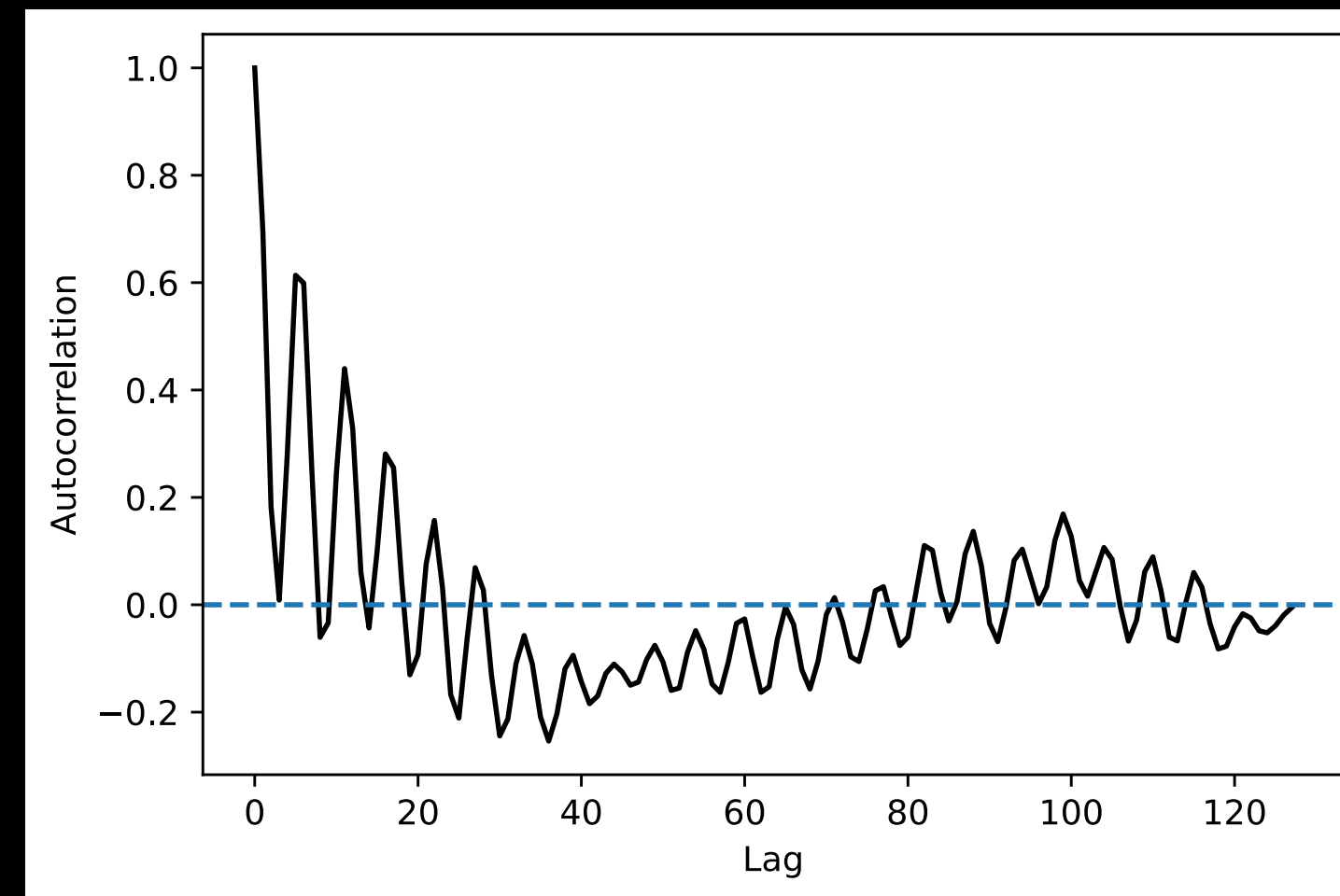
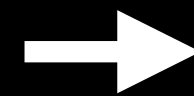
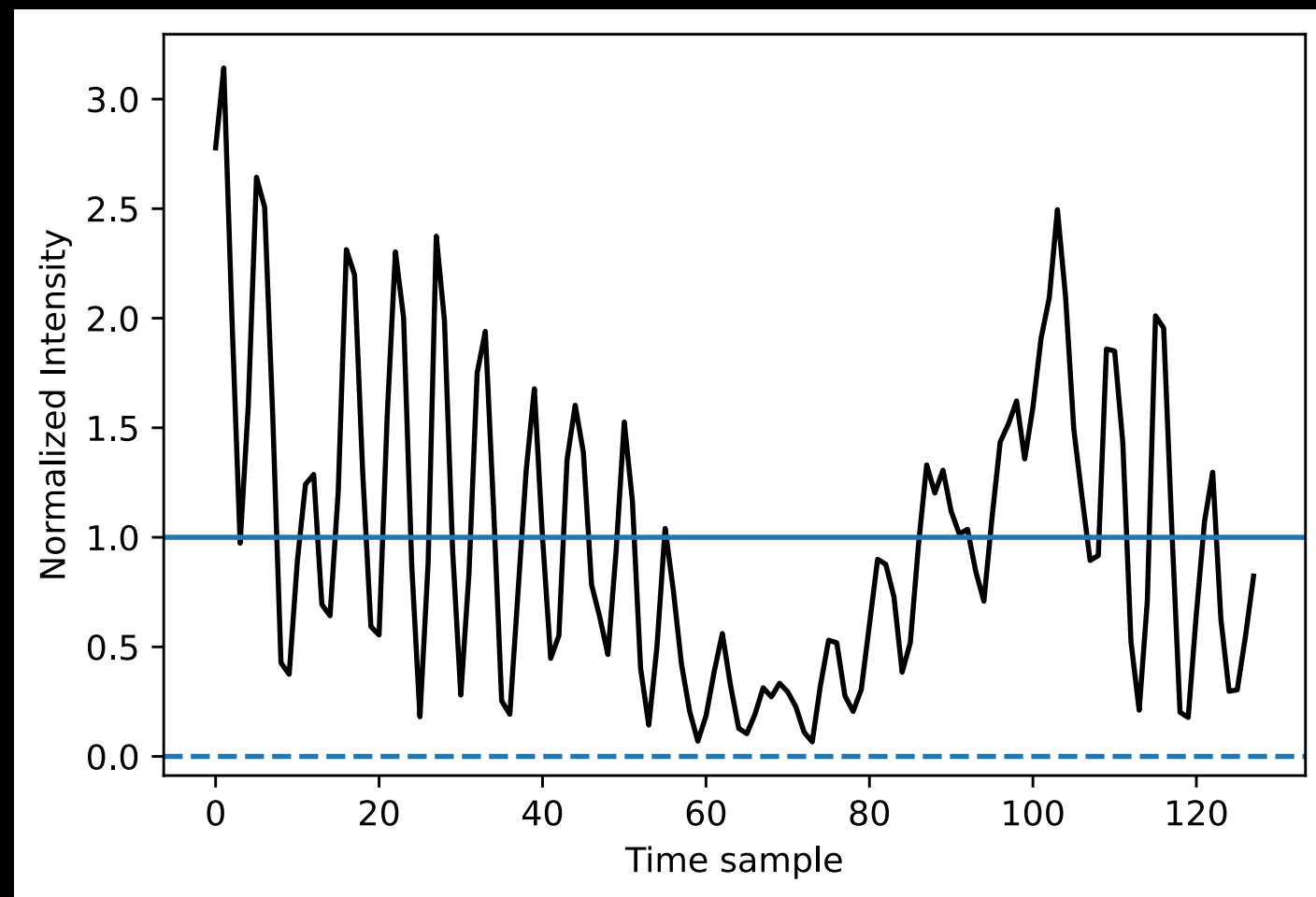
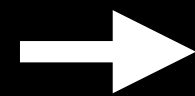
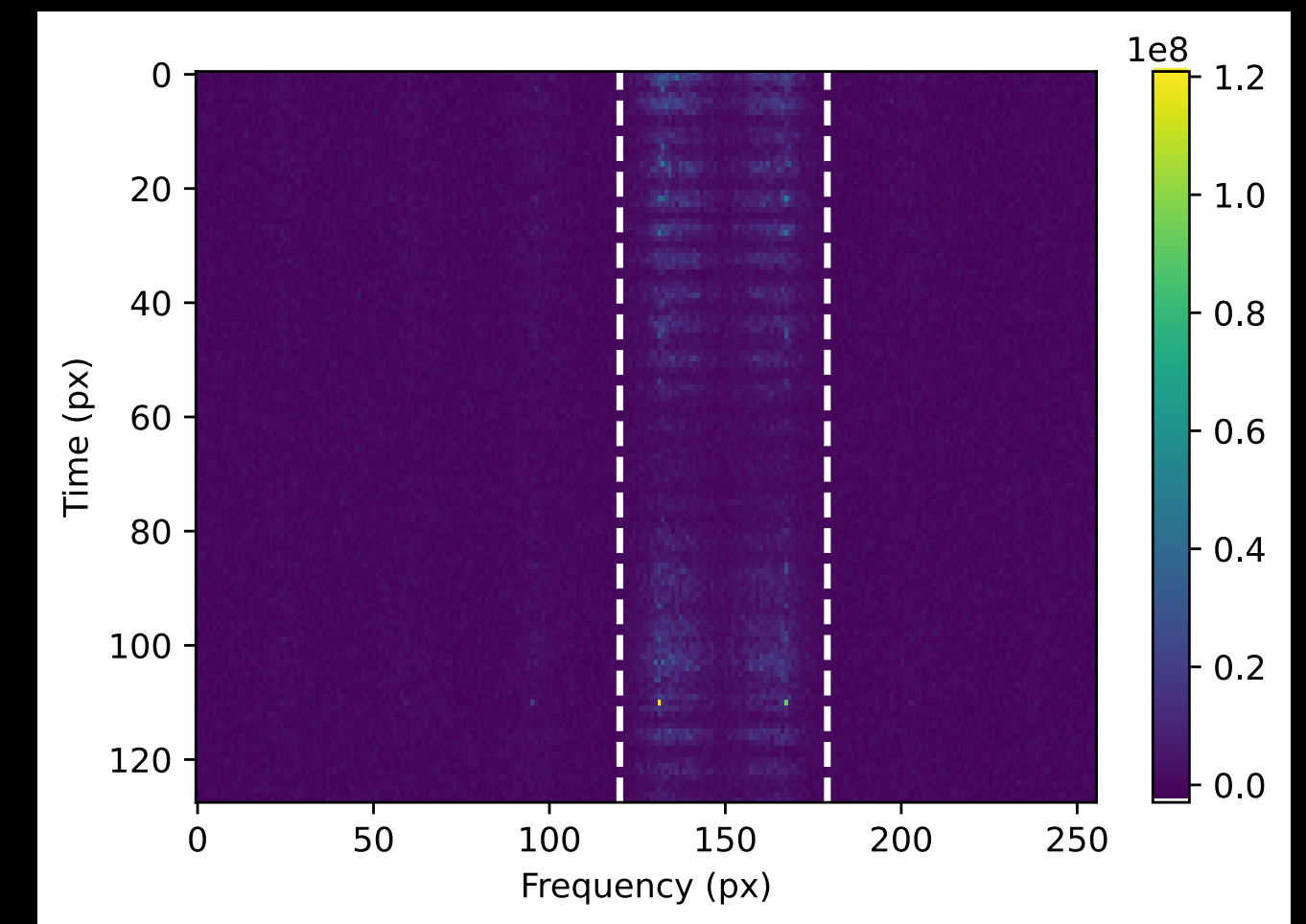
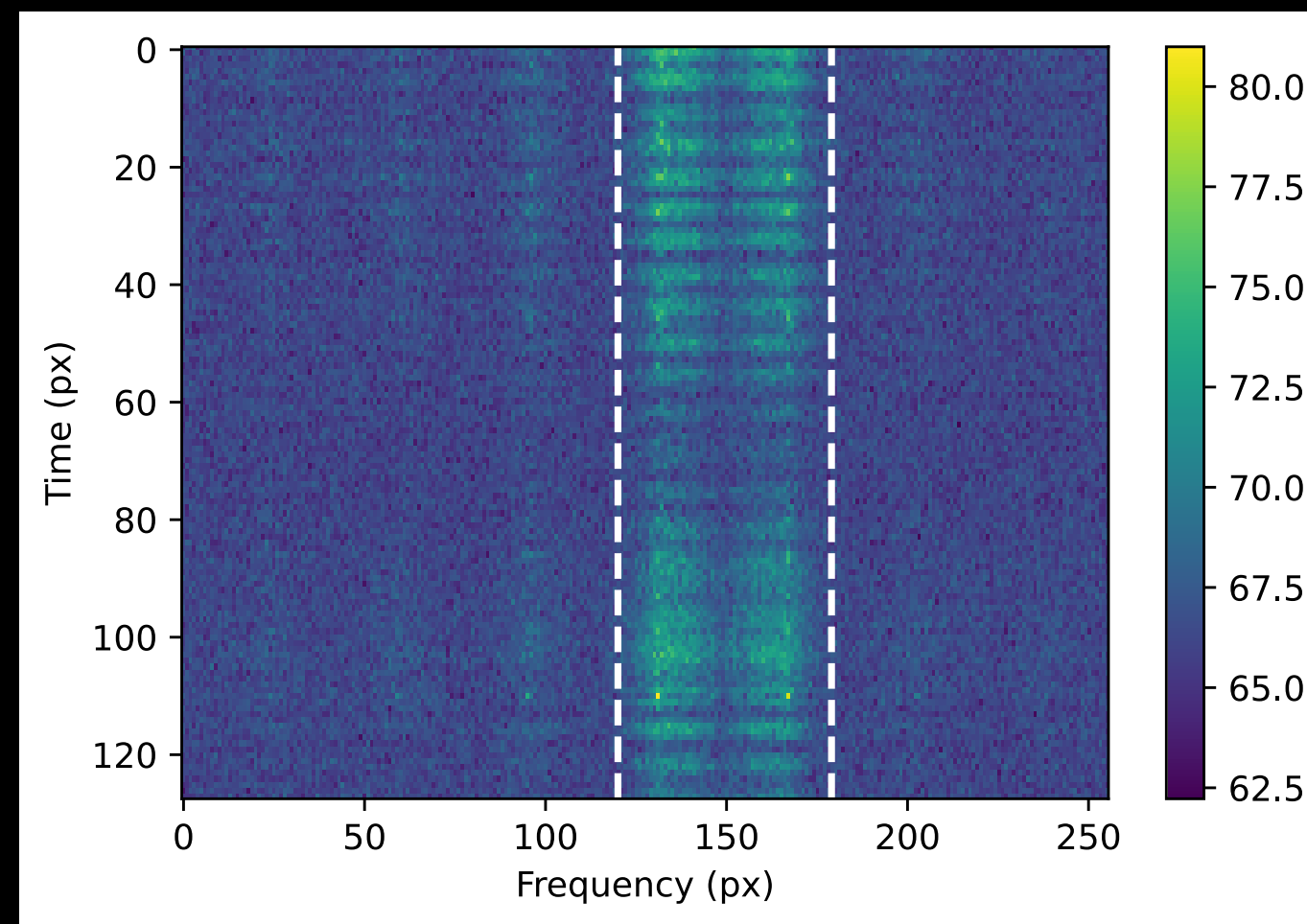
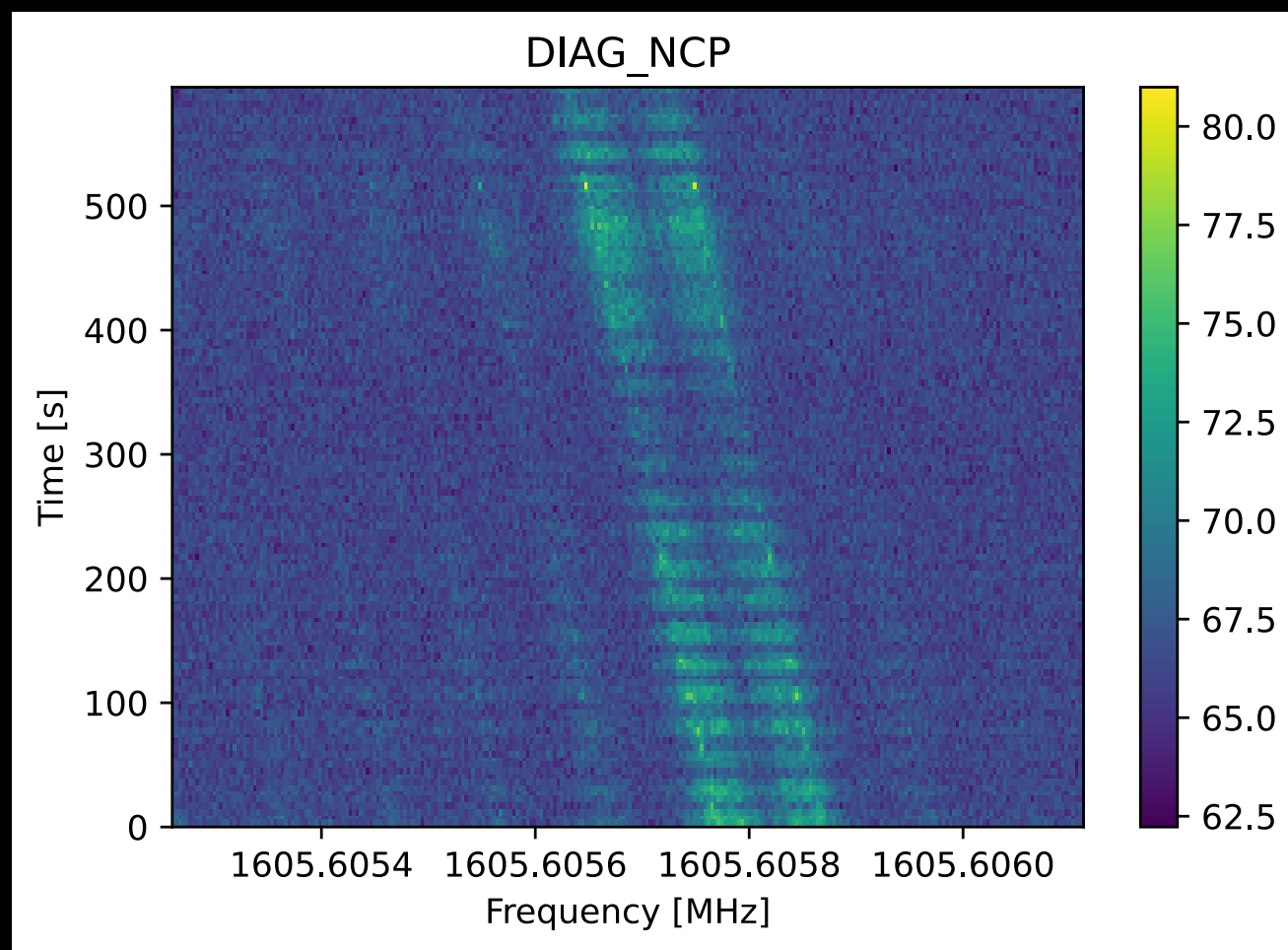




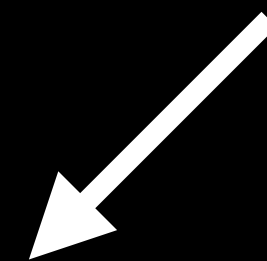








Diagnostic statistics



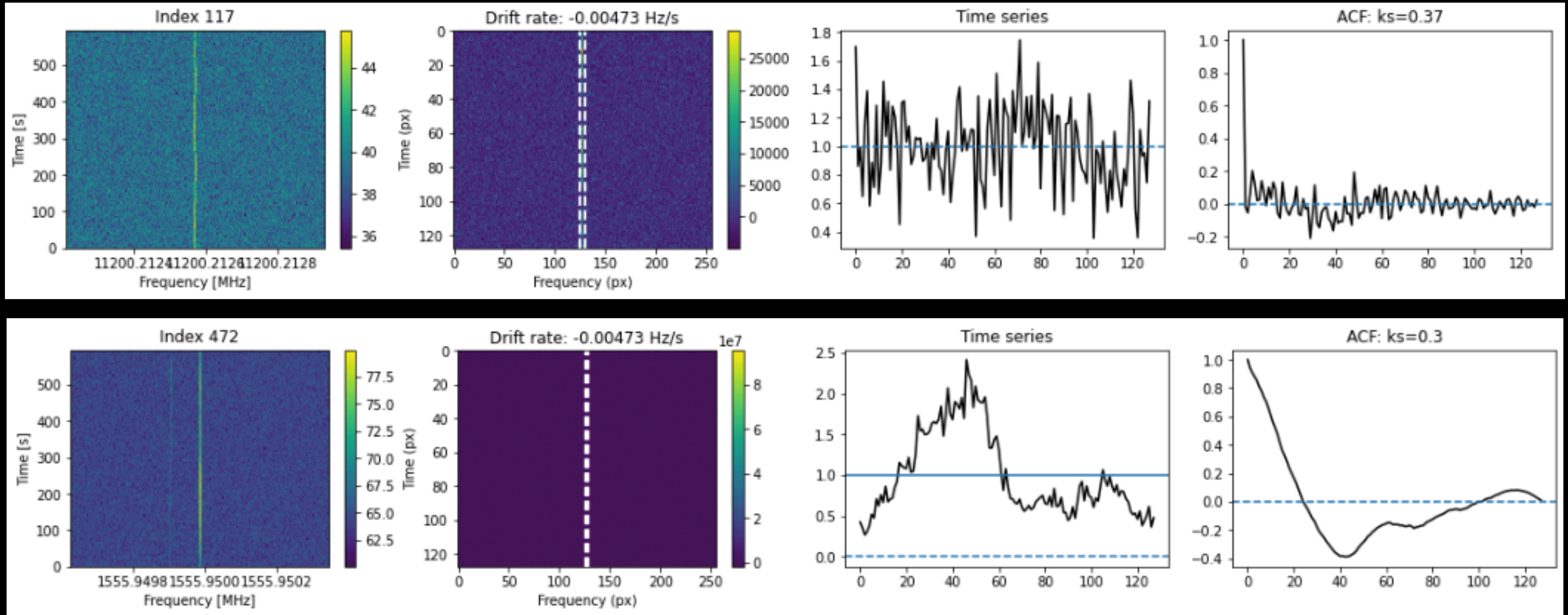
Some examples

Original

De-drifted

Intensity time series

Autocorrelation



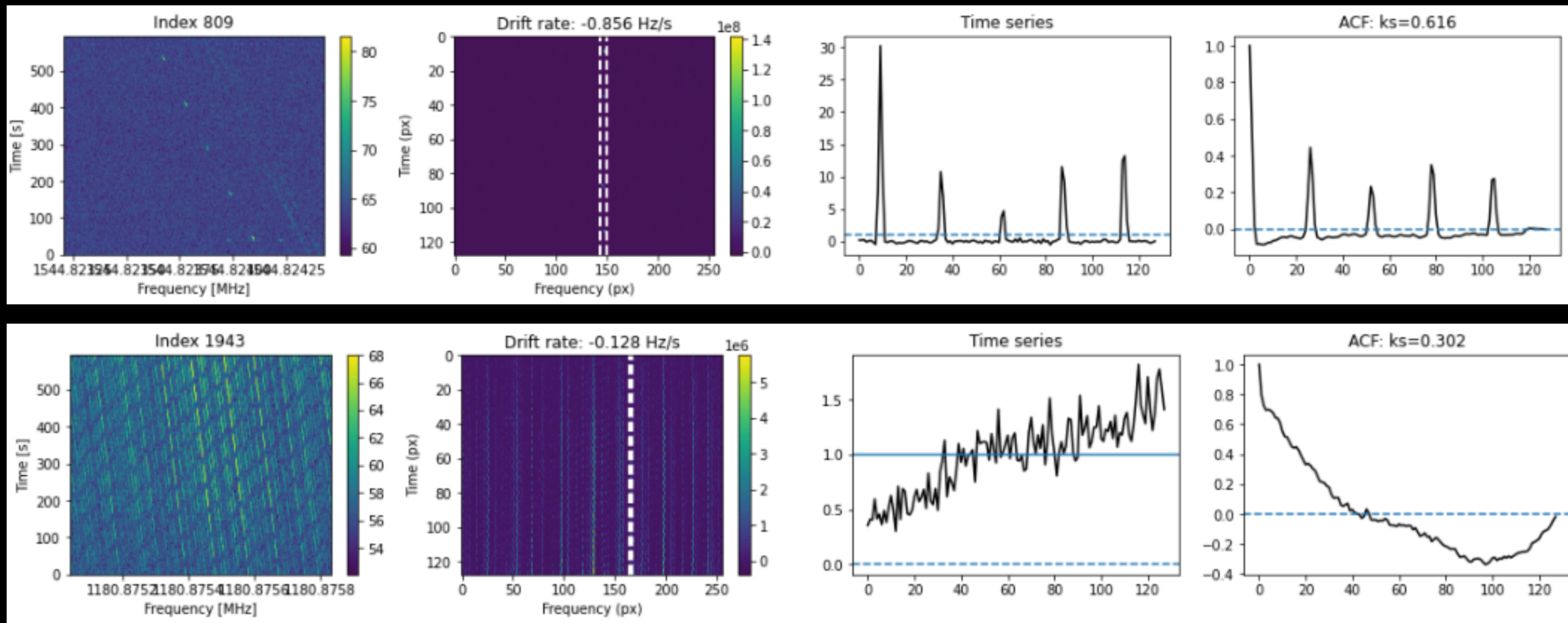
Some more examples

Original

De-drifted

Intensity time series

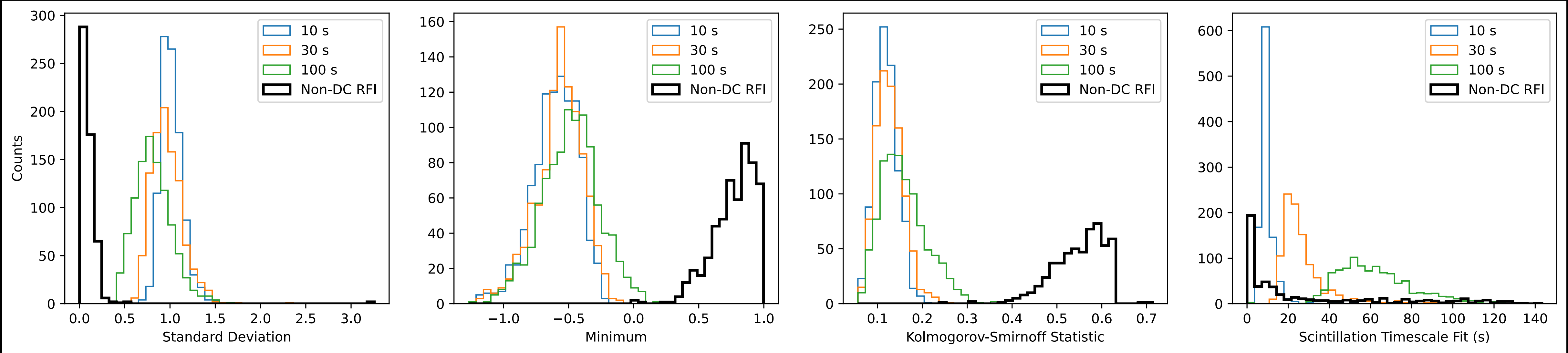
Autocorrelation



GBT RFI vs. *synthetic* scintillated signals

C band

$S/N > 25$



Standard Deviation

Minimum

Kolmogorov-Smirnoff Statistic

Scintillation Timescale Fit

GBT RFI vs. synthetic scintillated signals

 **setigen** Public

Python library for generating and injecting artificial narrow-band signals into radio frequency data

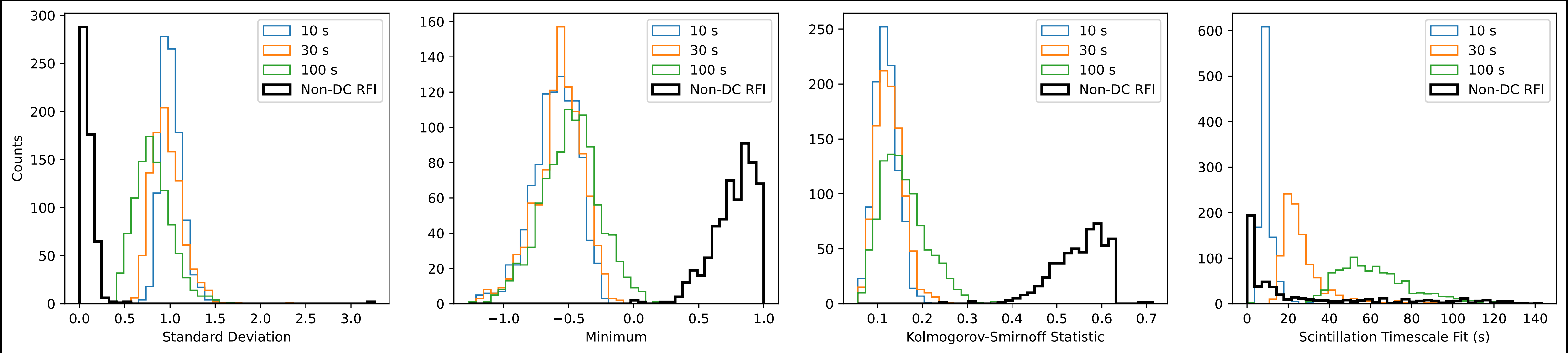
 Jupyter Notebook

 23

 12

C band

$S/N > 25$



Standard Deviation

Minimum

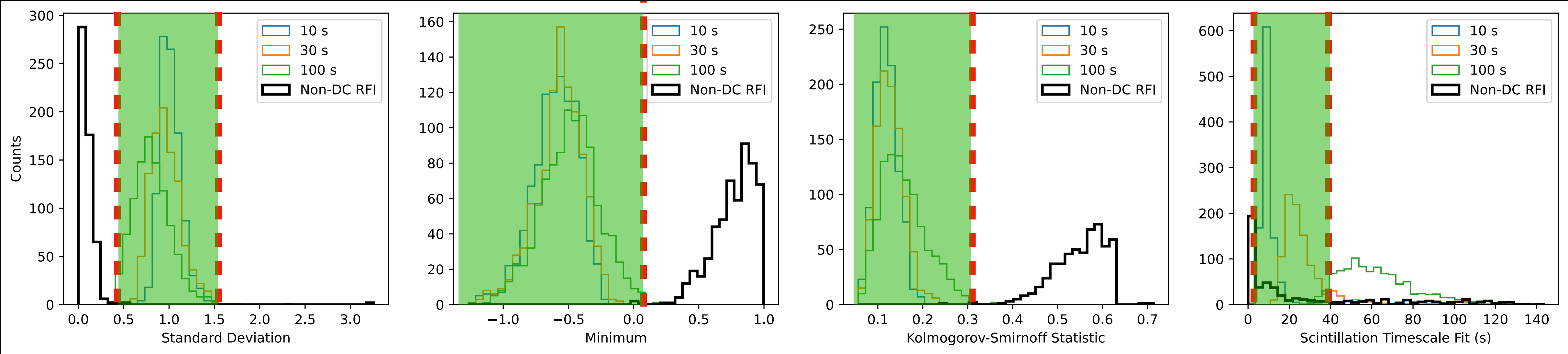
Kolmogorov-Smirnoff Statistic

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C band

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Standard Deviation

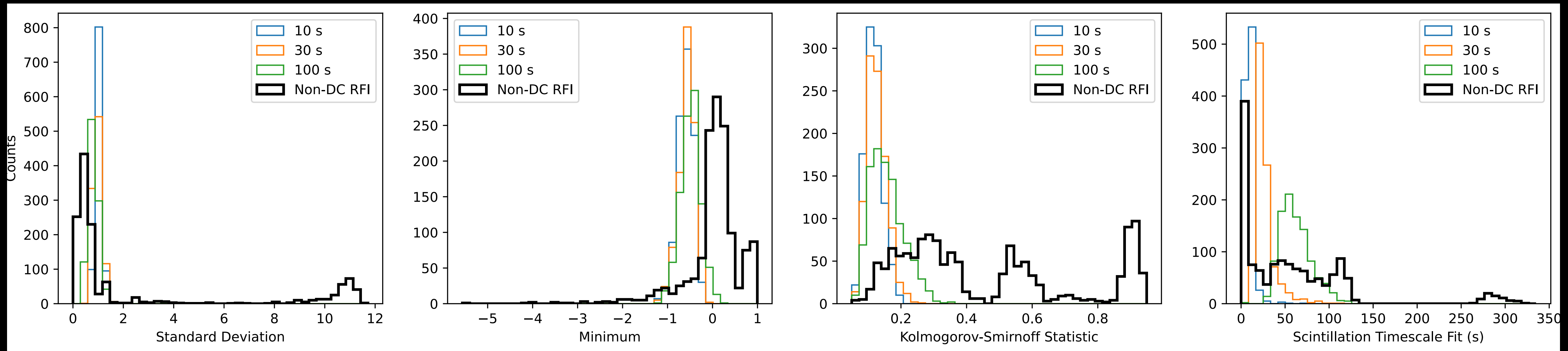
Minimum

Kolmogorov-Smirnoff Statistic

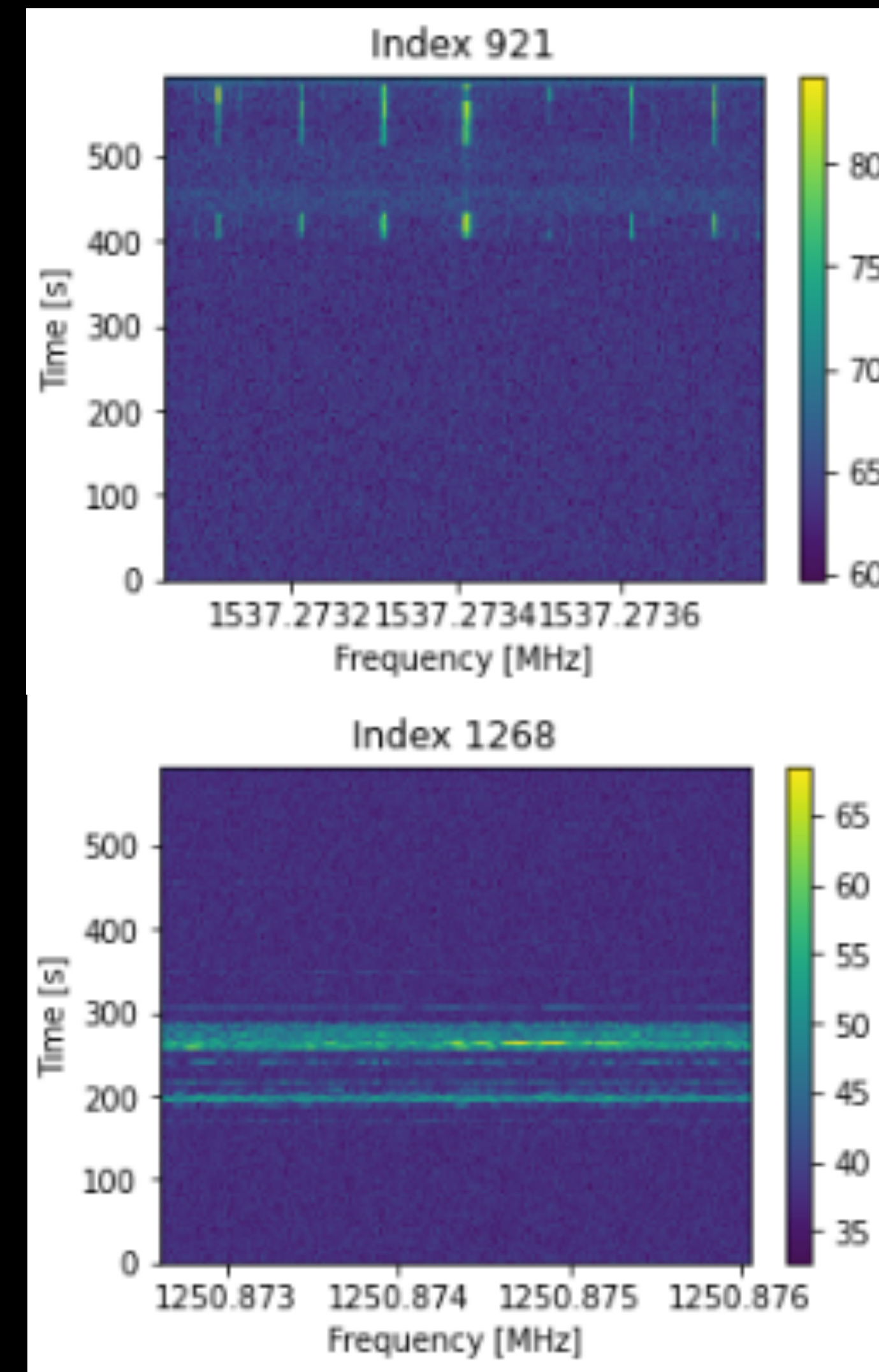
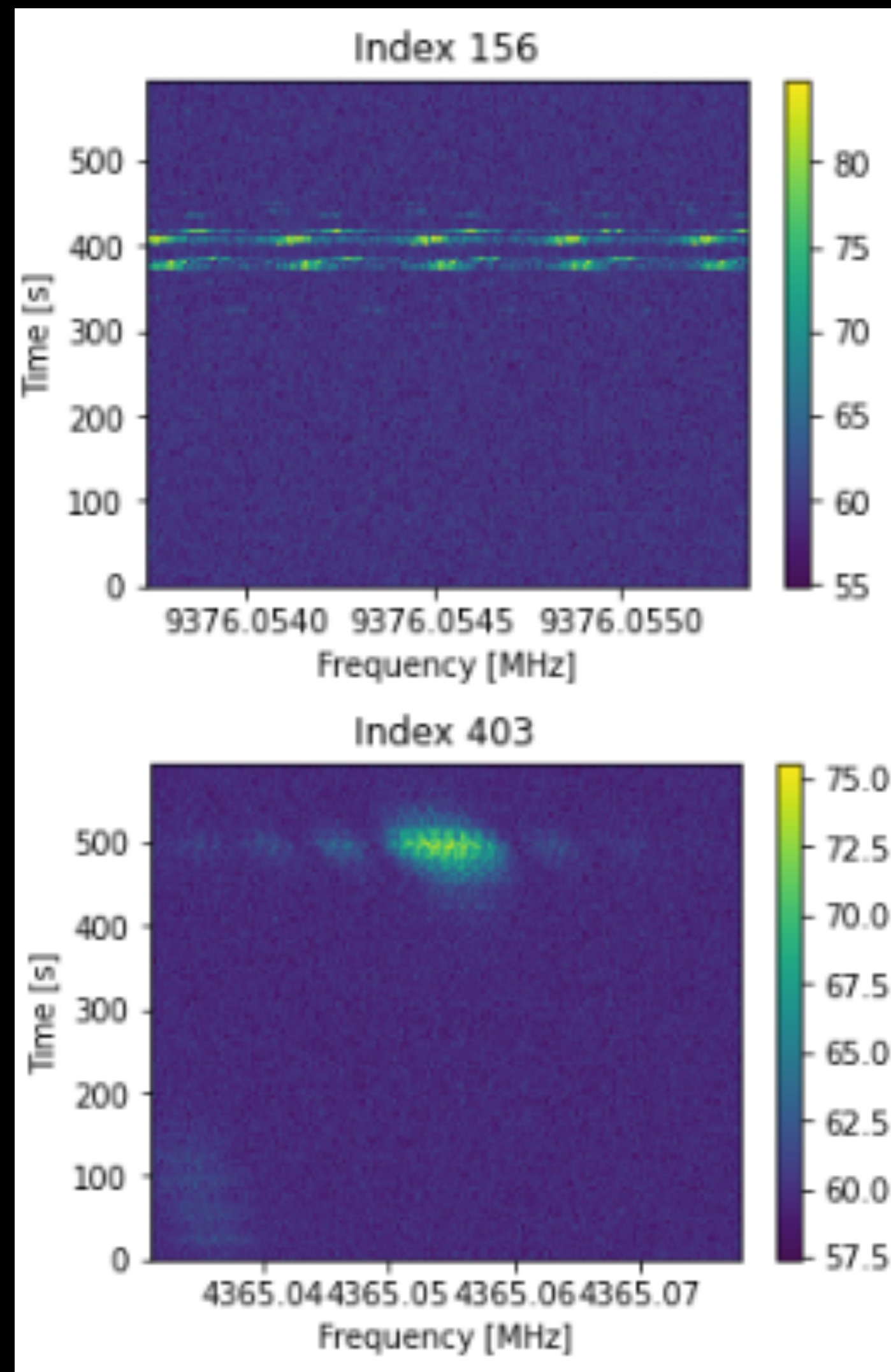
Scintillation Timescale Fit

L band

$S/N = 25$

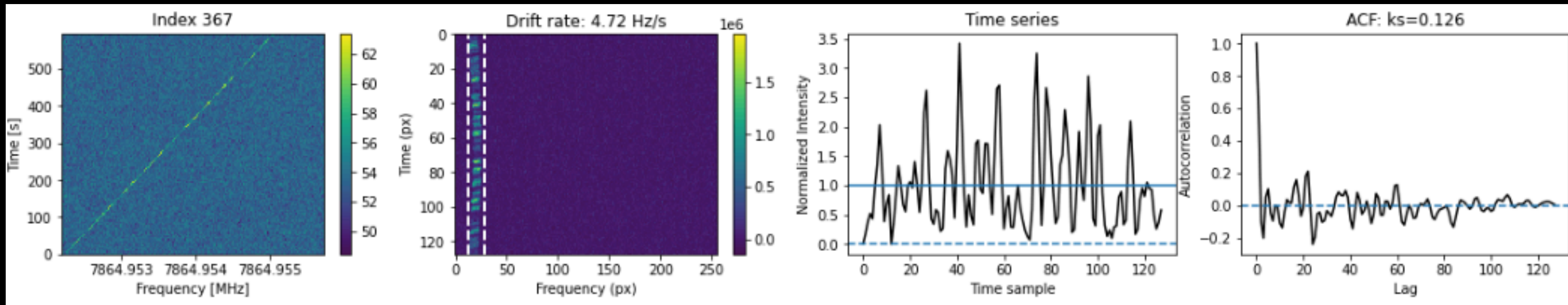


High standard deviation (RMS) signals are pulsed and/or broadband

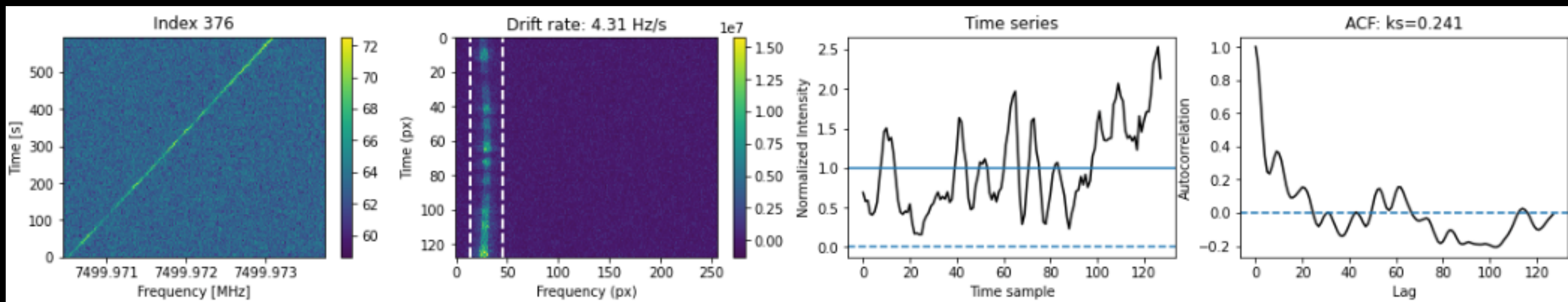


What signals pass the threshold?

- At C-band, $S/N > 25$, 3 out of 1102



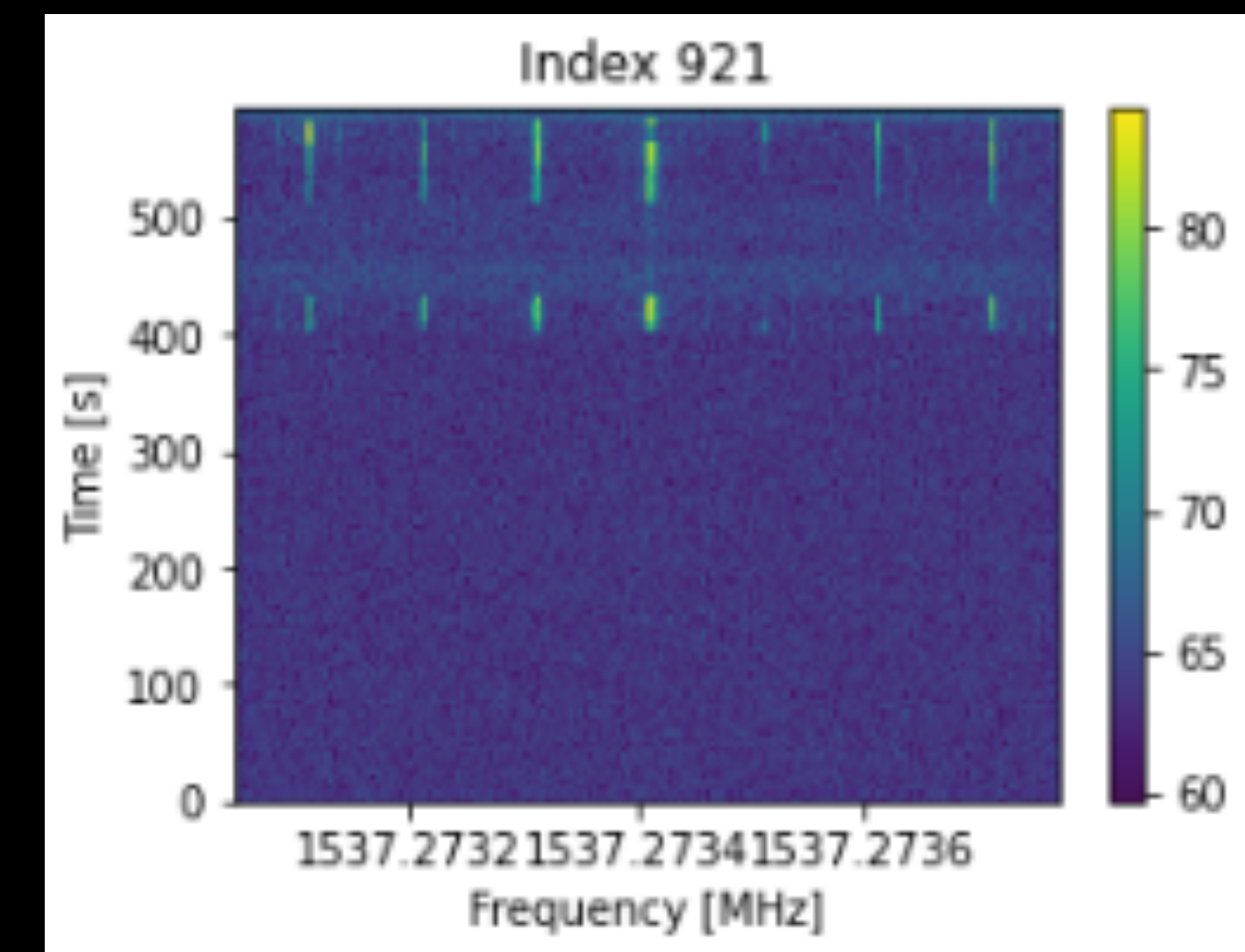
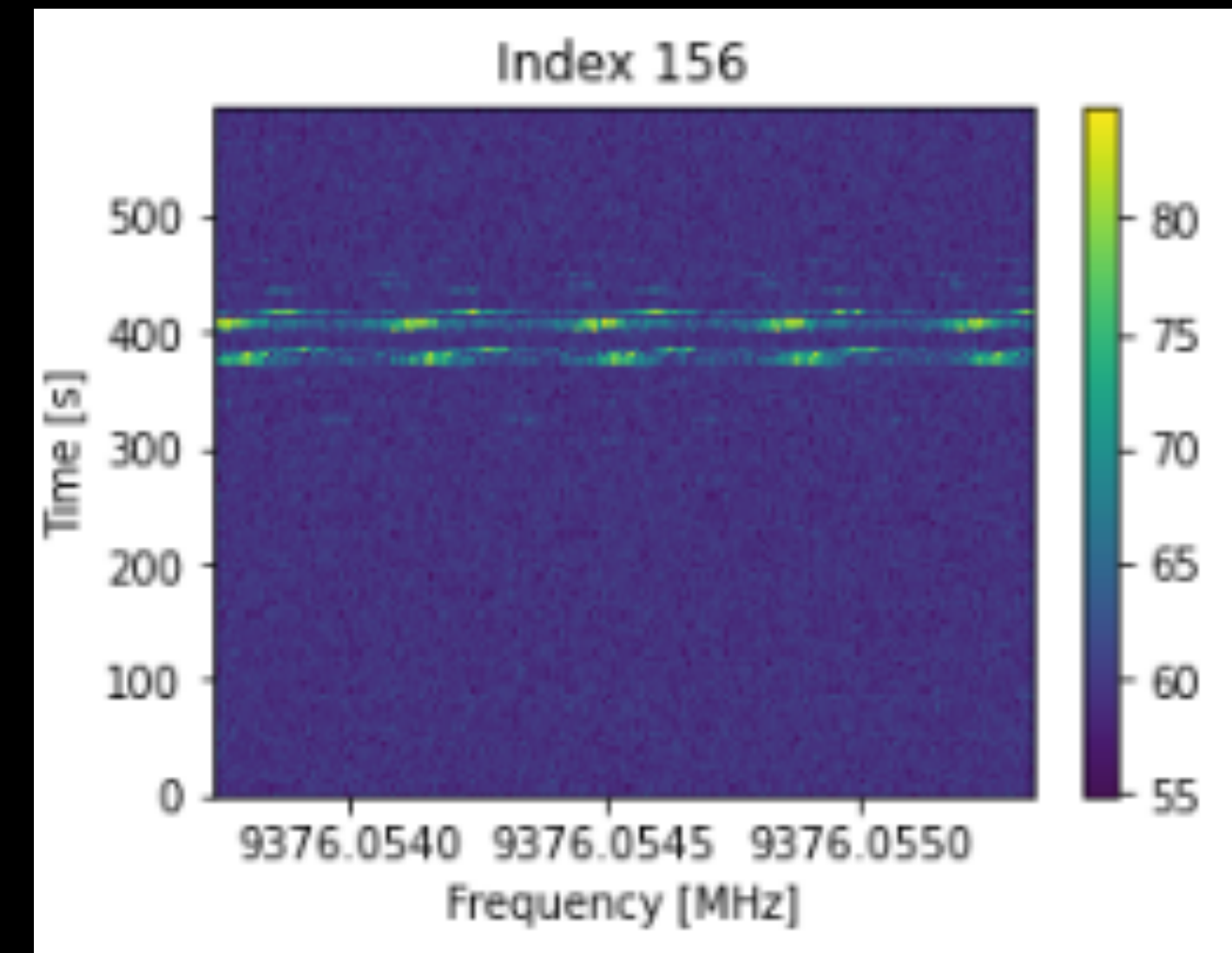
Timescale fit ~ 2 s



Timescale fit ~ 60 s

Limitations from RFI analysis?

- L and S bands in particular are very noisy
- Non-narrowband signals detected just because they are above the SNR threshold
- Difficult to apply a one-size-fits-all bounding box method
- Perhaps ML can help!

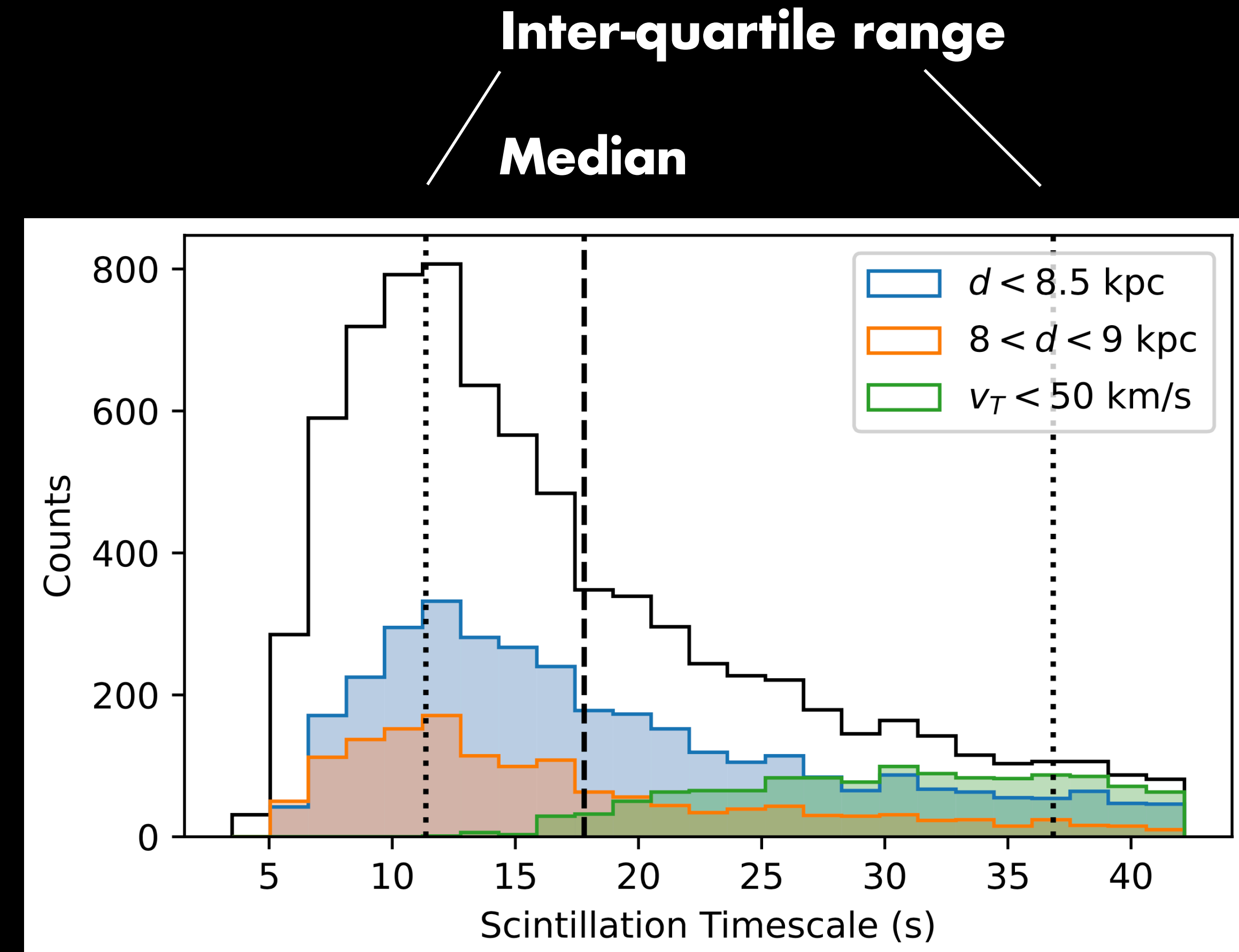


Can we detect scintillated narrowband technosignatures?

1. What scintillation timescales should we expect?
2. How can we probe asymptotic statistics?
3. Can we differentiate scintillated signals from existing RFI?
- 4. How can we design a survey to search for scintillated technosignatures?**

Planning Galactic Center observations — Monte Carlo sims with NE2001

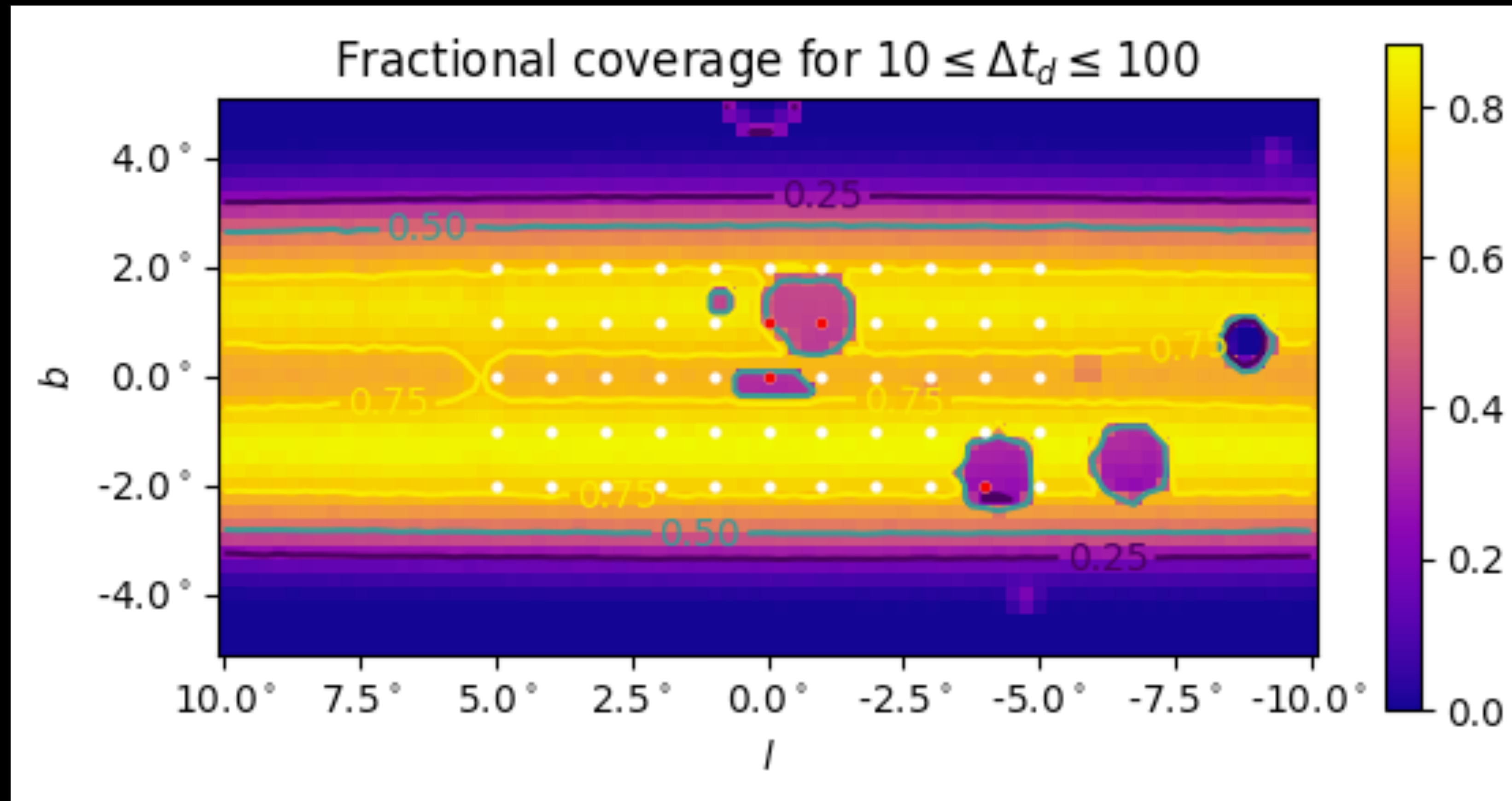
- Estimate scintillation timescales with NE2001 (Cordes & Lazio 2002) and scale with different sets of parameters
 - Galactic coordinates
 - Distance
 - Frequency
 - Transverse velocities
- Monte Carlo sample to estimate most probable scintillation timescales



$(l, b) = (5, 0)$ at C-band

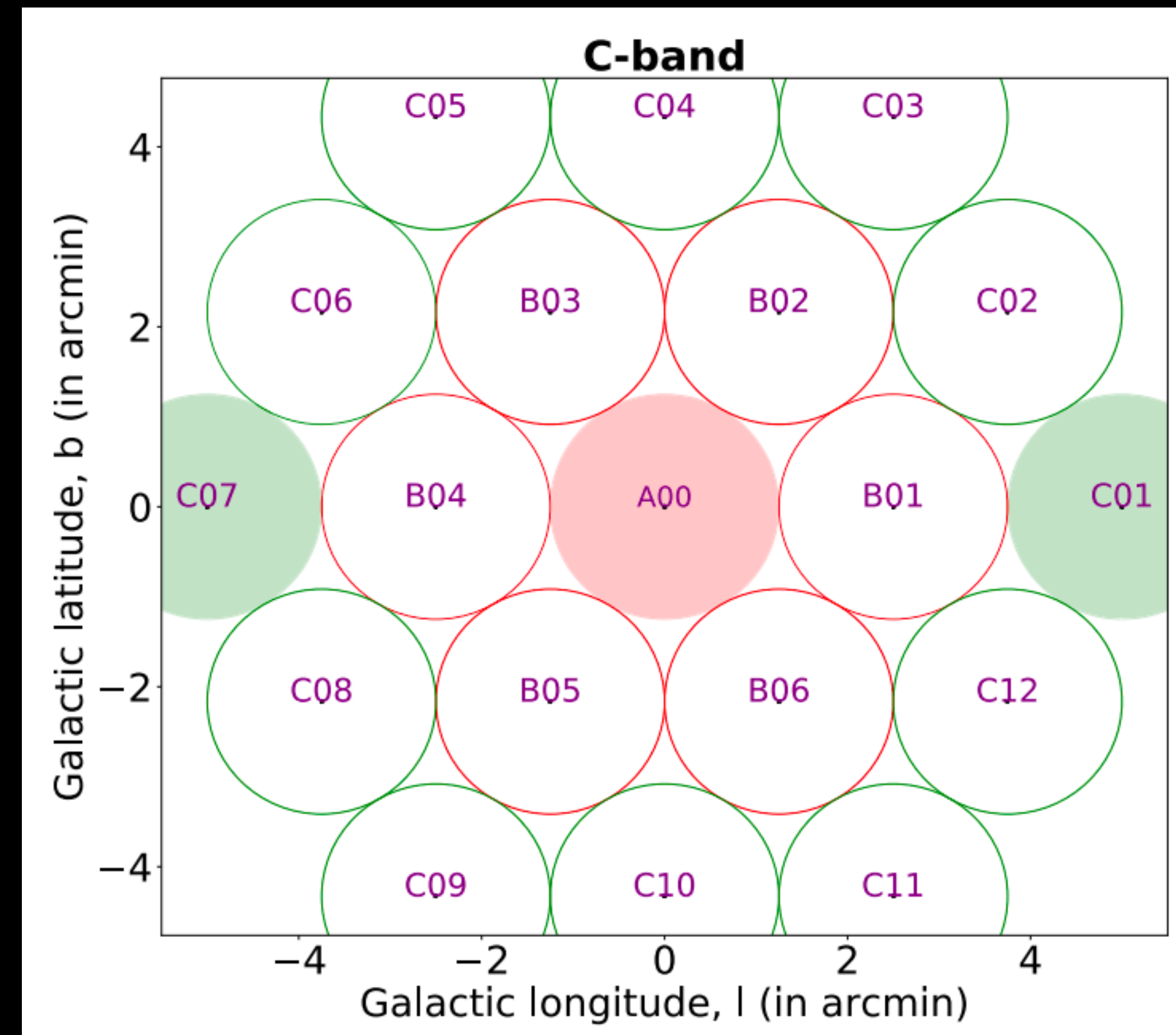
Current observing plan for scintillation survey of the Galactic center

- Galactic plane survey: 54 pointings, with $|l| < 5$ deg, $|b| < 2$ deg



Current observing plan for scintillation survey of the Galactic center

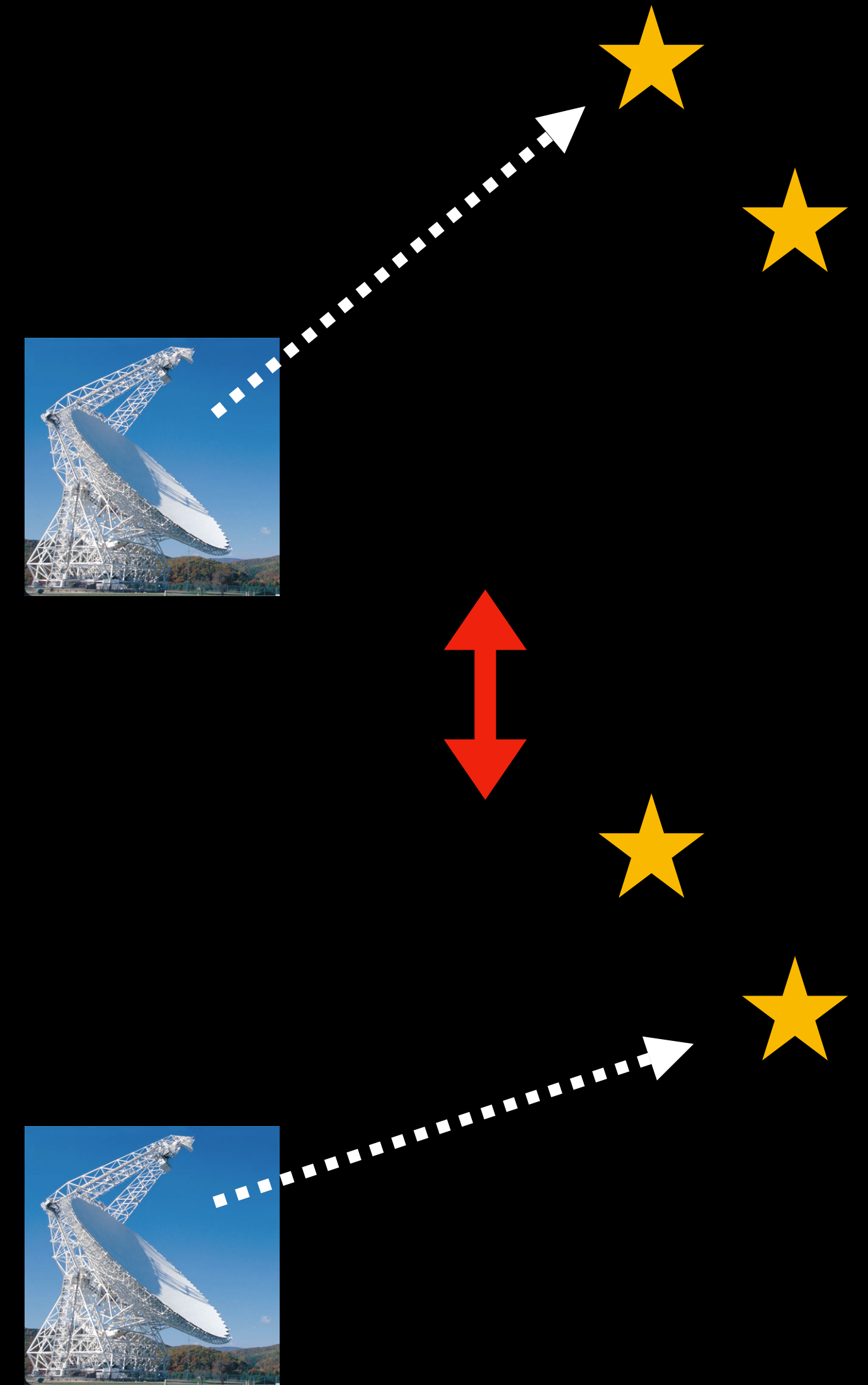
- Galactic center survey: 19 pointings (following Gajjar et al. 2021)



Gajjar et al. 2021

Current observing plan for scintillation survey of the Galactic center

- ABAB cadences
- 10 minutes per observation, so each pointing gets 20 minutes total
- 2.5 s, 2.8 Hz resolution
- Start each observing session with single pointing of North Galactic Pole as probe of local RFI environment



NRAO

Next Steps

- Currently, we have data for 16 out of 27 cadences of the Galactic plane survey, about 12 hours of data
 - 11 GP cadences and 9 GC cadences remain
- Filter collected data using established ON-OFF search methods and perform scintillation analysis
- Ultimate goal is to comment on the prevalence of scintillated technosignatures, as well as the prevalence of RFI that might pass the scintillation thresholds

Summary

- We developed a scintillation analysis framework, with accompanying codebase
- We can set statistical filter thresholds based on synthetic signals and the local RFI environment
- We've planned a survey to search for scintillated signals towards the Galactic center / plane, which is well under way

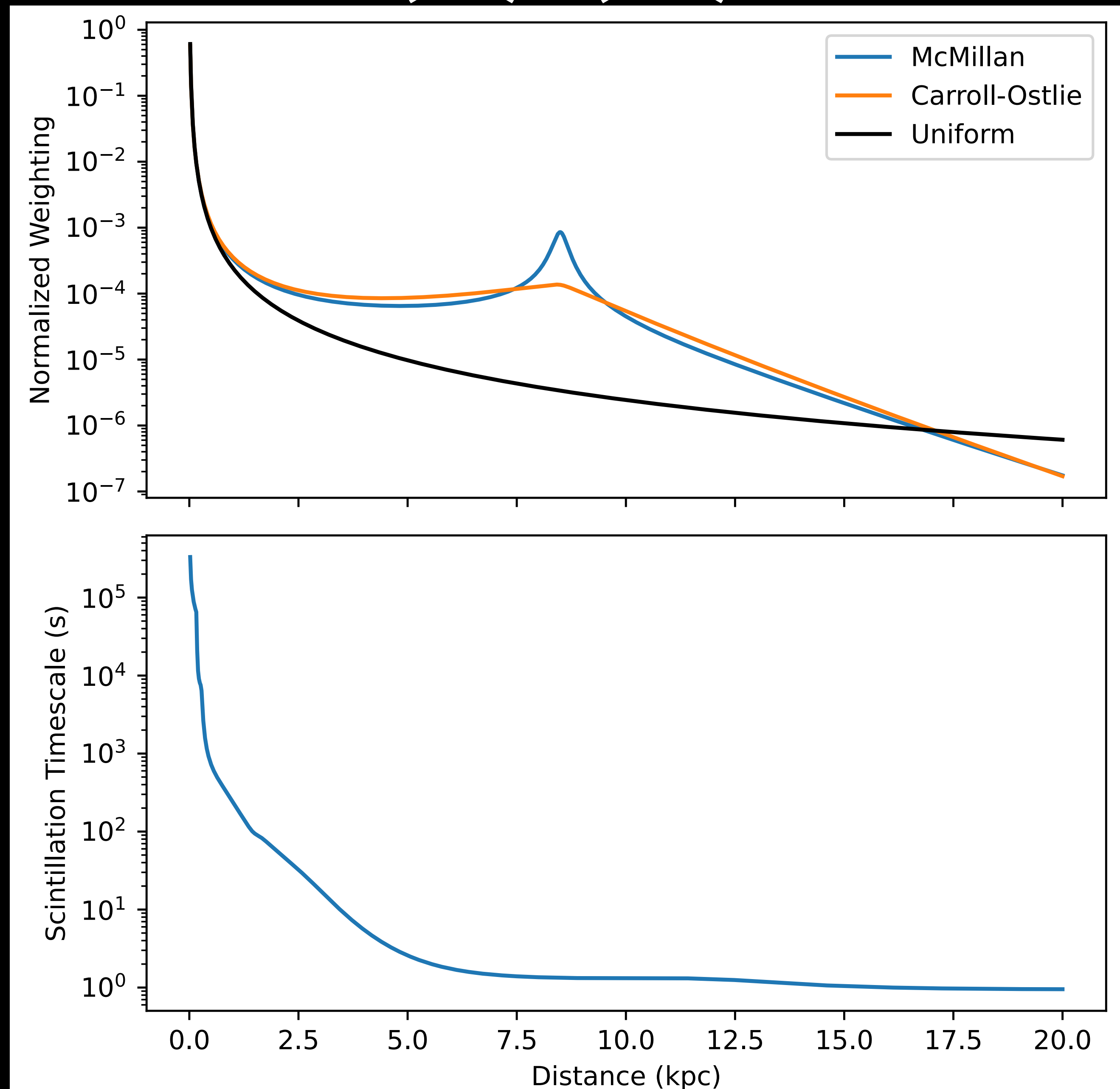
Extra Slides

Density-based sampling

Modulating by the inverse square-law for detectability:

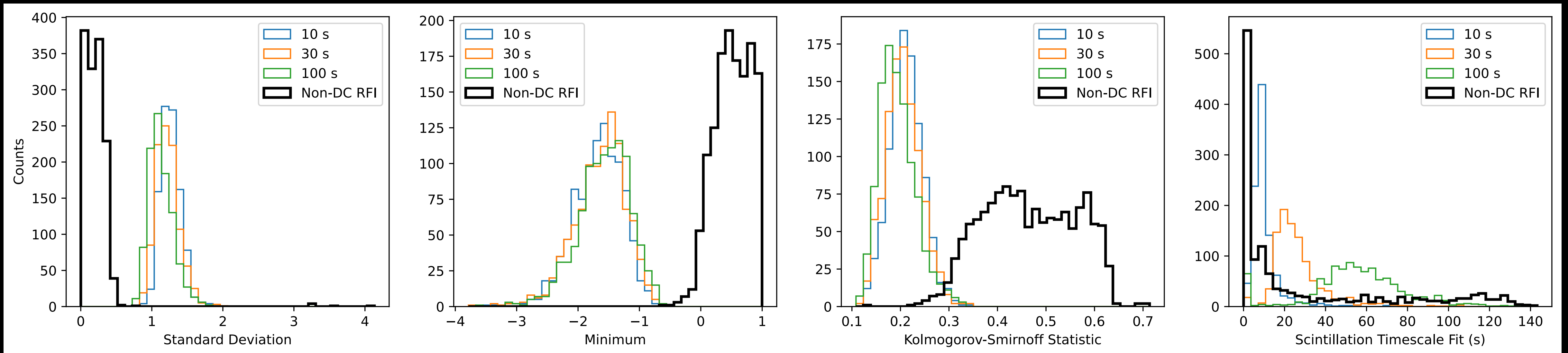
Depends on the assumptions made about transmission power and resources.

$$(l, b) = (1, 0)$$



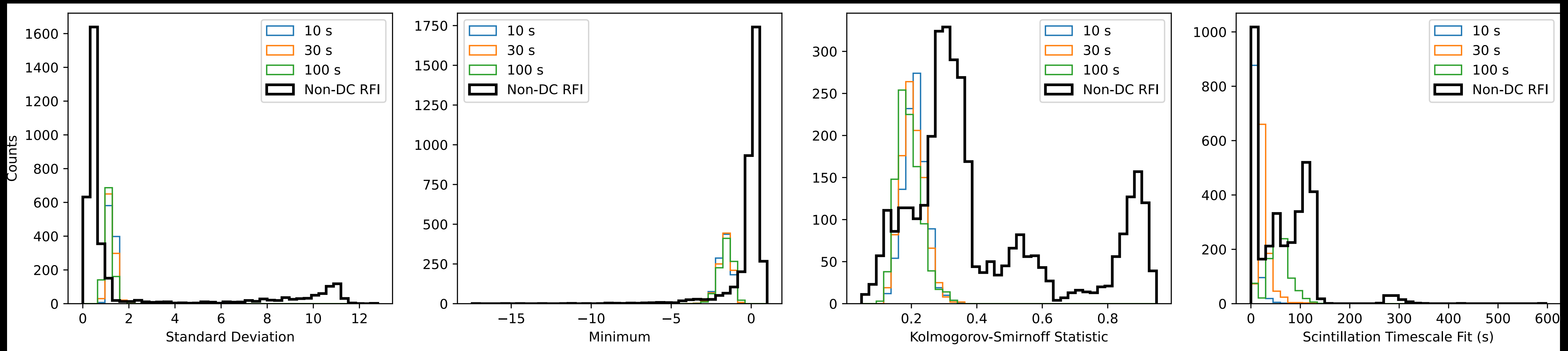
C band

$S/N = 10$



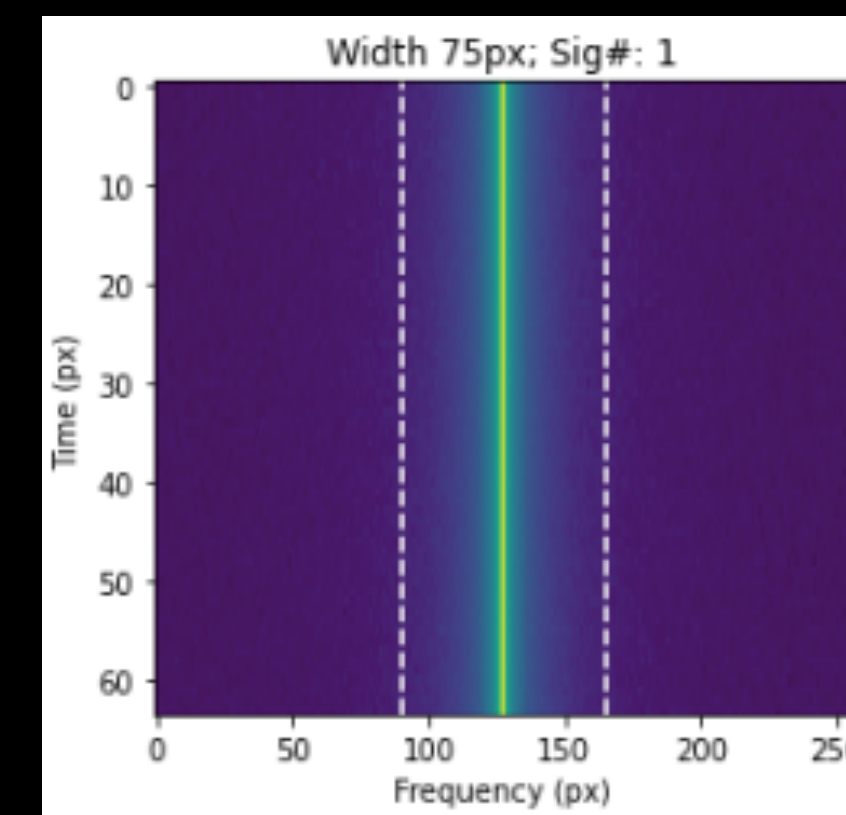
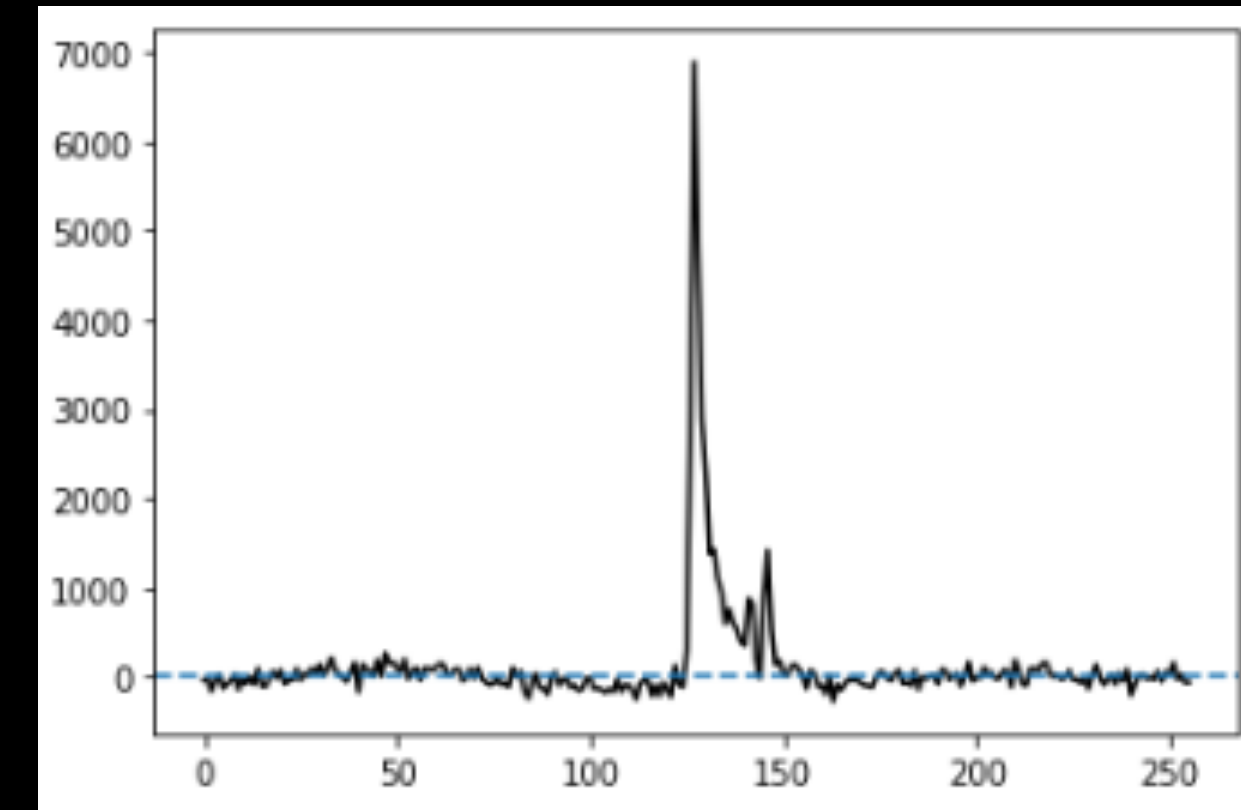
L band

$S/N = 10$

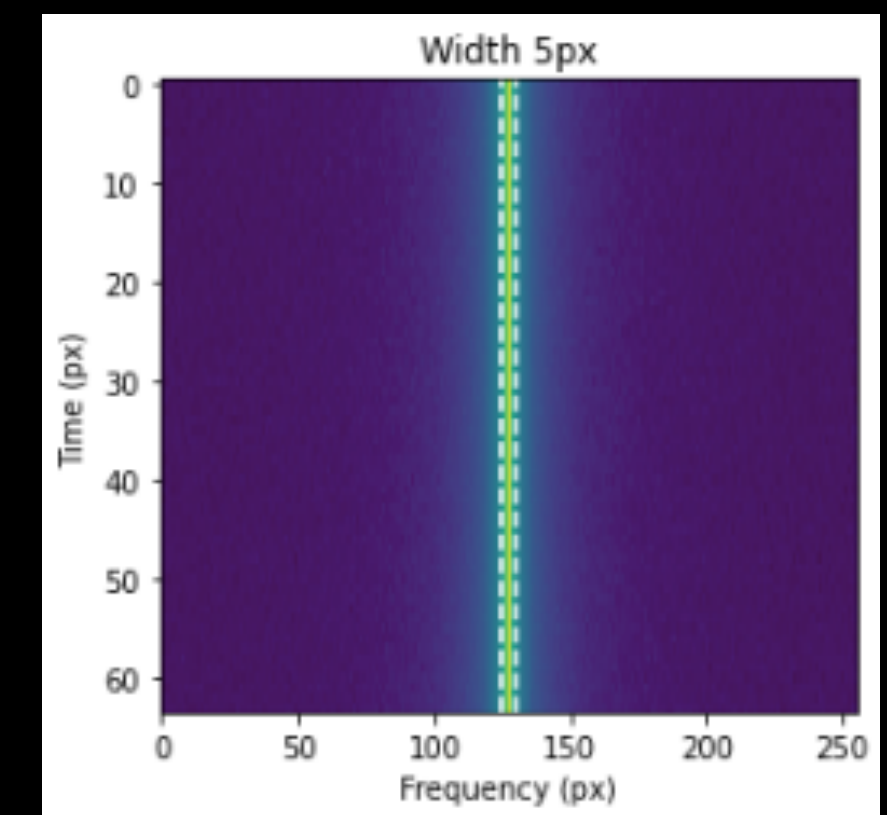


Selecting bounding boxes

- After experimentation with various methods, the final pipeline uses a combination of baseline fitting and peak detection to calculate the right size of frame to use
- The final bounds are created using a thresholding method, similar to PSRCHIVE
- Take the final bounded signal and integrate in the frequency direction to derive our raw time series — then we normalize to mean of 1 before calculating our scattering statistics



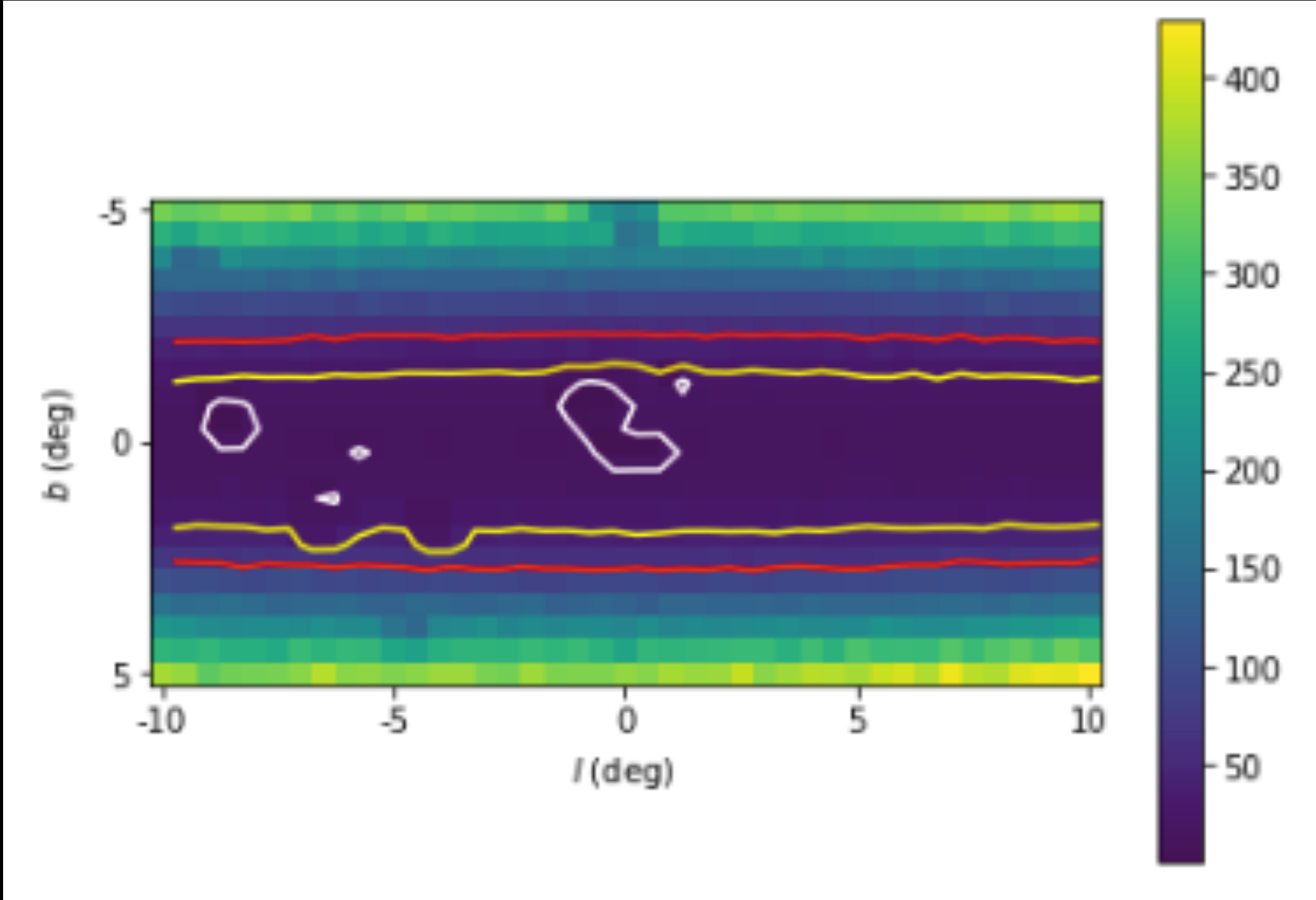
Polynomial fit



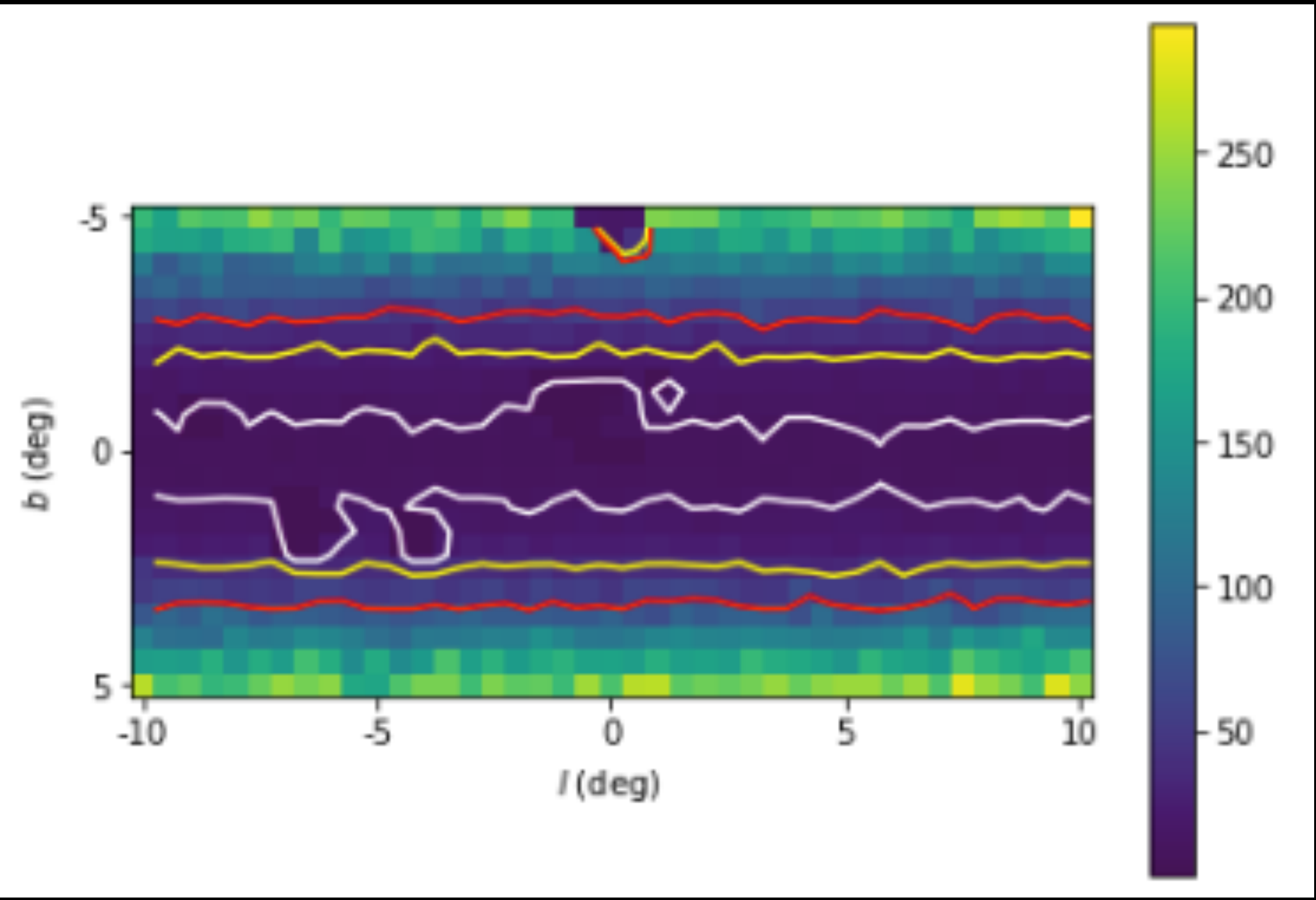
Threshold fit

Scintillation maps around the GC at C-band

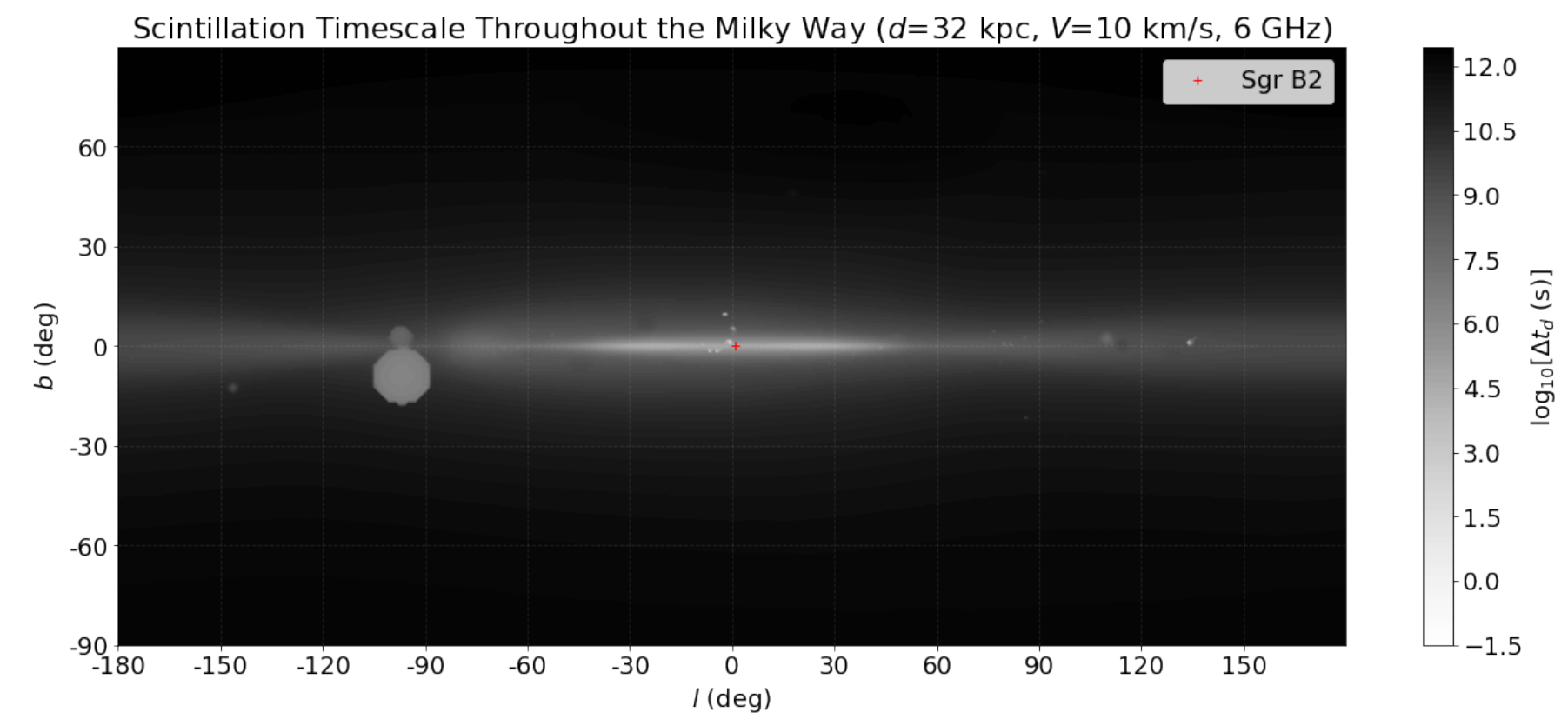
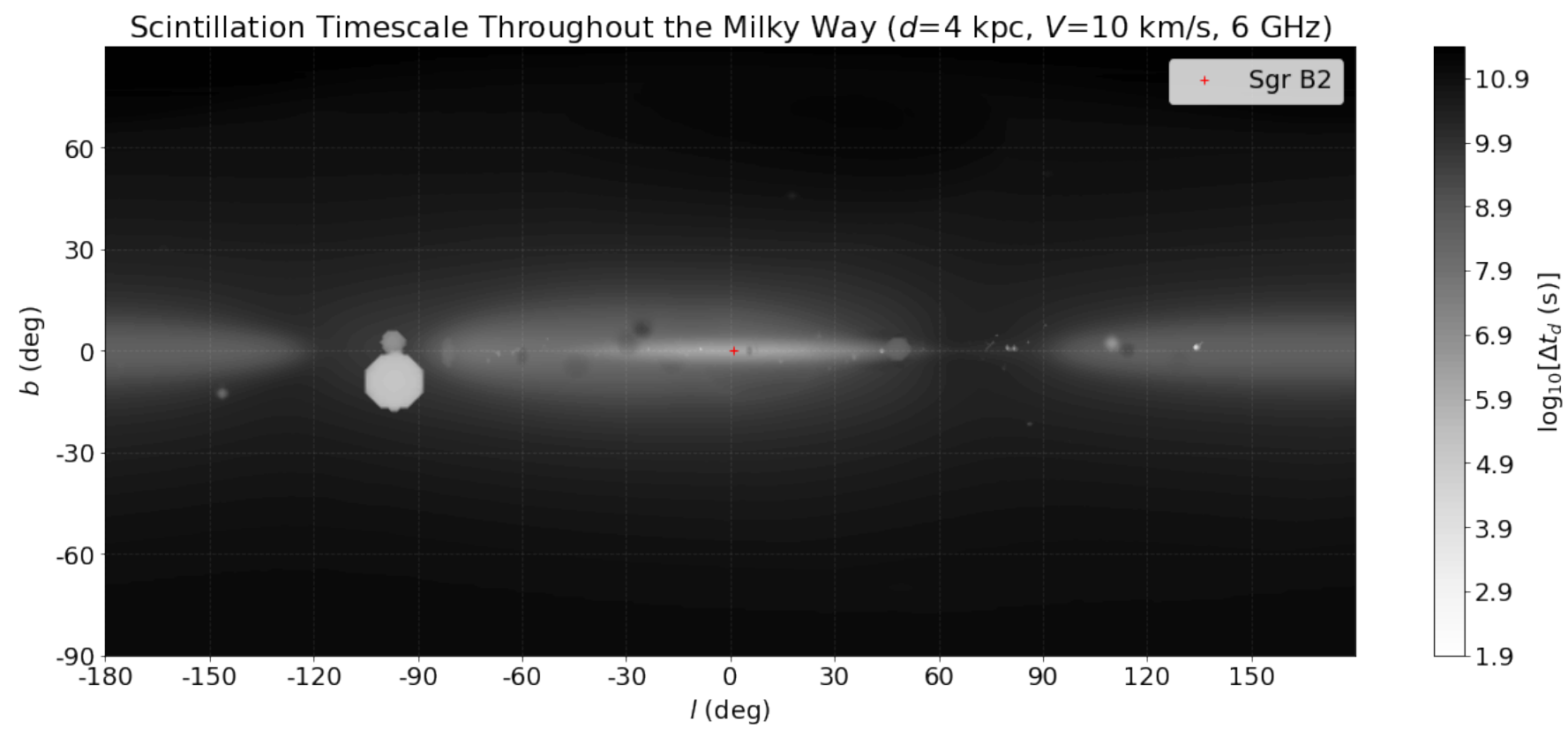
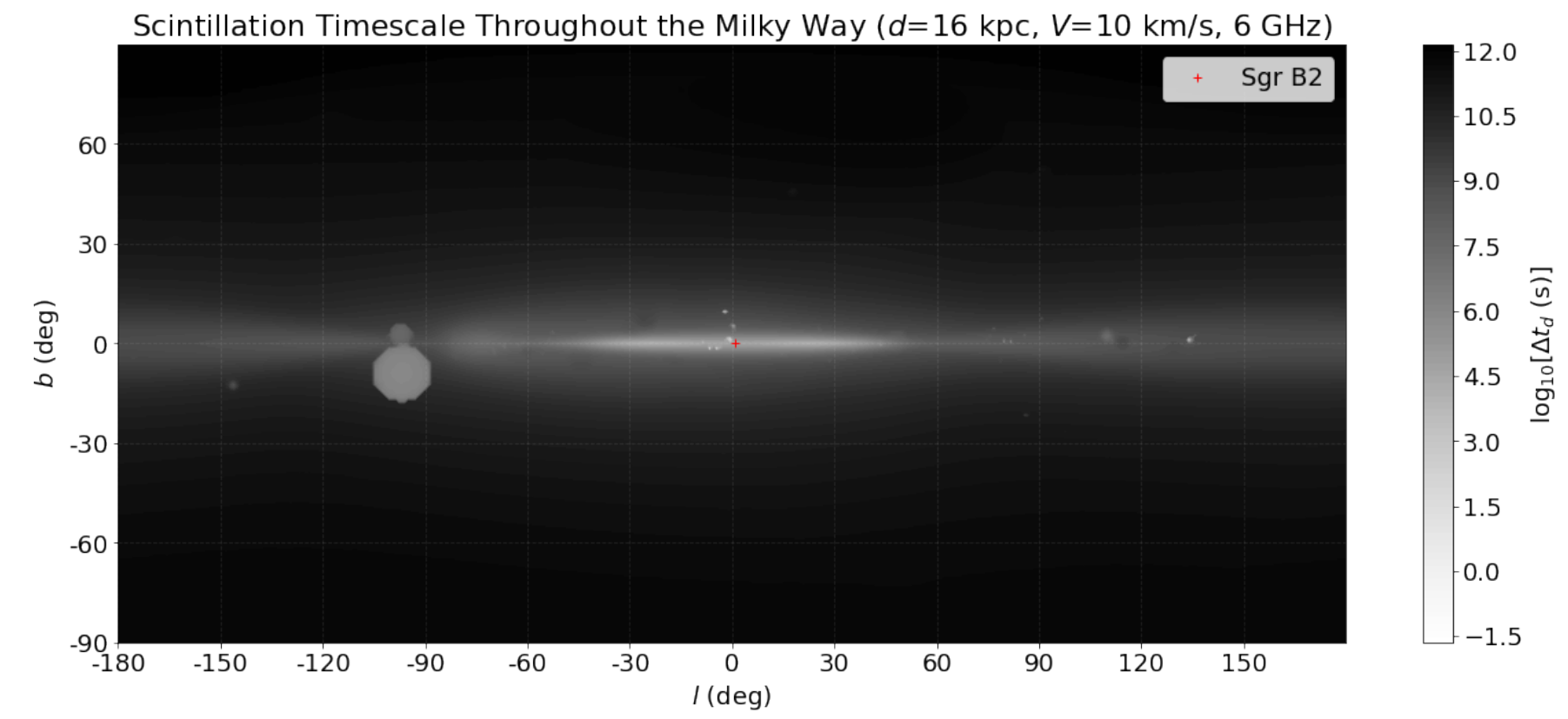
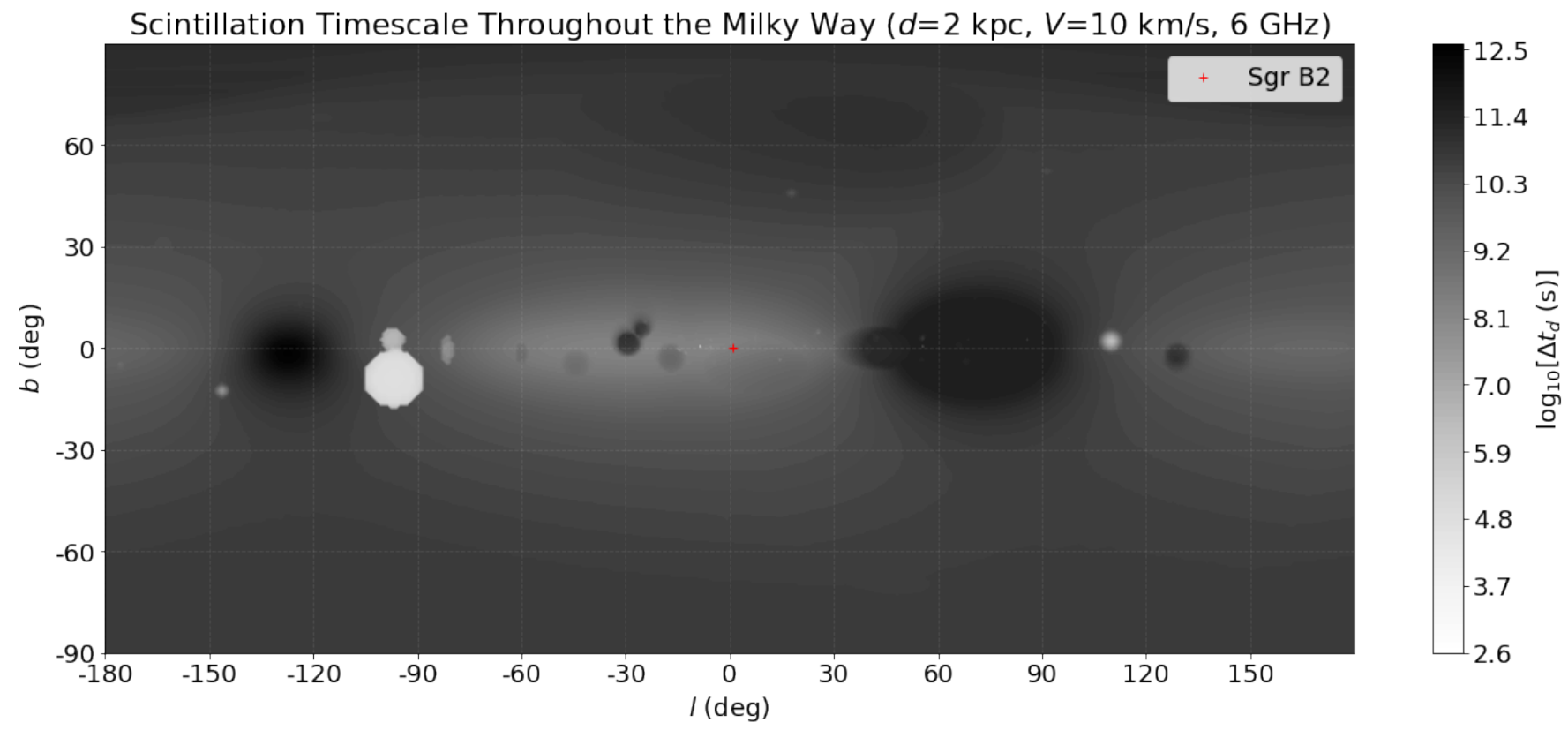
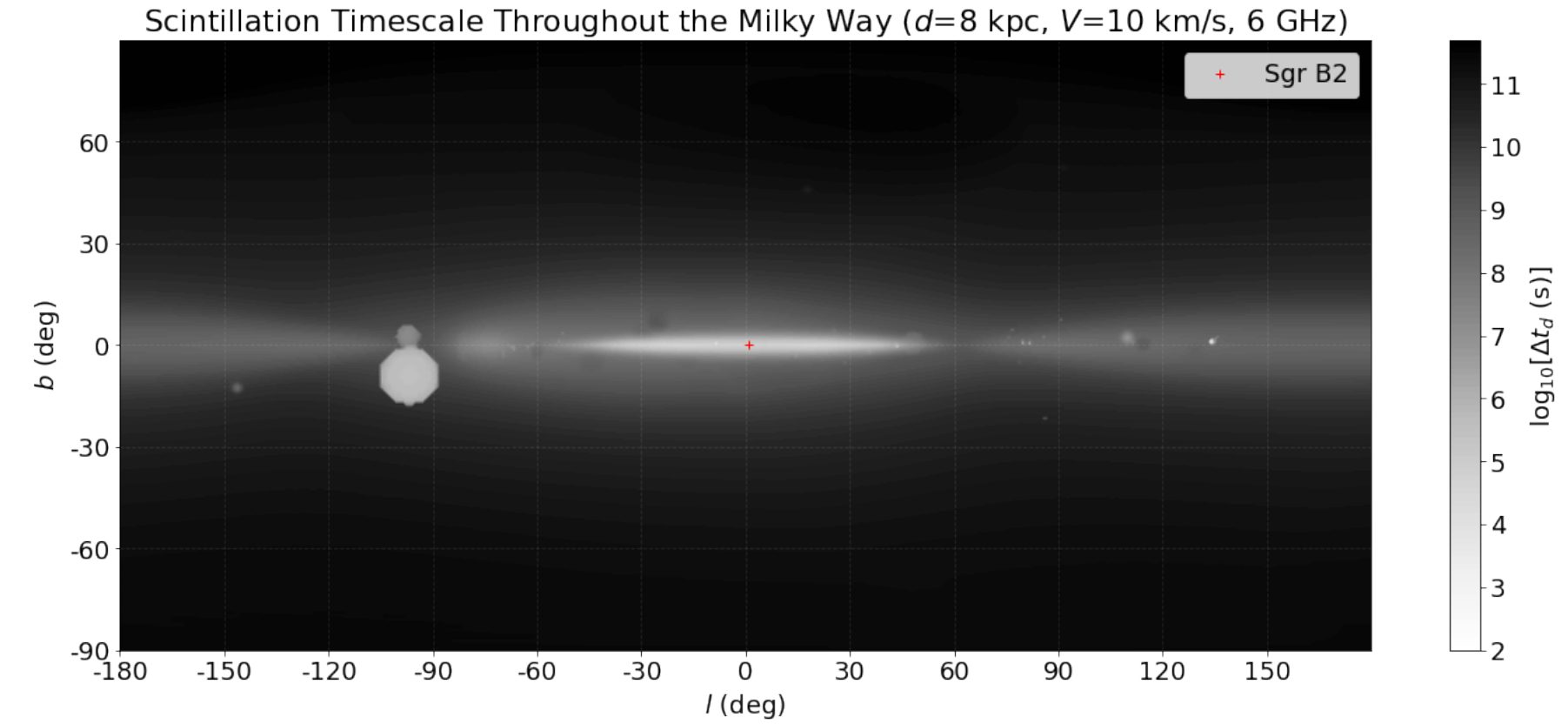
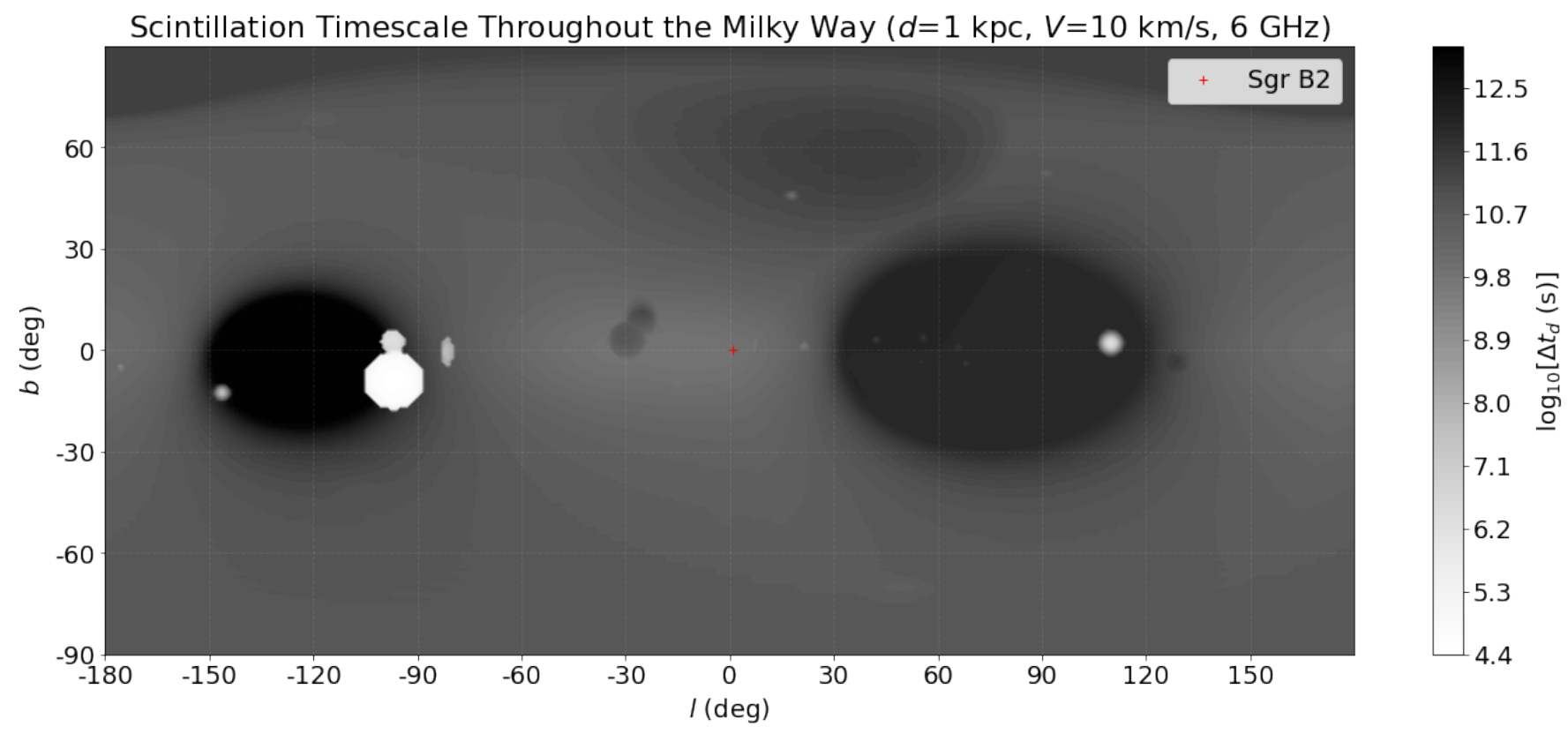
Median



Mode

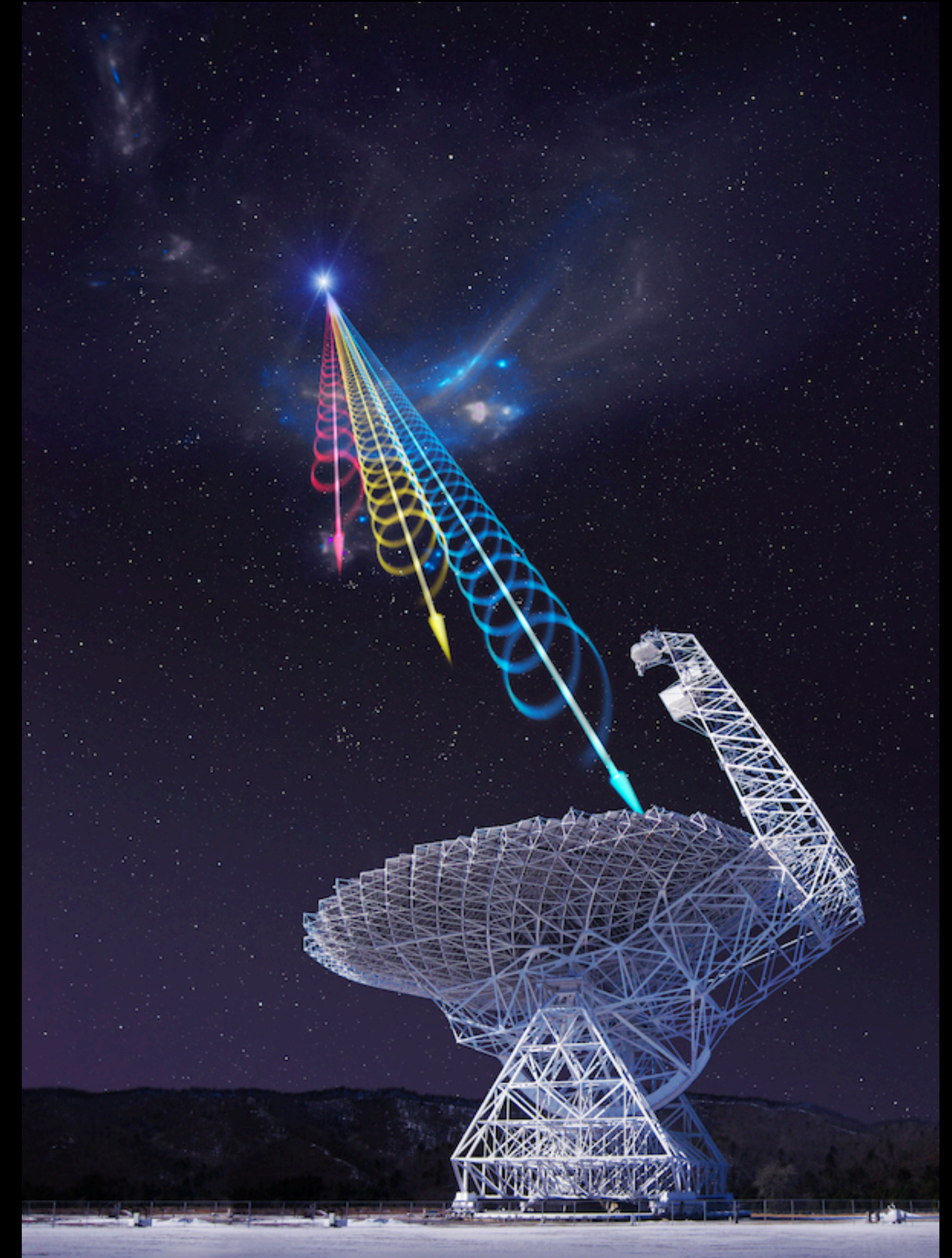


10 s, 30 s, 60 s



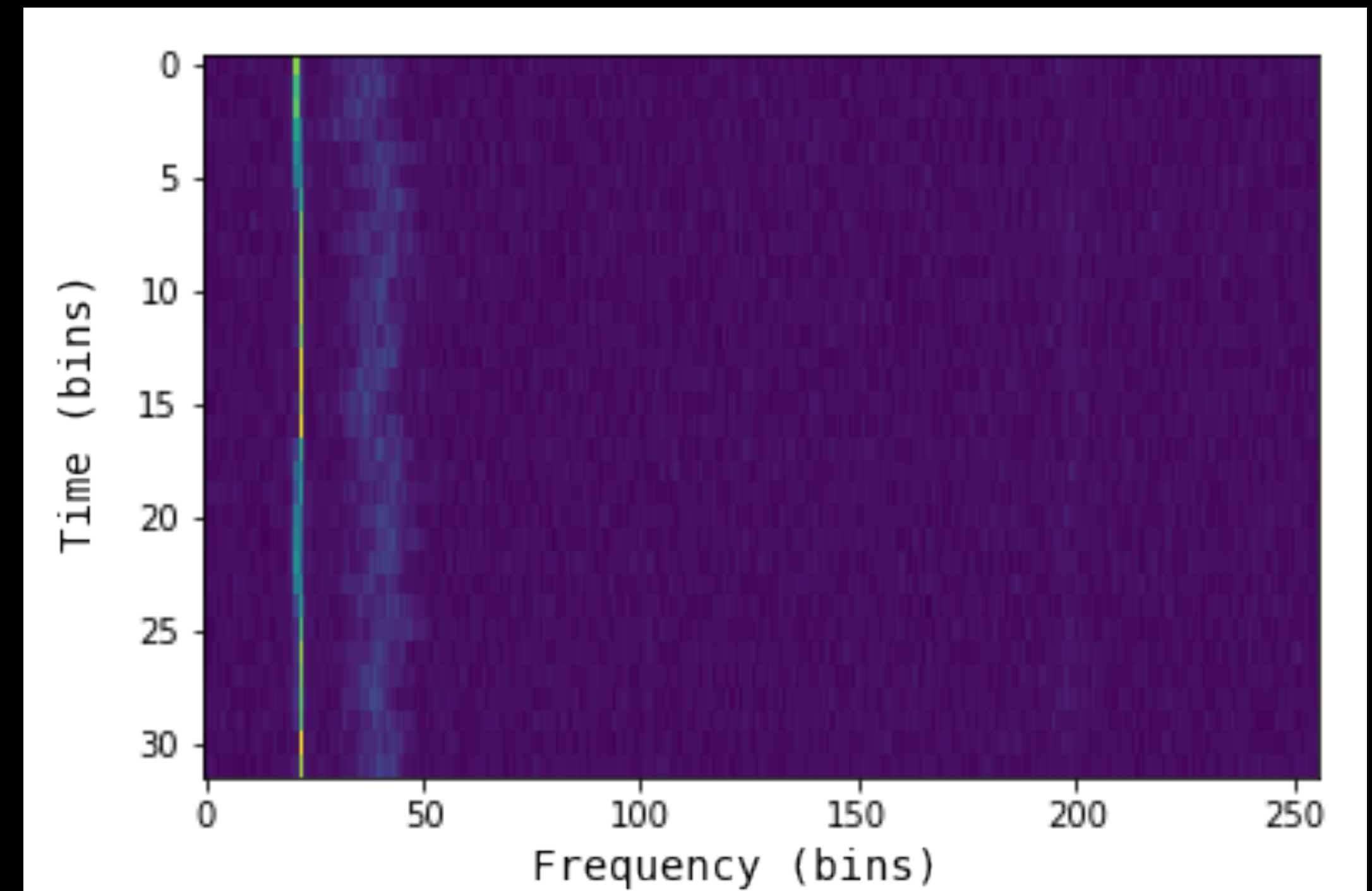
My goal: develop search methods for SETI from both angles

- Machine learning and software tools to support more complex detections
- Investigate astrophysical effects imprinted on technosignatures themselves



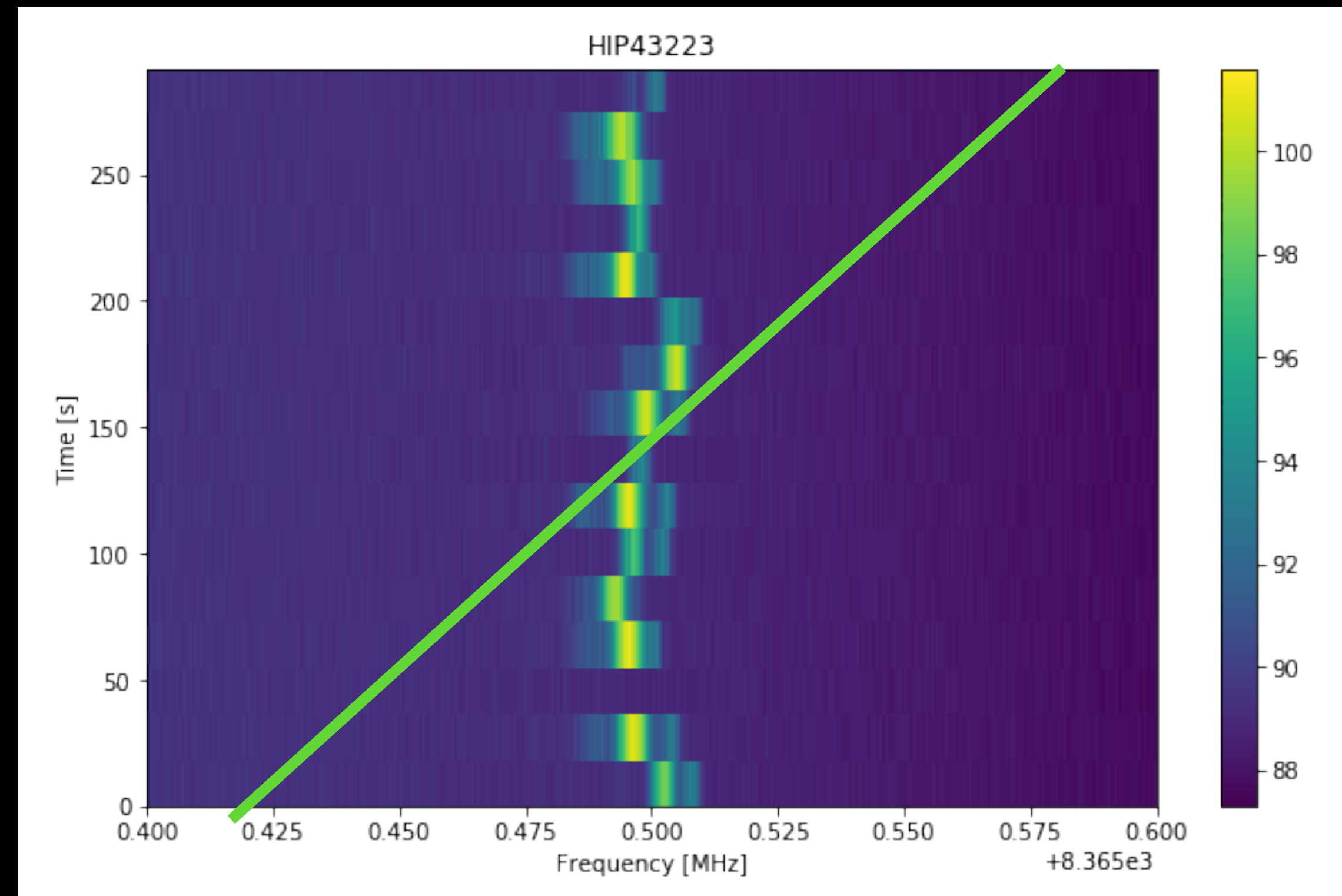
Narrowband signal localization with machine learning

- Standard deDoppler pipeline:
 - Dim signals concealed by nearby bright signals
 - Computationally expensive to search high drift rates

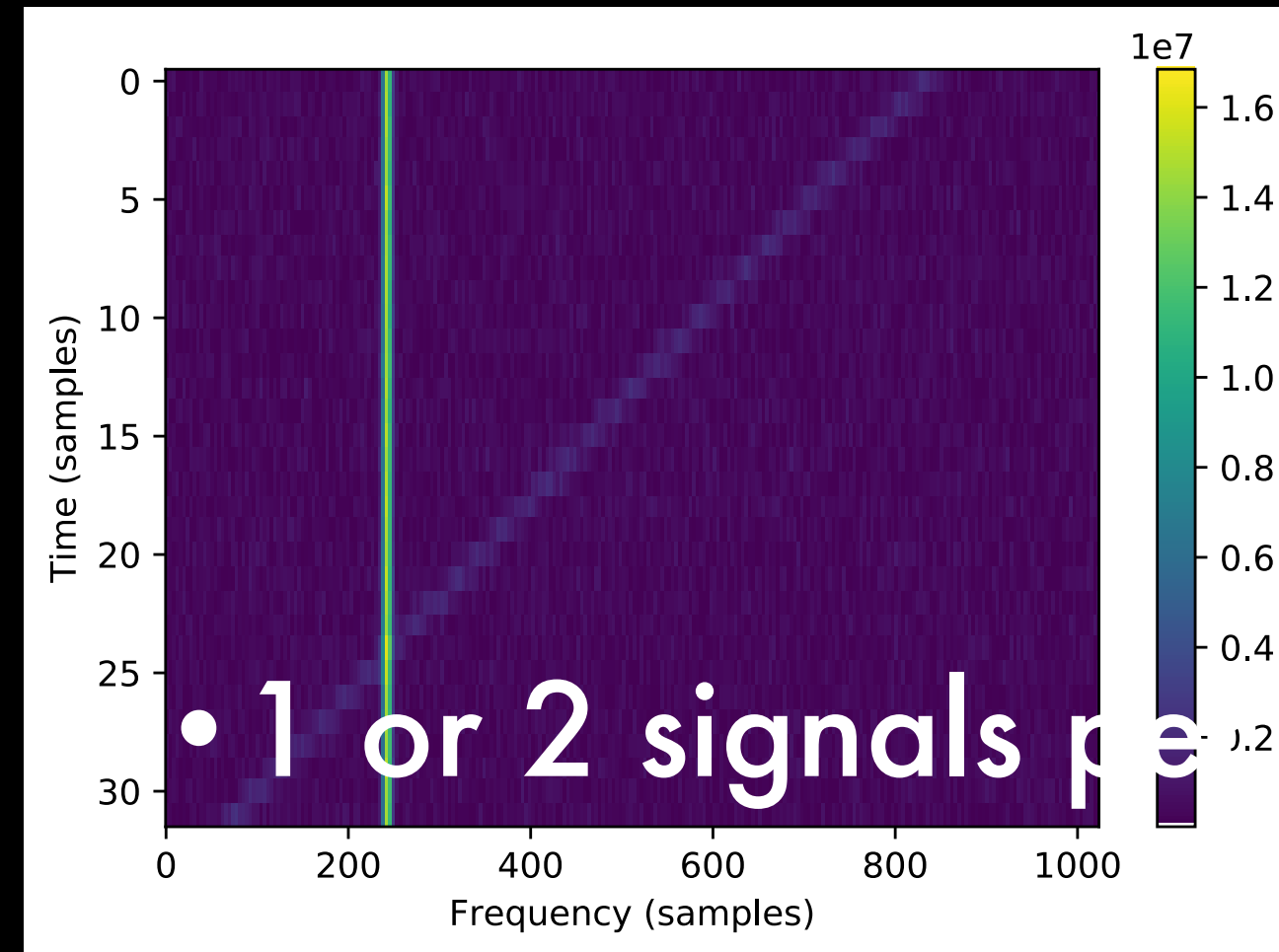


Small snippet of GBT data at C-band

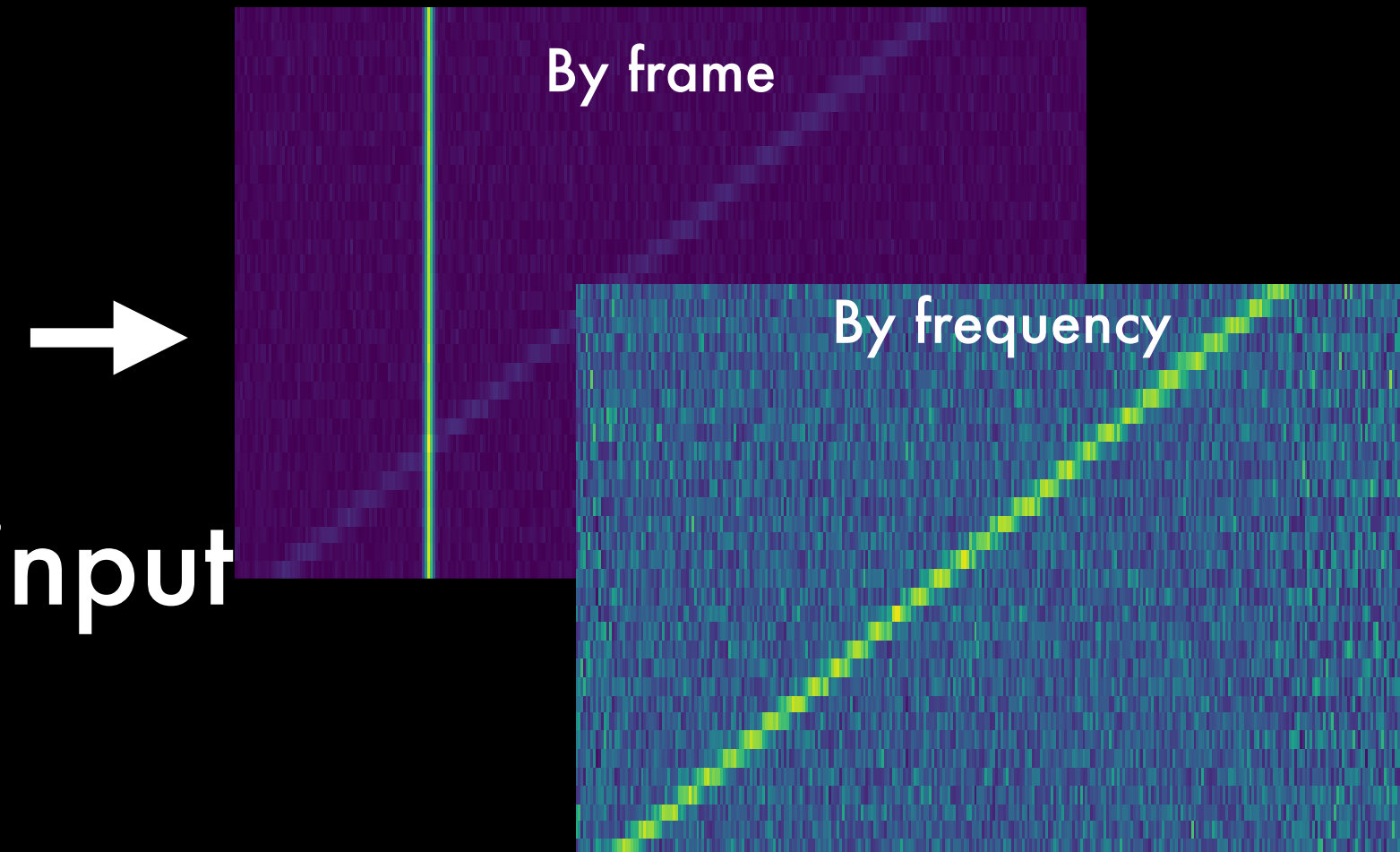
- Masking?



Synthetic training data



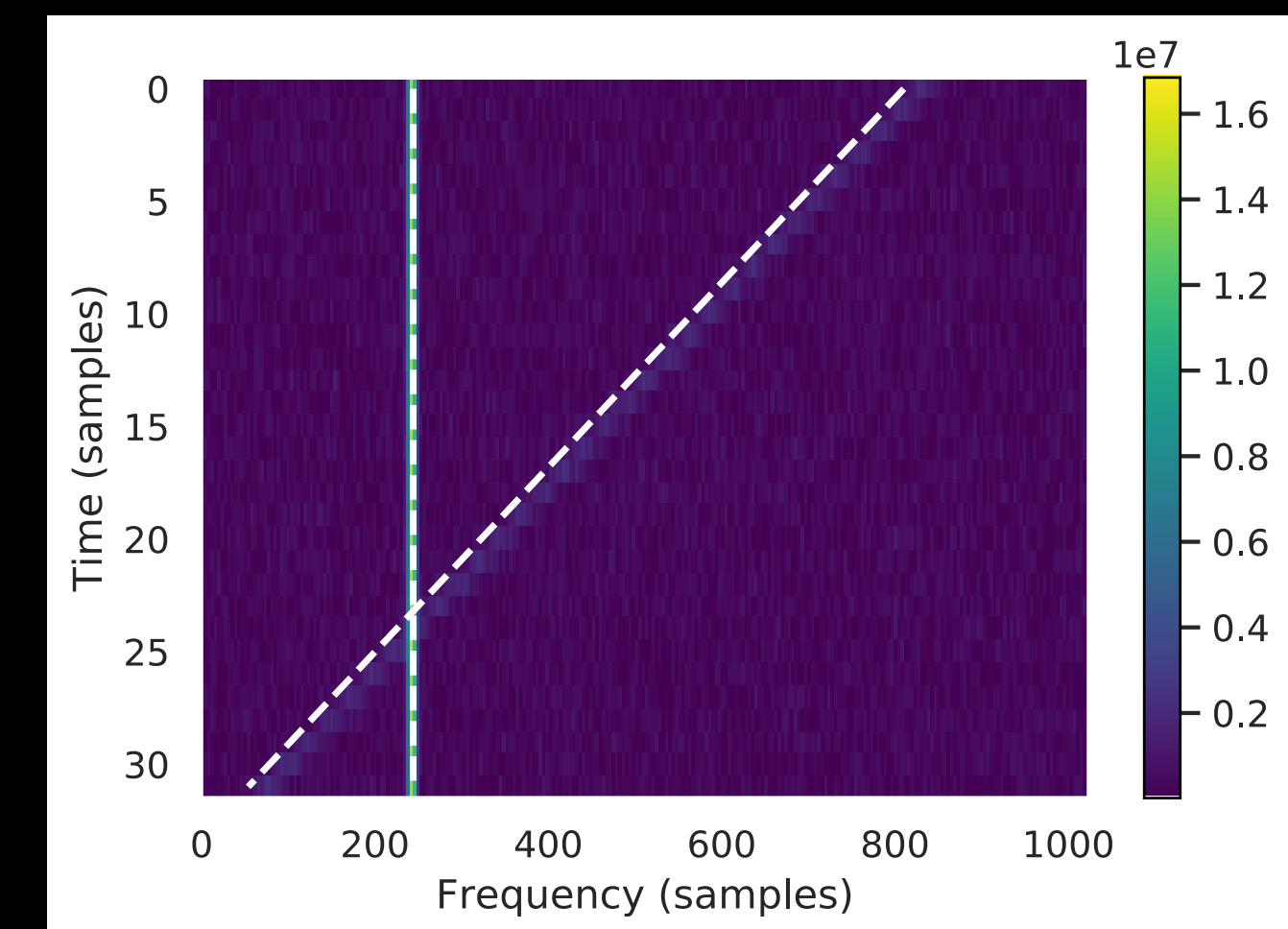
Normalization



Neural Network

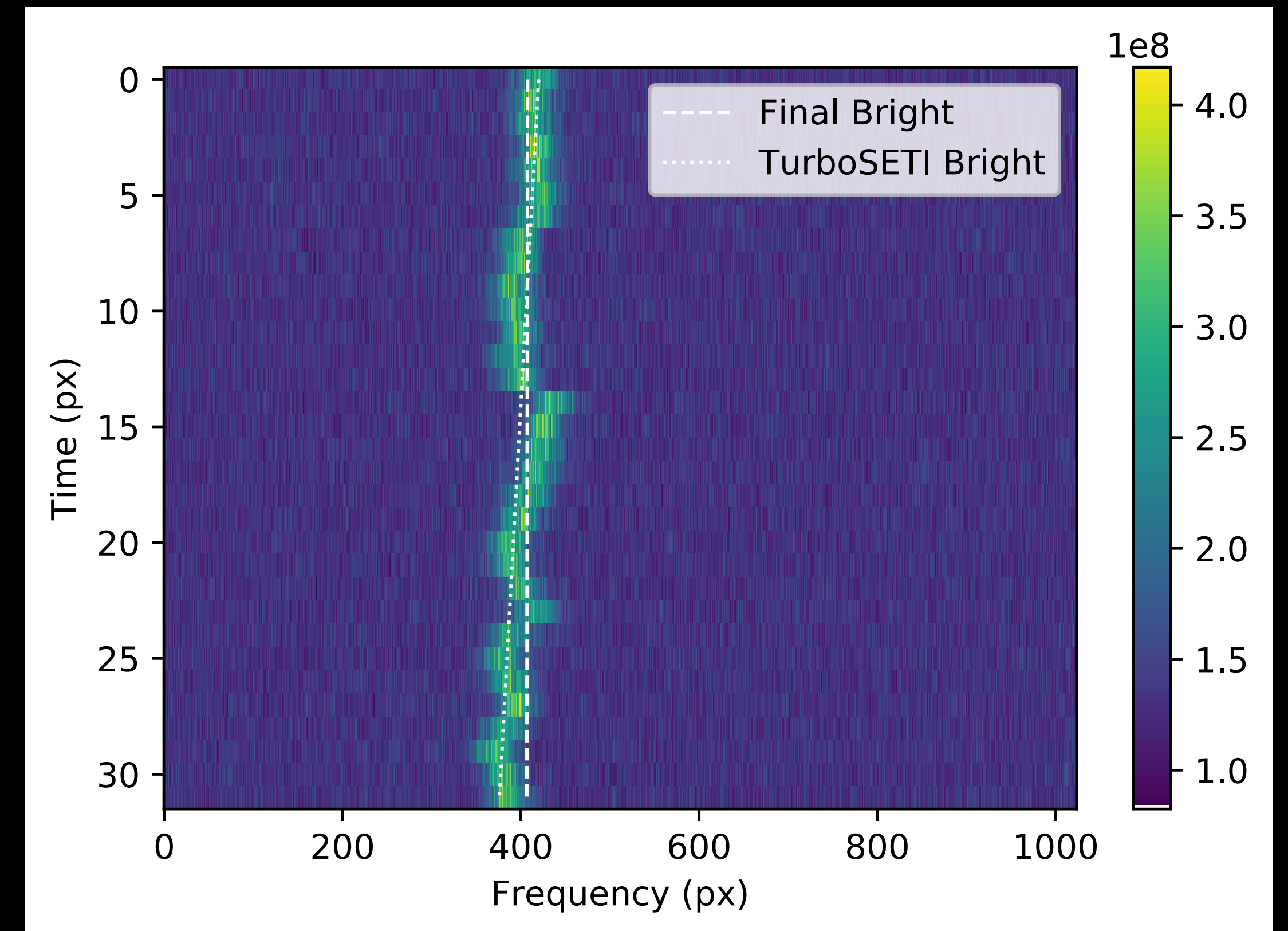


Predicted locations



Takeaways

- Less accurate than deDoppler methods, but generally 20-40x faster
- Trained on ideal signals but still relatively robust
- For production use, would need to extend to variable number of signals



C-band RFI signal, with ML prediction dashed and TurboSETI localization dotted.

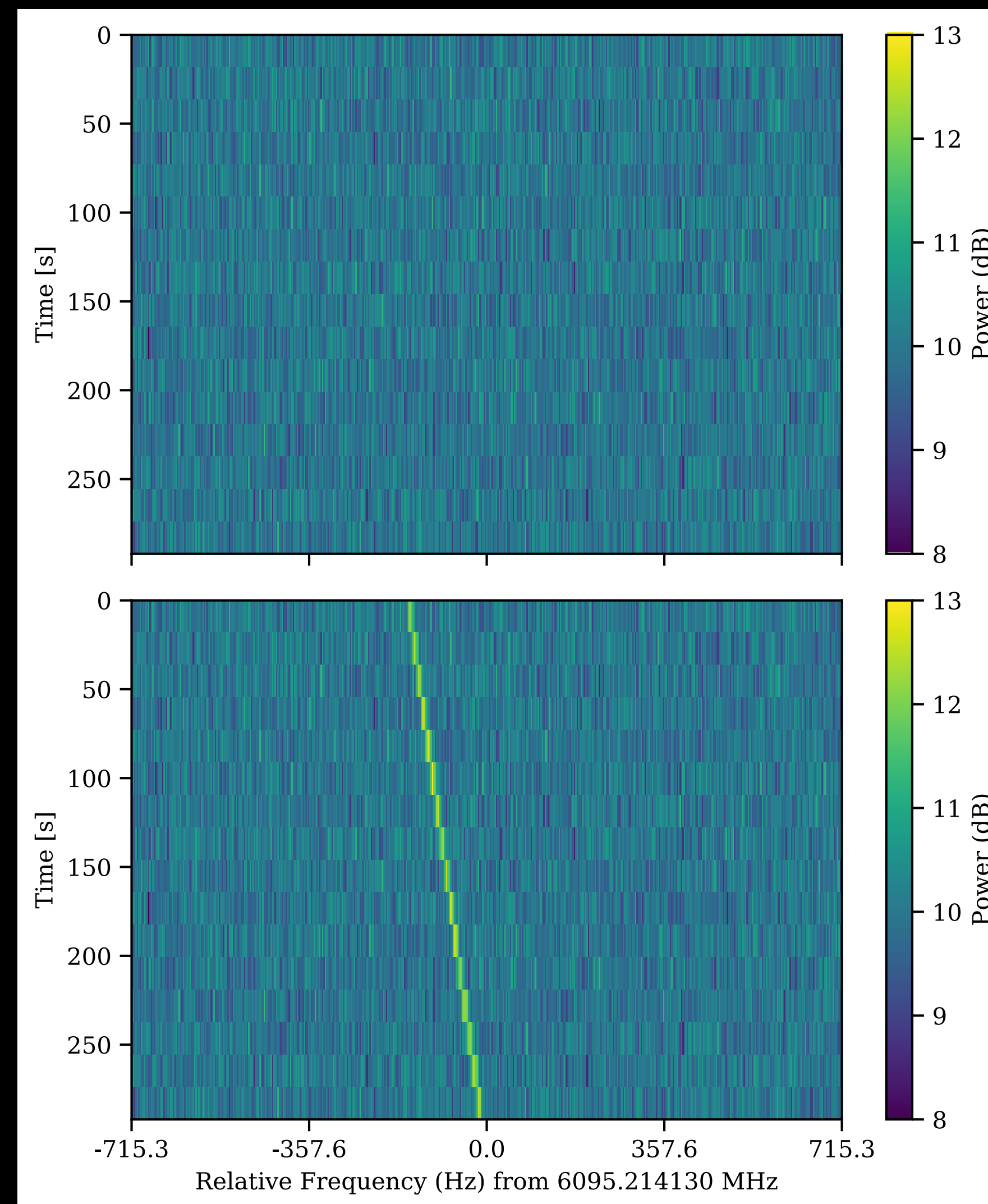
Setigen

- Python library for synthetic spectrogram and voltage data
- Specific focus on narrowband signal generation and injection

 **setigen** Public

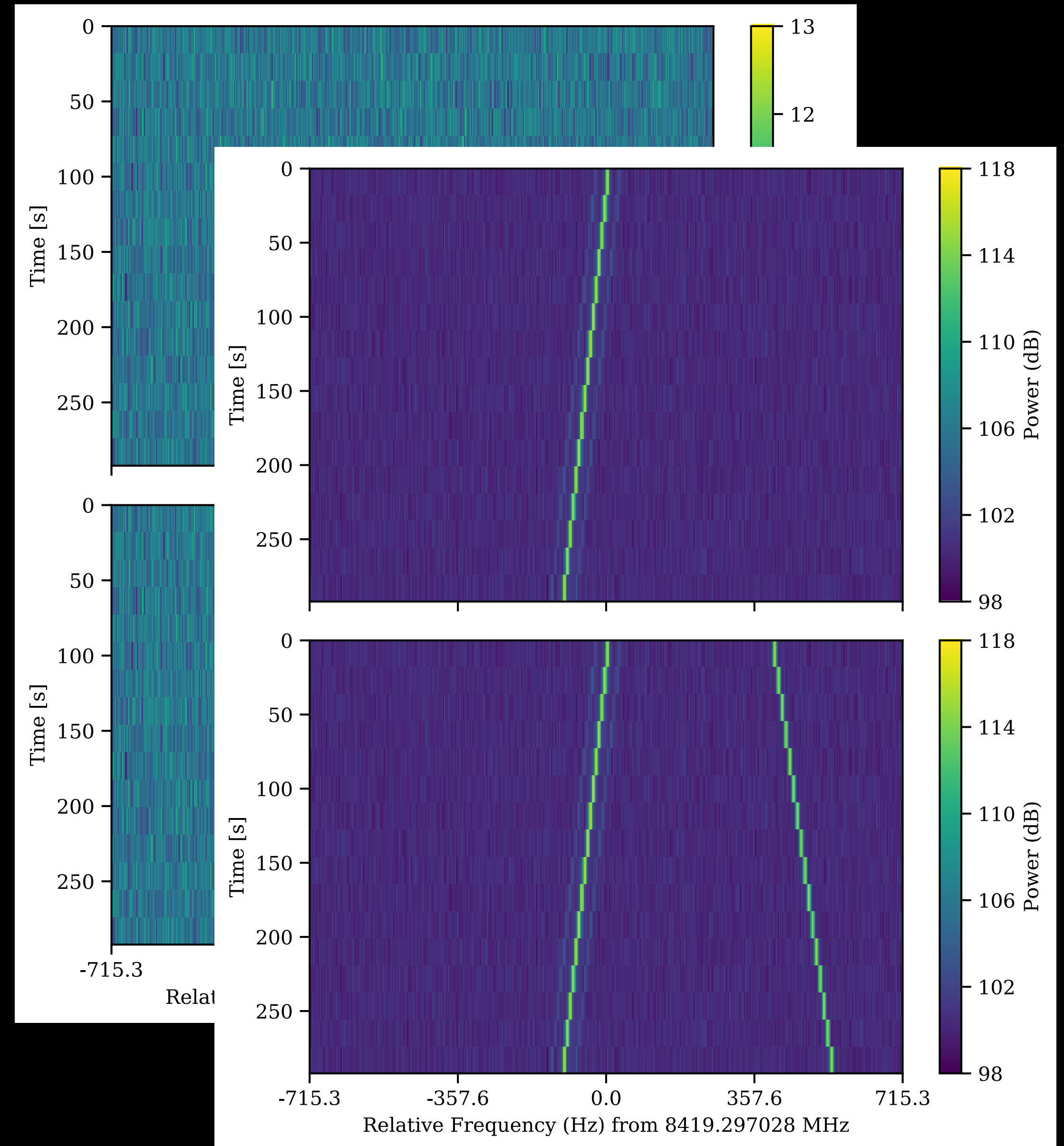
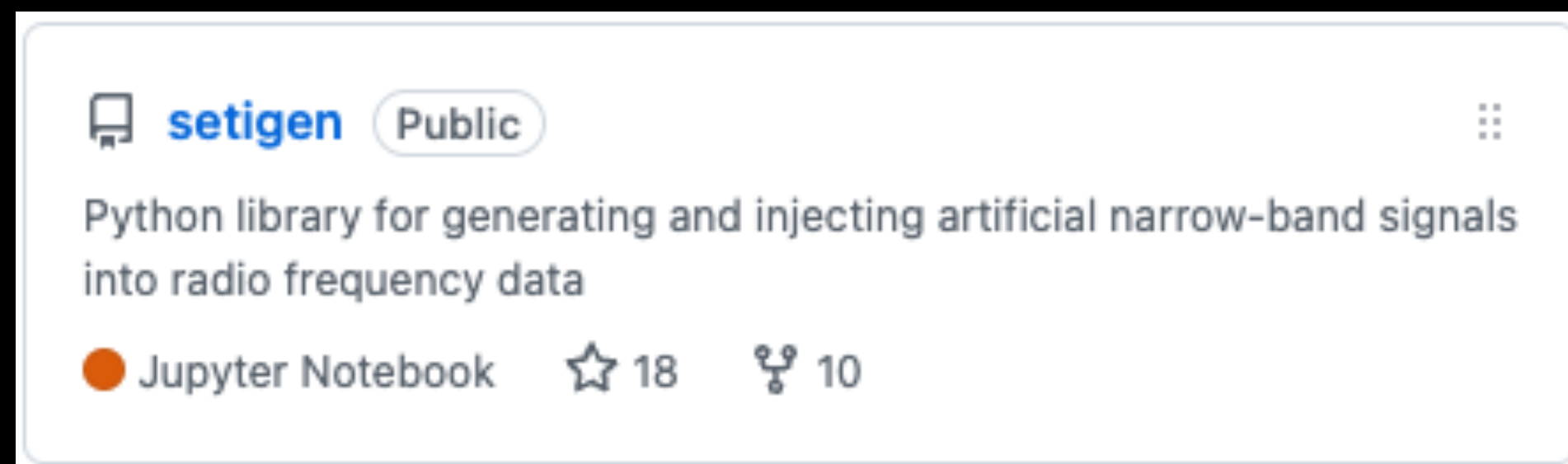
Python library for generating and injecting artificial narrow-band signals into radio frequency data

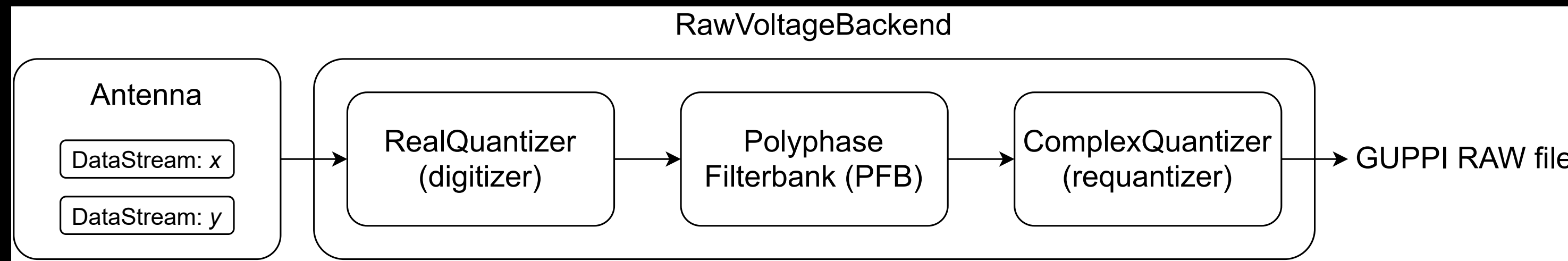
 Jupyter Notebook  18  10



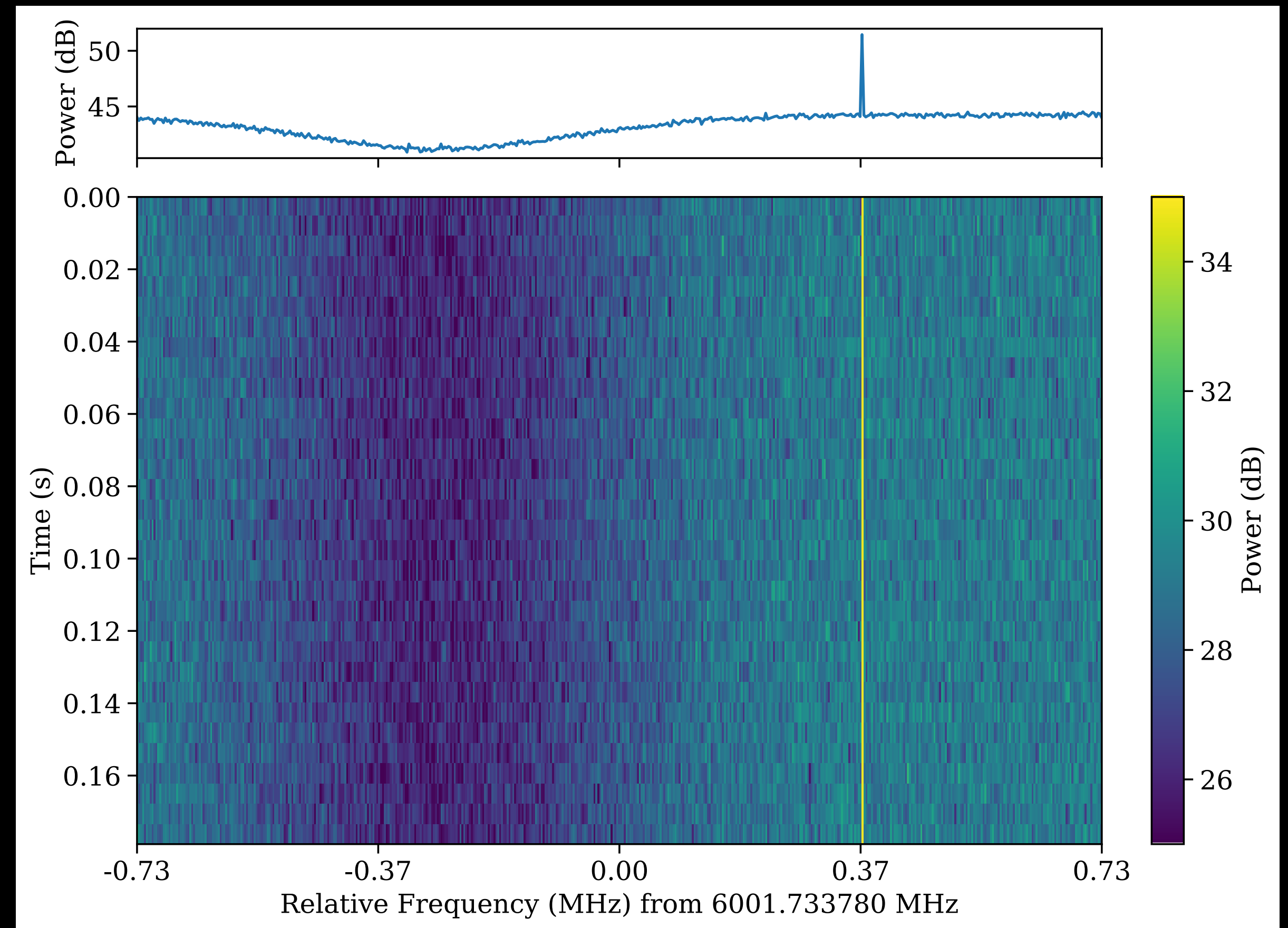
Setigen

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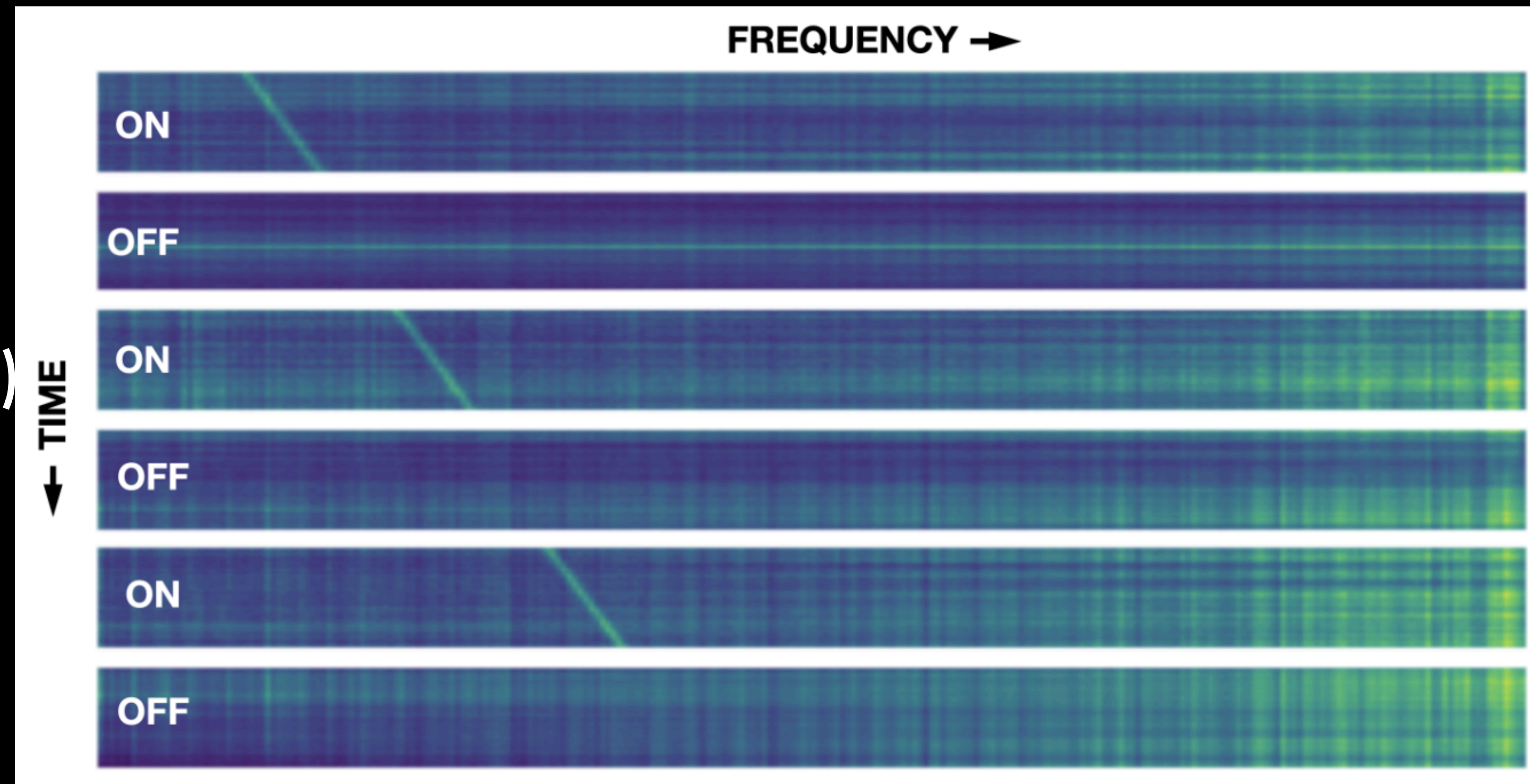


- Synthetic complex voltage data
- Simple models of backend components, such as a polyphase filterbank



Applications of Setigen beyond my research

- Injection — recovery testing
- ML dataset production (e.g. Kaggle)
- Multibeam search surveys
- Development of software for the Allen Telescope Array



Breakthrough Listen x Kaggle 2021

Scattering intensity

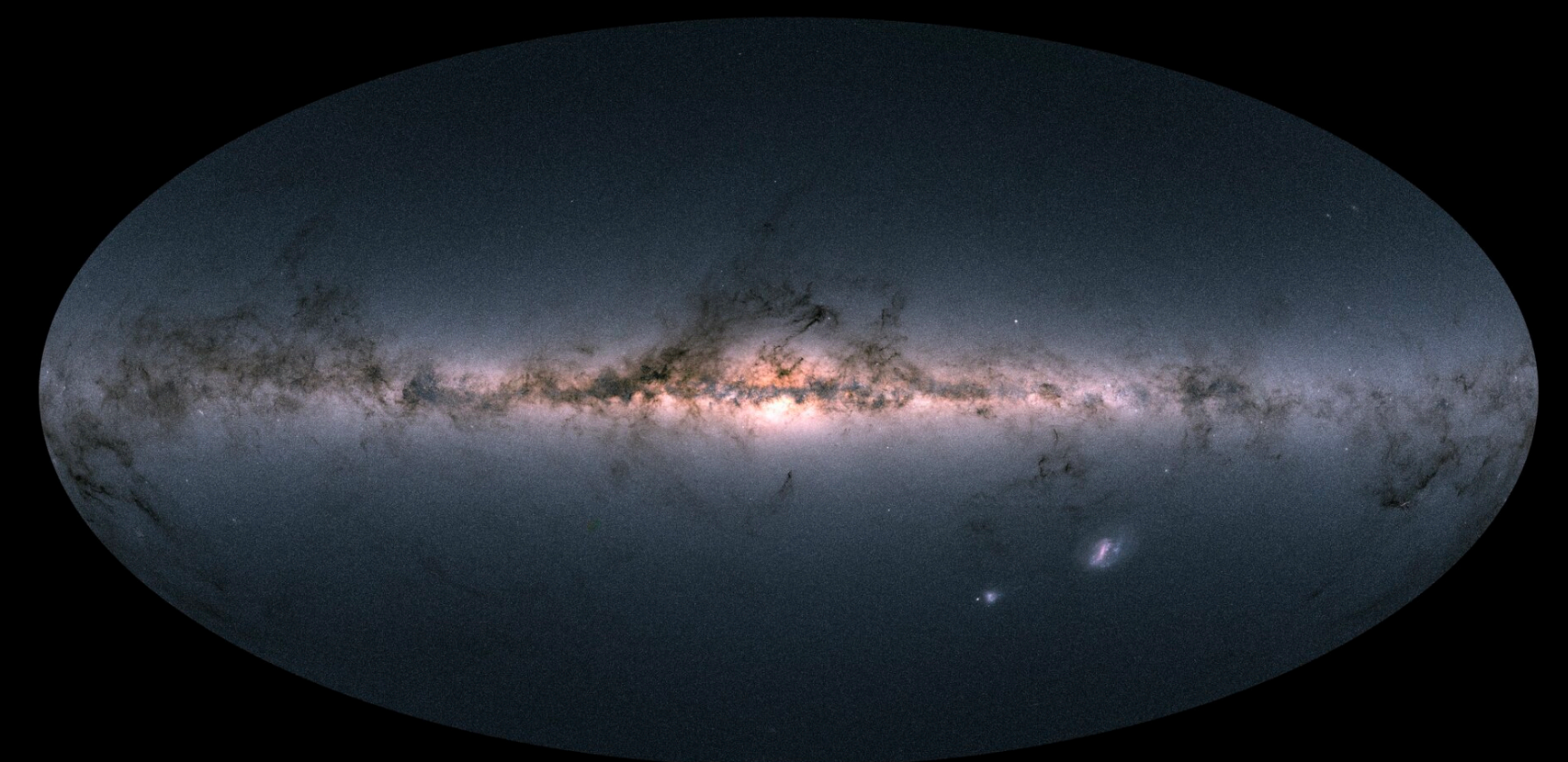
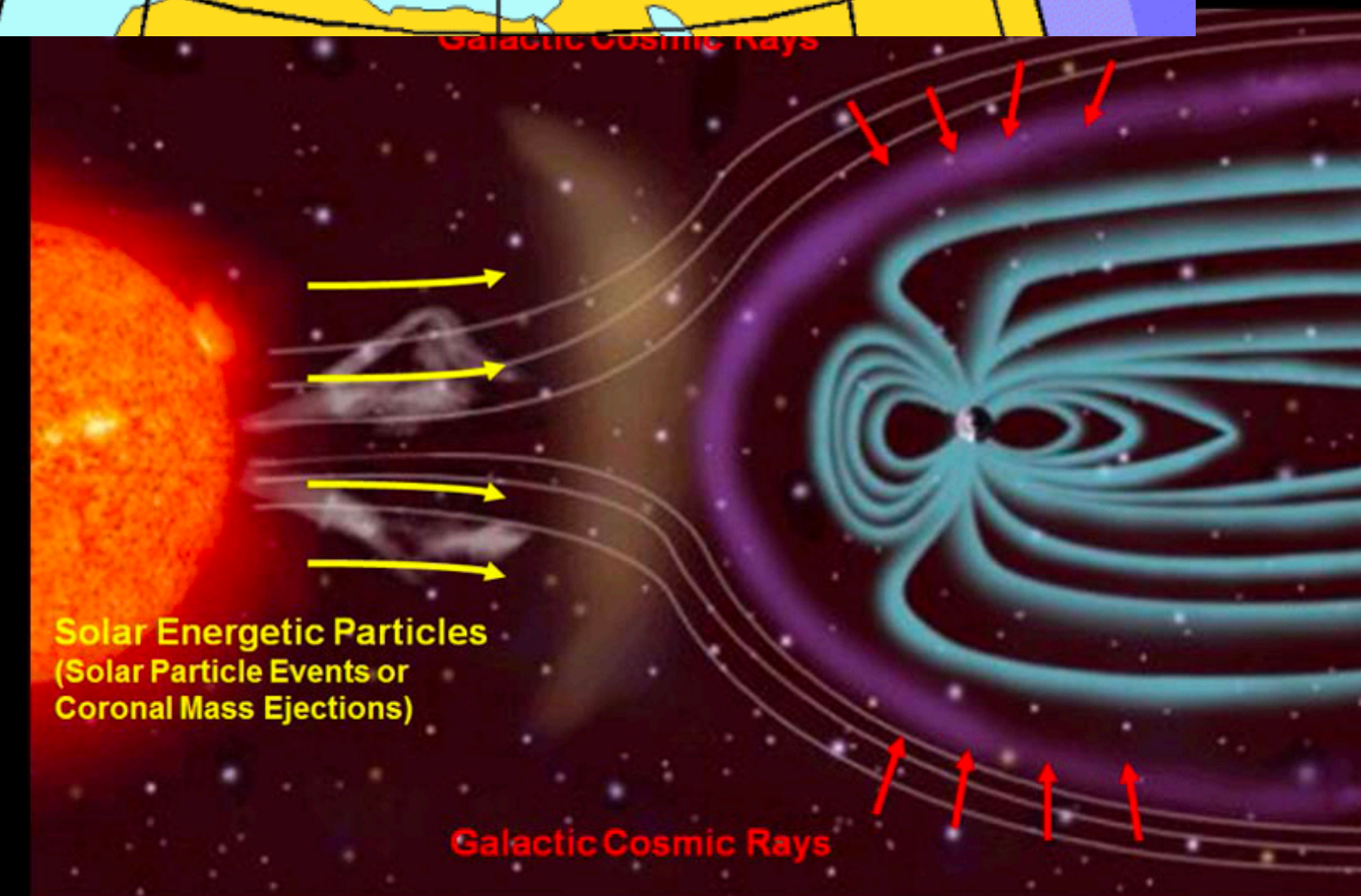
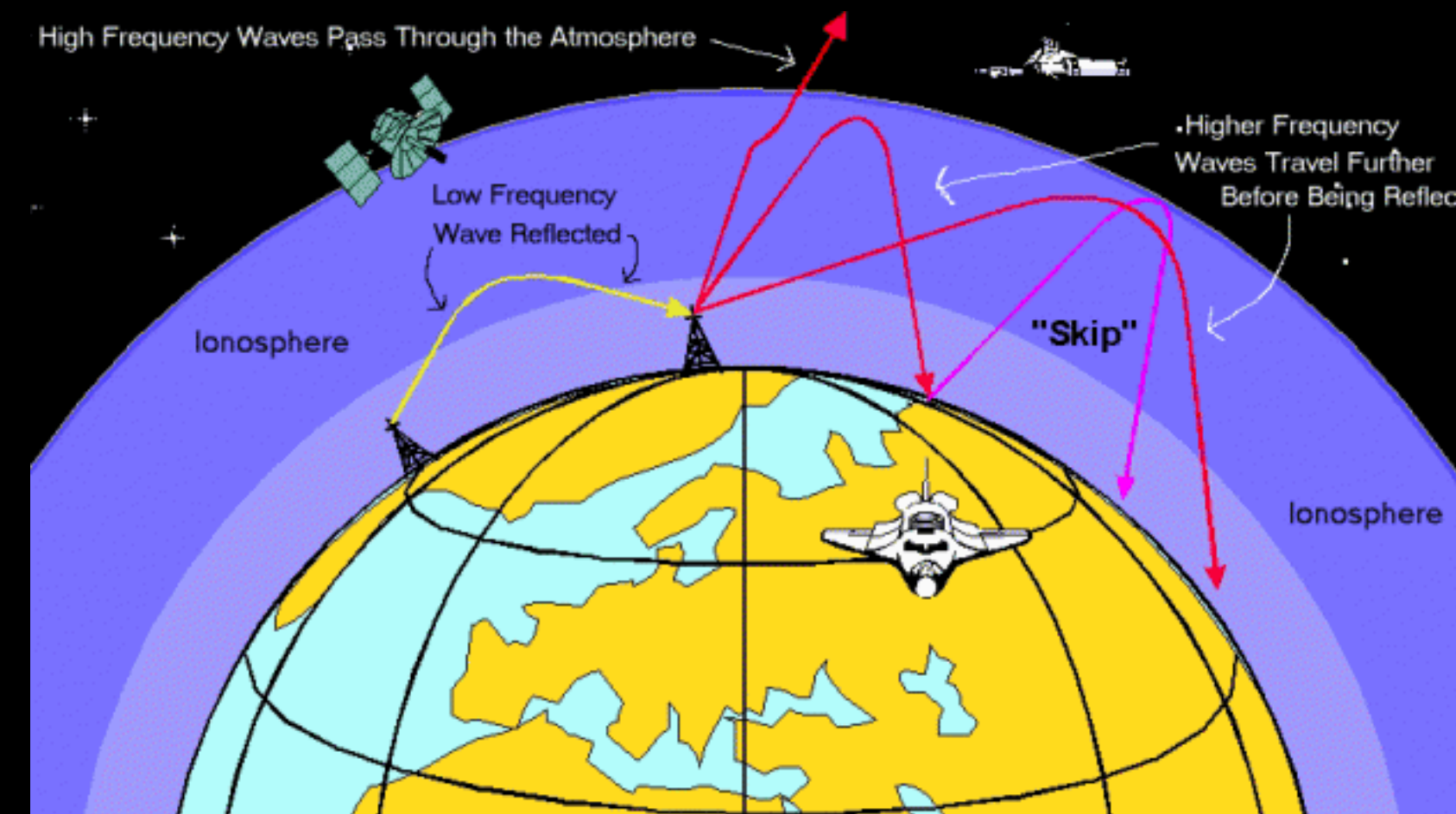
- Ionosphere — weak

$$m_d \ll 1$$

- IPM — mostly weak

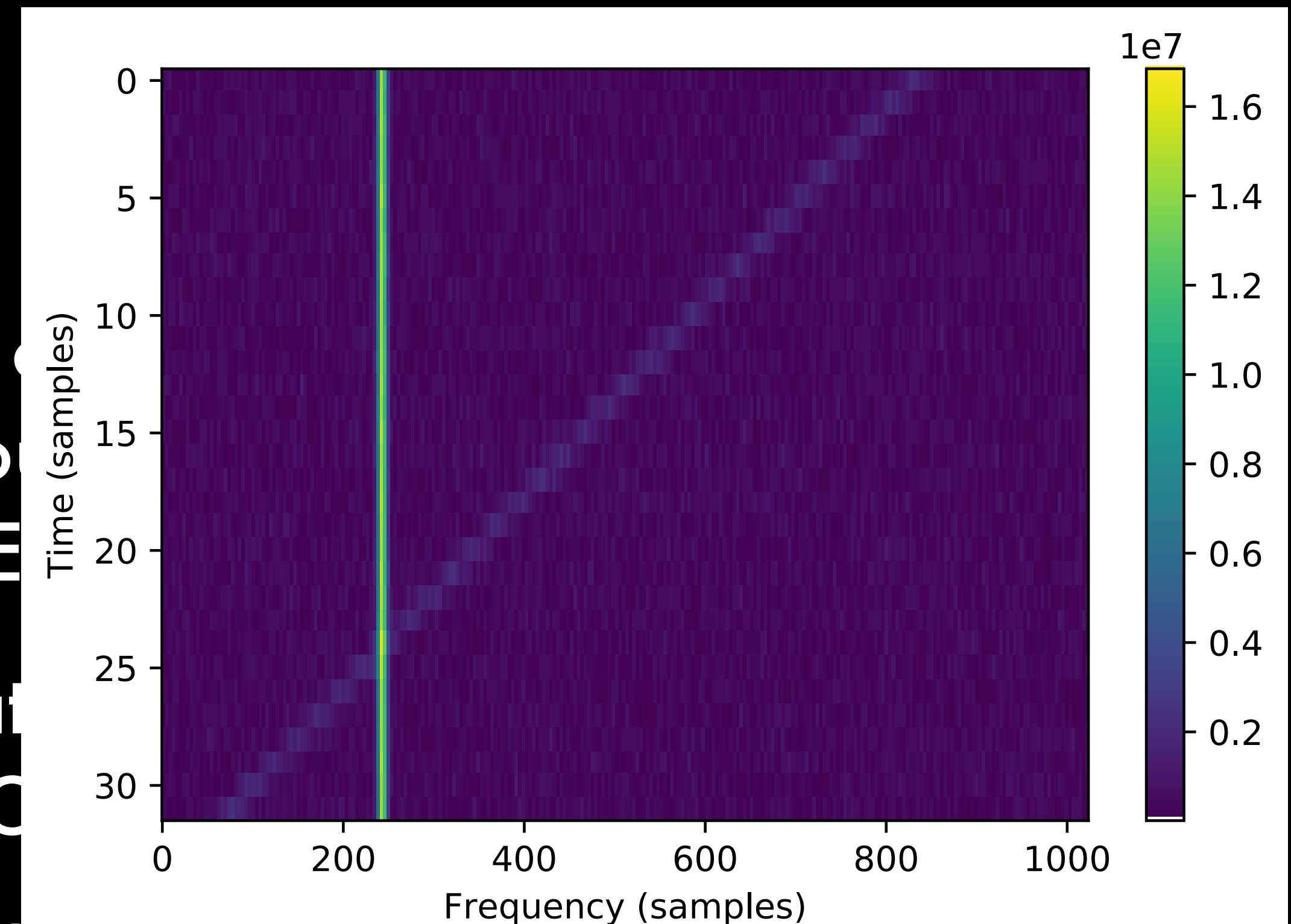
- ISM — can be strong!

$$m_d \approx 1$$



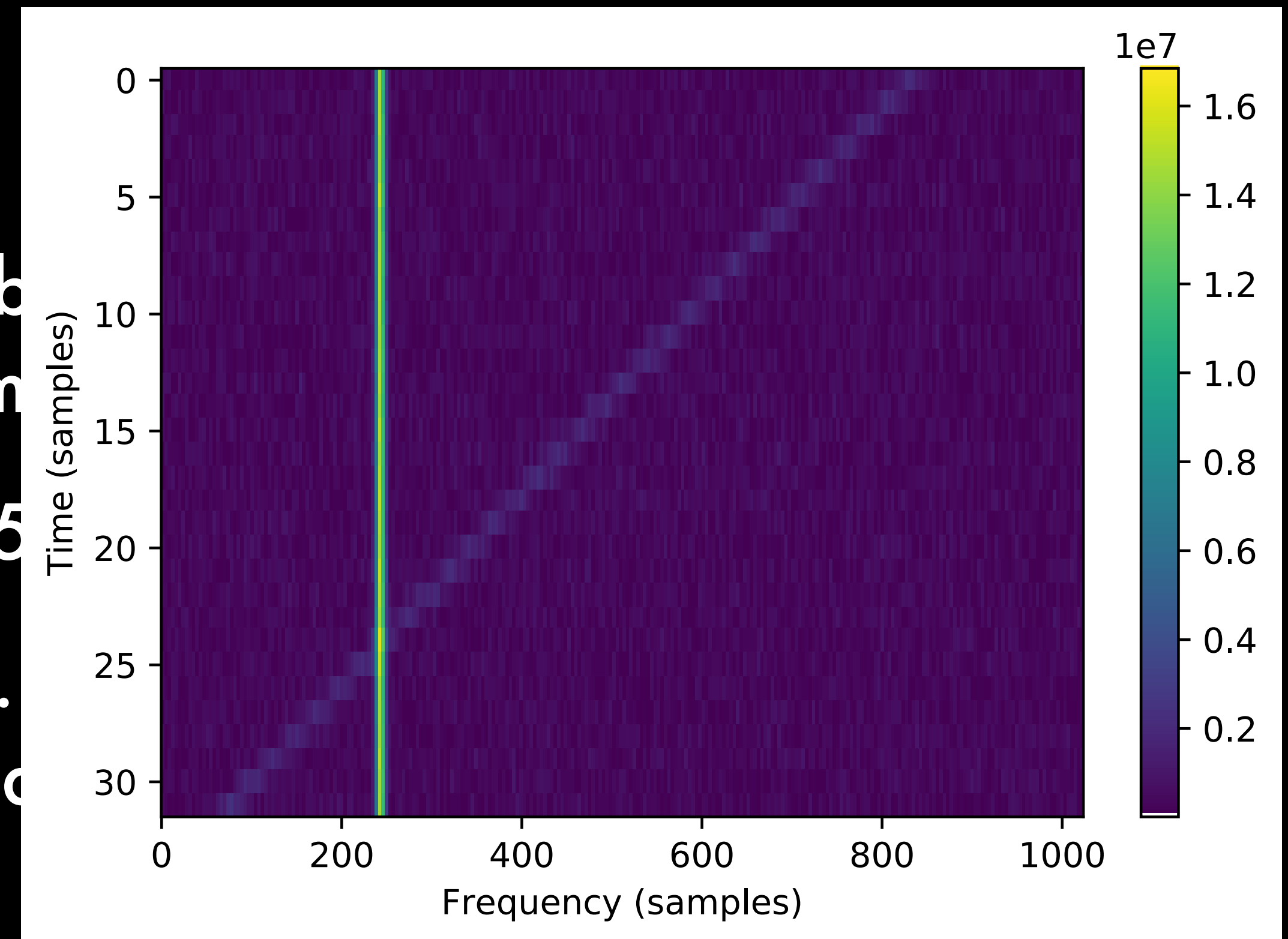
NARROW-BAND SIGNAL LOCALIZATION (BRZYCKI ET AL. 2020)

- With a means of simulating signals, we can produce frames that would be “organically” using TurboSE
- Importantly, we can generate synthetic data, and train a CNN
- Localization of narrow-band signals is a good ML problem because it’s a relatively simple task; predict 2 numbers per signal



NARROW-BAND SIGNAL LOCALIZATION (BRZYCKI ET AL. 2020)

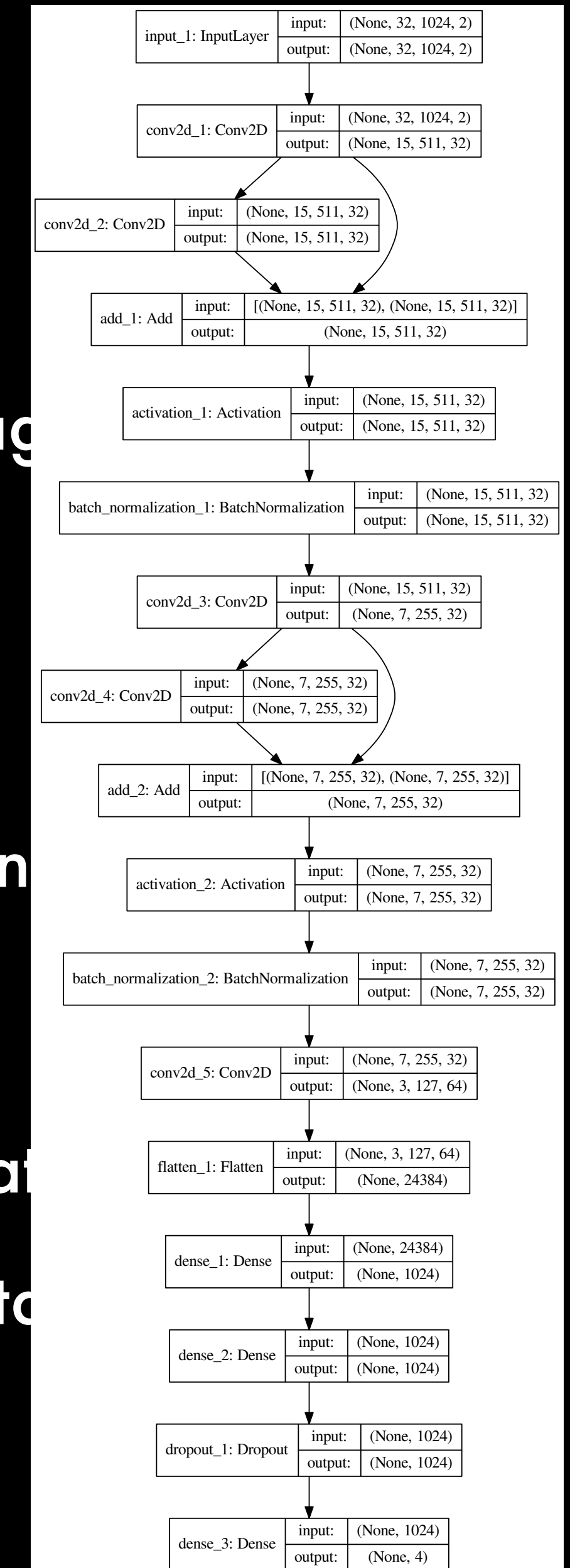
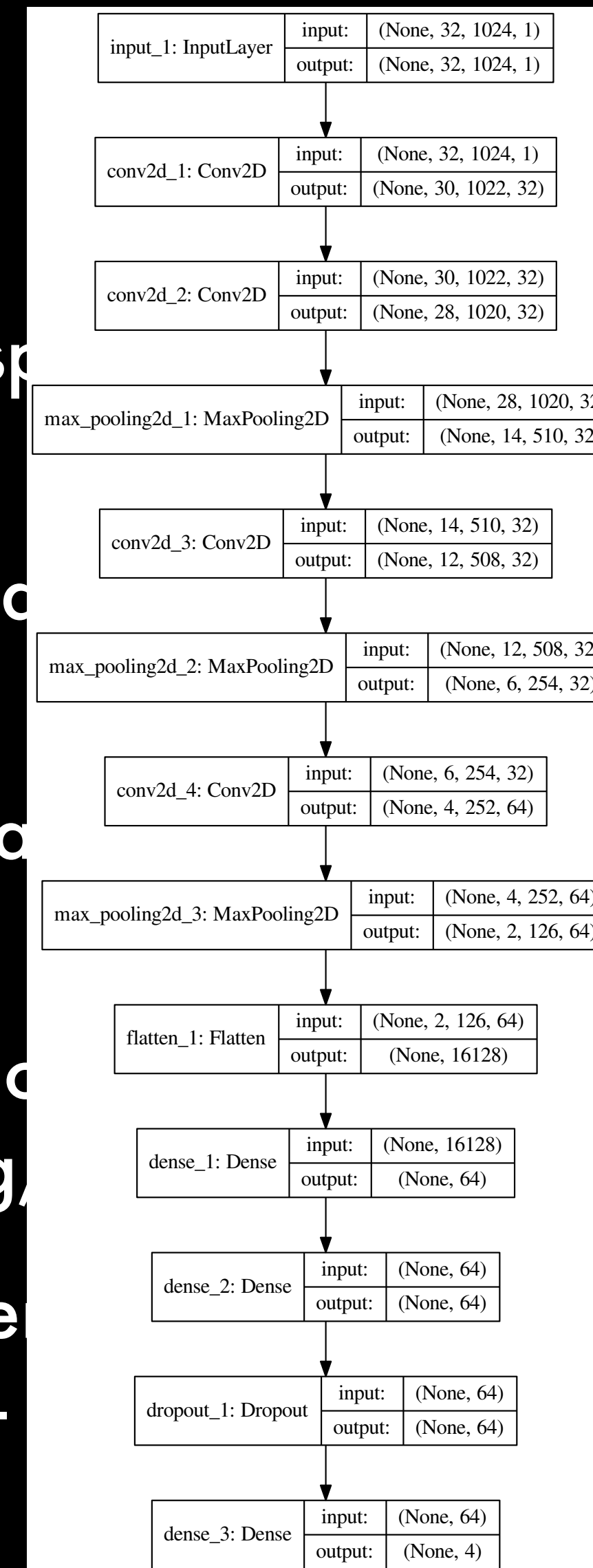
- Created two main datasets, both with 10,000 training samples and 24,000 test samples
 - One signal, at 0, 5, ..., 25 Hz
 - Two signals, one at 0, 5, ..., 25 Hz and the other at 25, 30, ..., 50 Hz to simulate “bright” RFI)
- The one signal dataset allows for direct comparison with TurboSETI; the two signal dataset tests the effectiveness of localizing multiple signals simultaneously



Example of a frame with 2 synthetic signals.

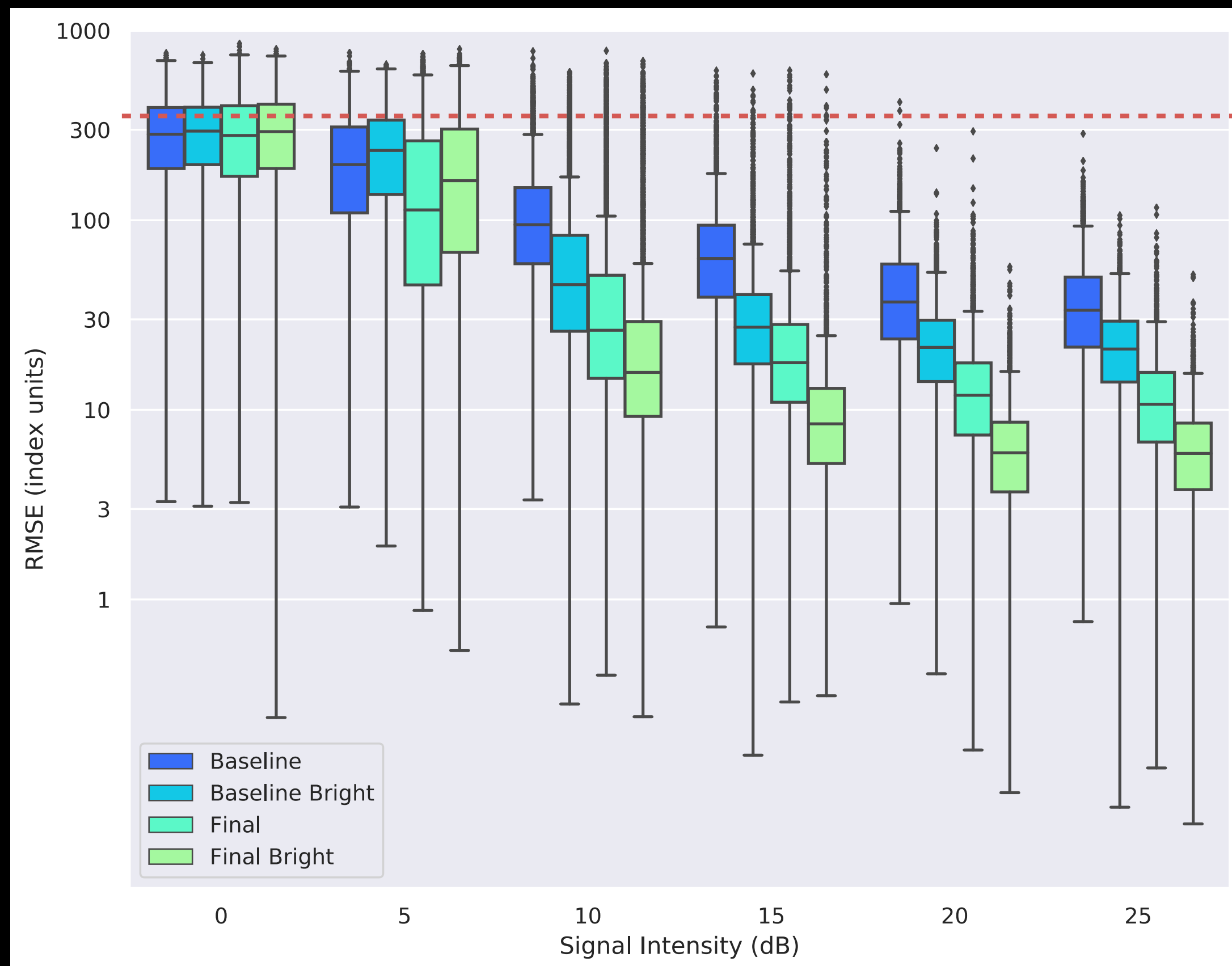
MODEL ARCHITECTURES

- Used convolutional neural networks, especially for processing input data
- Created a “baseline” and a “final” model to compare performance:
 - Baseline model uses convolutional layers, max pooling, and fully connected layers
 - Final model includes residual connections and batch normalization
- In addition to training these models over 100 epochs, we did alternate training over only 10 - 20 epochs, labeling these as “bright” models

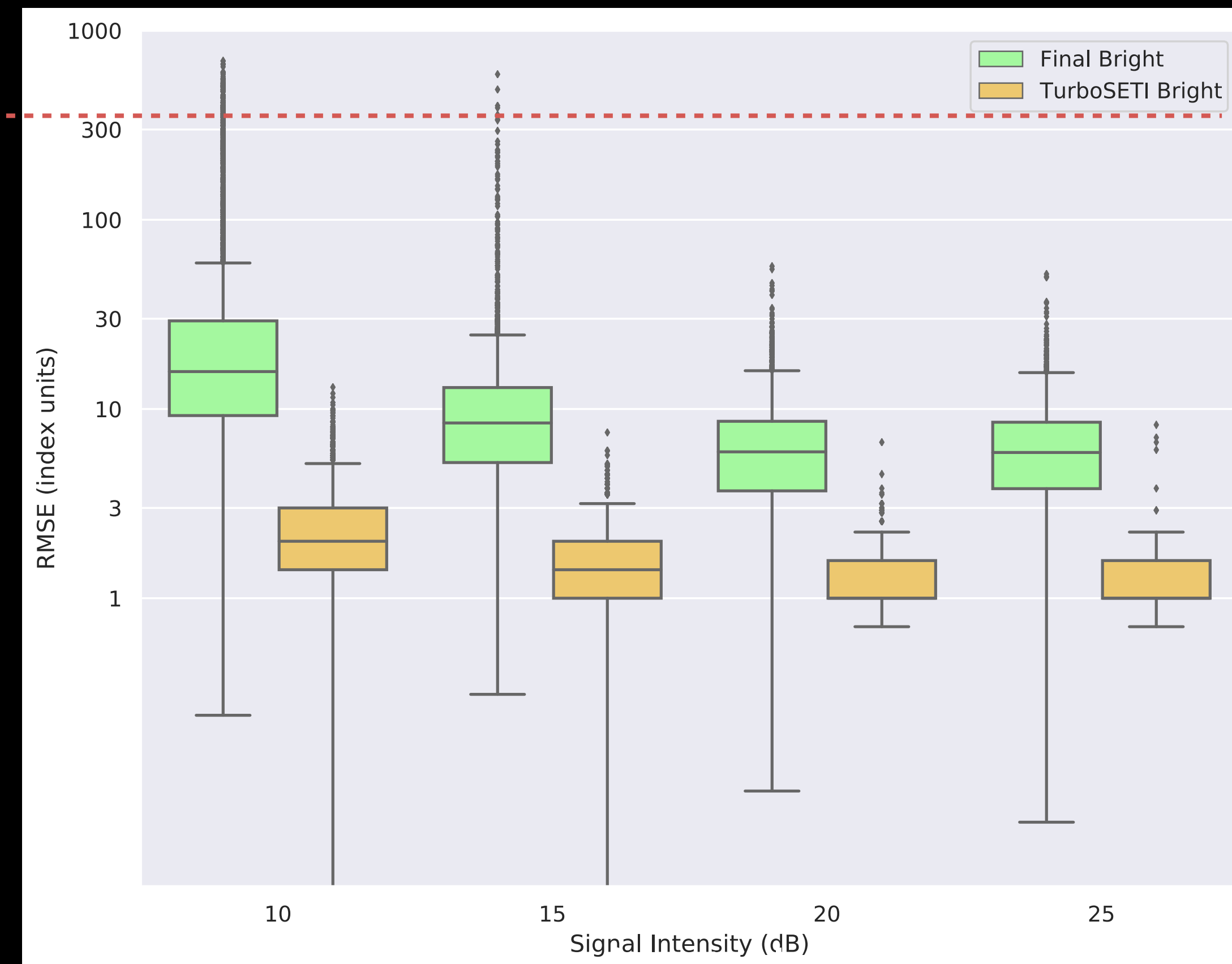


$$\text{RMSE (index units)} = 1024 \times \sqrt{\frac{1}{n} \sum_i (y_i - \hat{y}_i)^2}$$

ONE SIGNAL RESULTS ON TEST DATA



Root mean squared error (in pixels) across different models

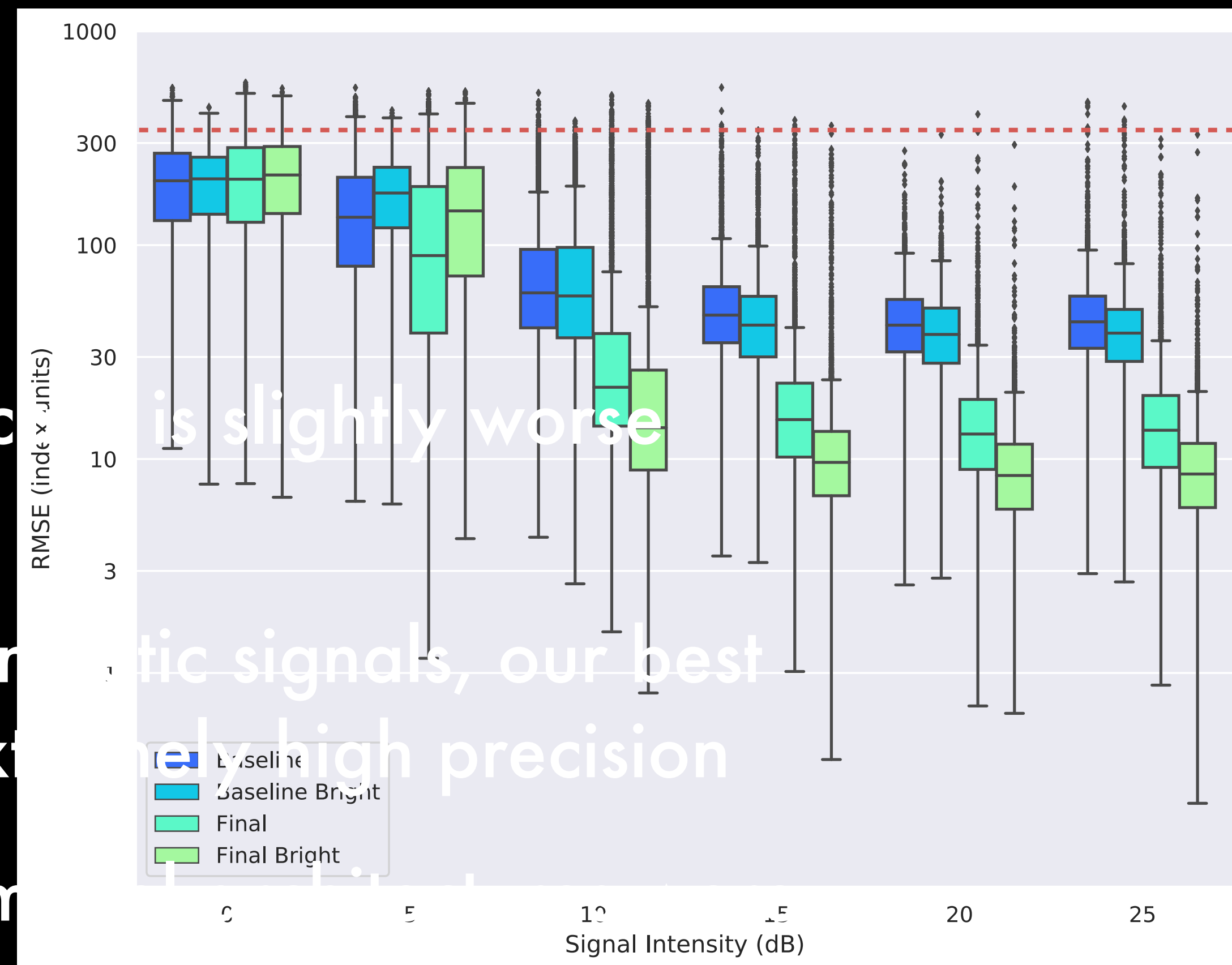


Root mean squared error (in pixels) for best ML model vs. TurboSETI. Only calculated for SNR > 10.

$$\text{RMSE (index units)} = 1024 \times \sqrt{\frac{1}{n} \sum_i^n (y_i - \hat{y}_i)^2}$$

TWO SIGNAL RESULTS ON TEST DATA

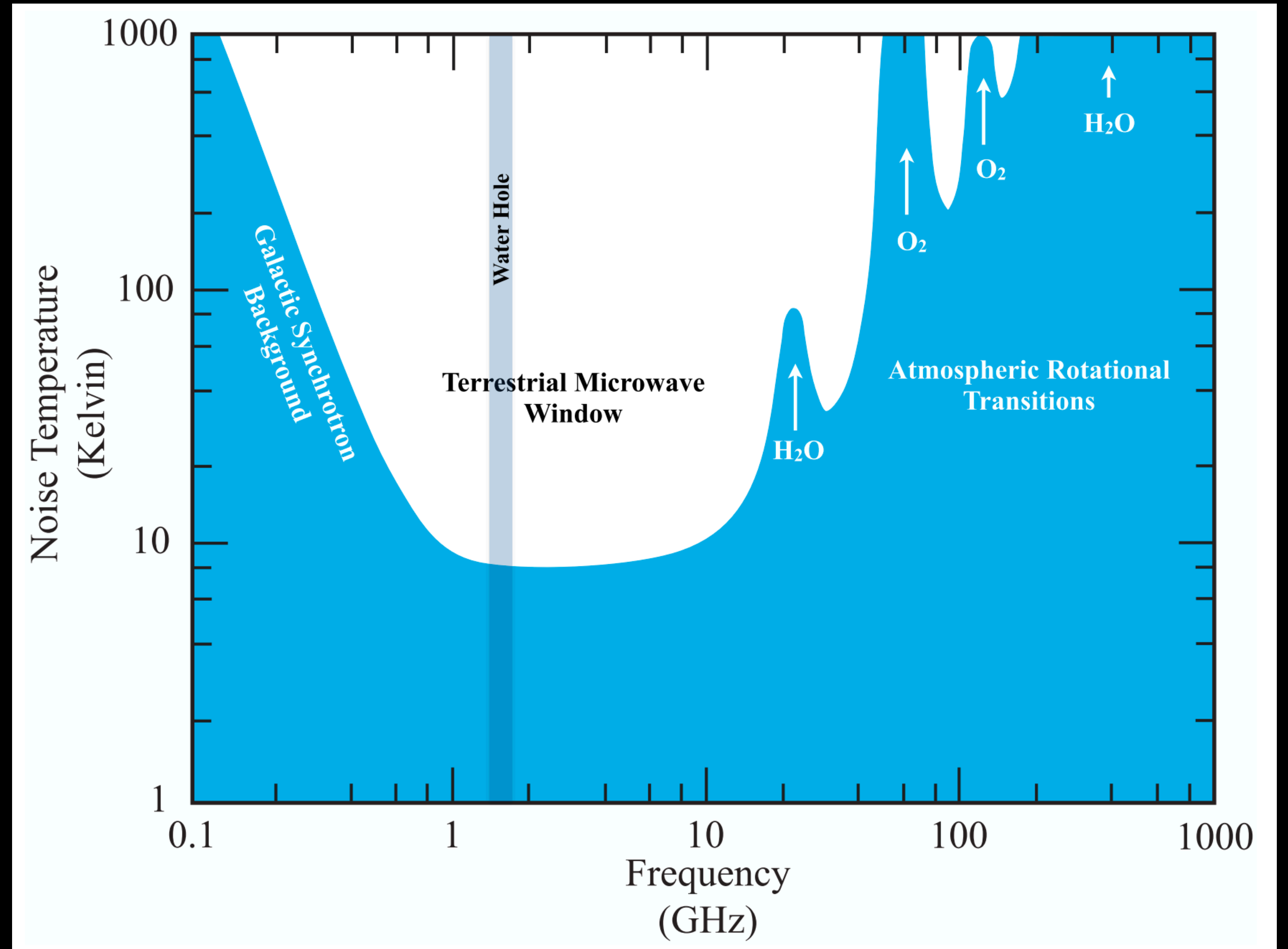
- Performance over two signal cases is slightly worse than in the one signal case
- Even though we used ideal synthetic signals, our best models failed to localize to extremely high precision
- Nevertheless, our two signal models are able to localize the dimmer signal better than random so these results are still encouraging



intensities, in pixels, for various neural network architectures in the 2 signal case.

Why radio?

- Low energy
- Low attenuation
- Produced by technology!



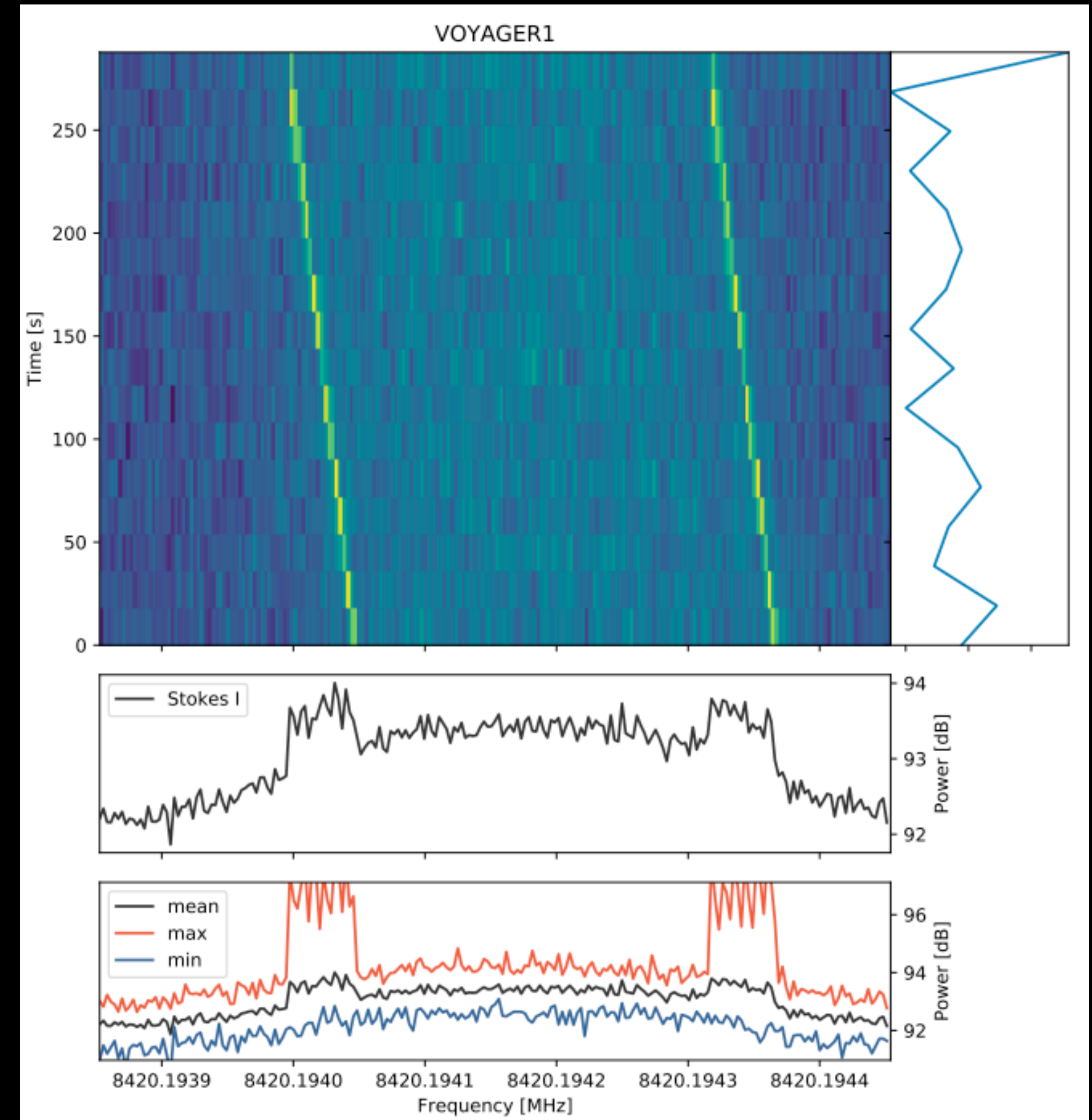
Siemion et al. 2014

Detection basis for SETI searches

- Raw signal detection
- Candidate identification and differentiation (filtering)

Raw signal detection

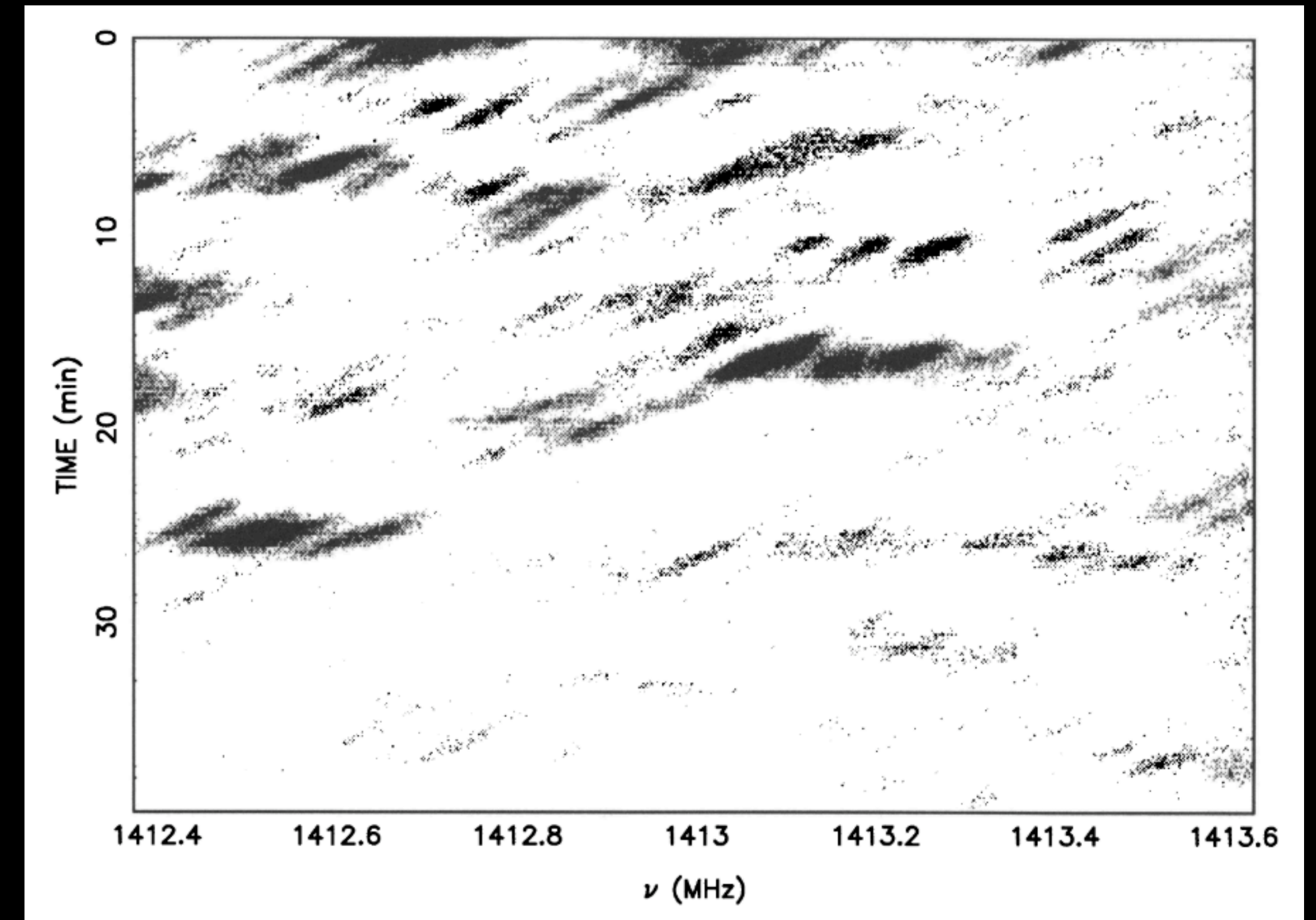
- Incoherent deDoppler (TurboSETI)
- Energy detection
- Machine learning (ML)



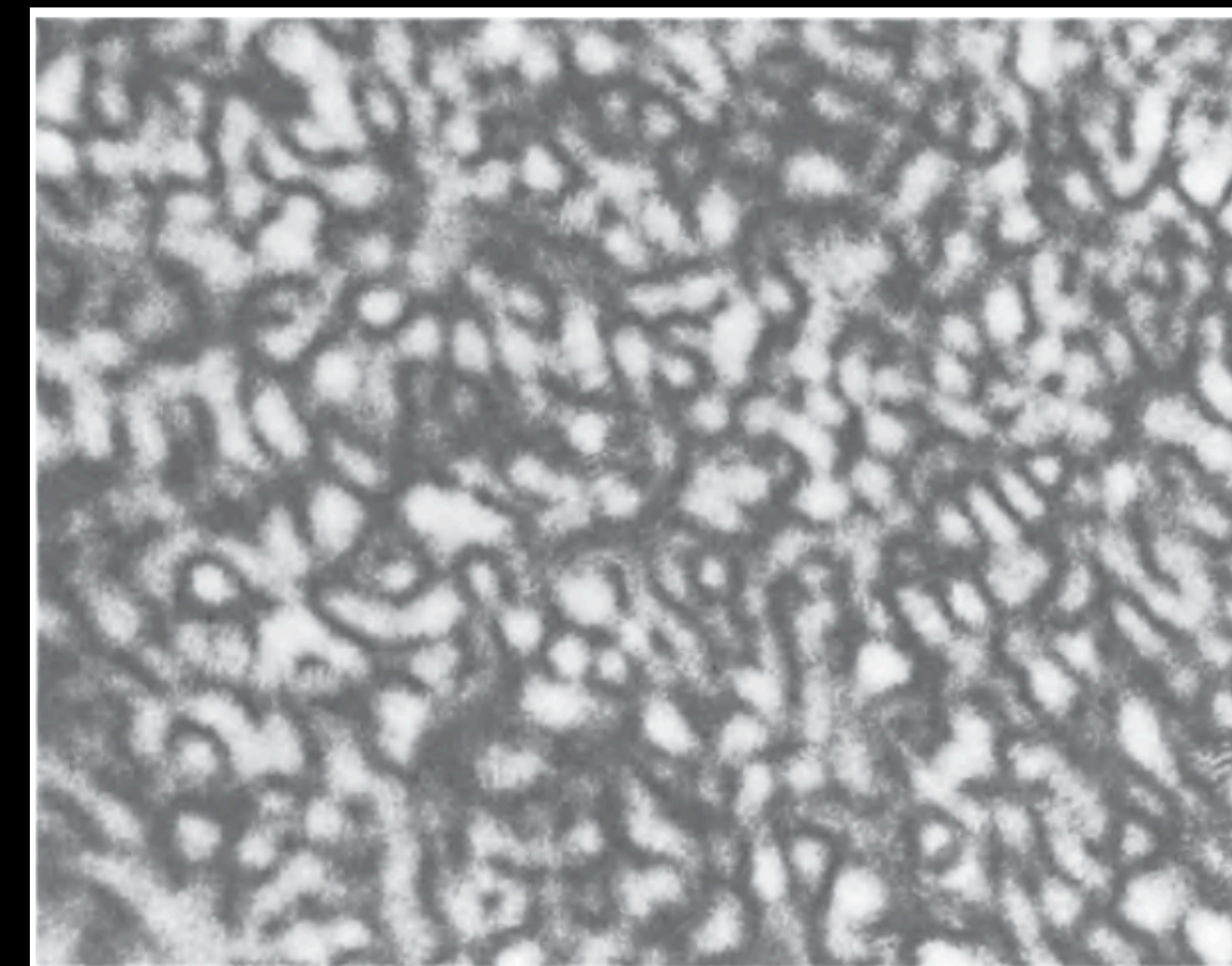
Lebofsky et al. 2019

Pulsar observations probe radio plasma effects

- Dispersion
- Scattering: scintillation and broadening
- Parallels with optical laser speckle



Cordes & Lazio 1991



Goodman 1984

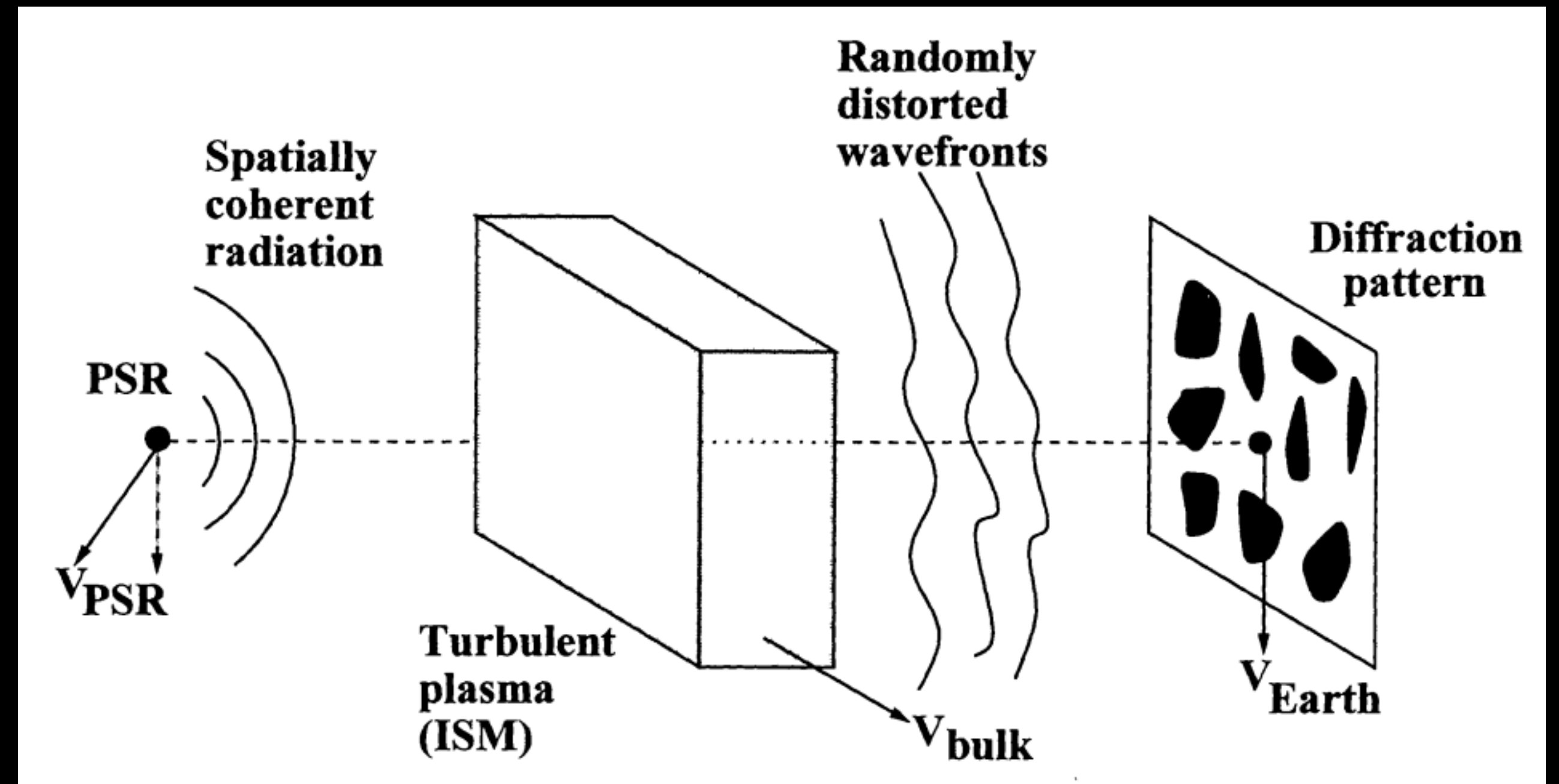
Bigger picture: research goals

- Where and how should we look to target scintillated narrowband sources?
Is this feasible and worth trying?
- Develop a methodology and analysis framework for evaluating interesting signals and studies on a case-by-case basis



Diffraction scintillation in the ISM

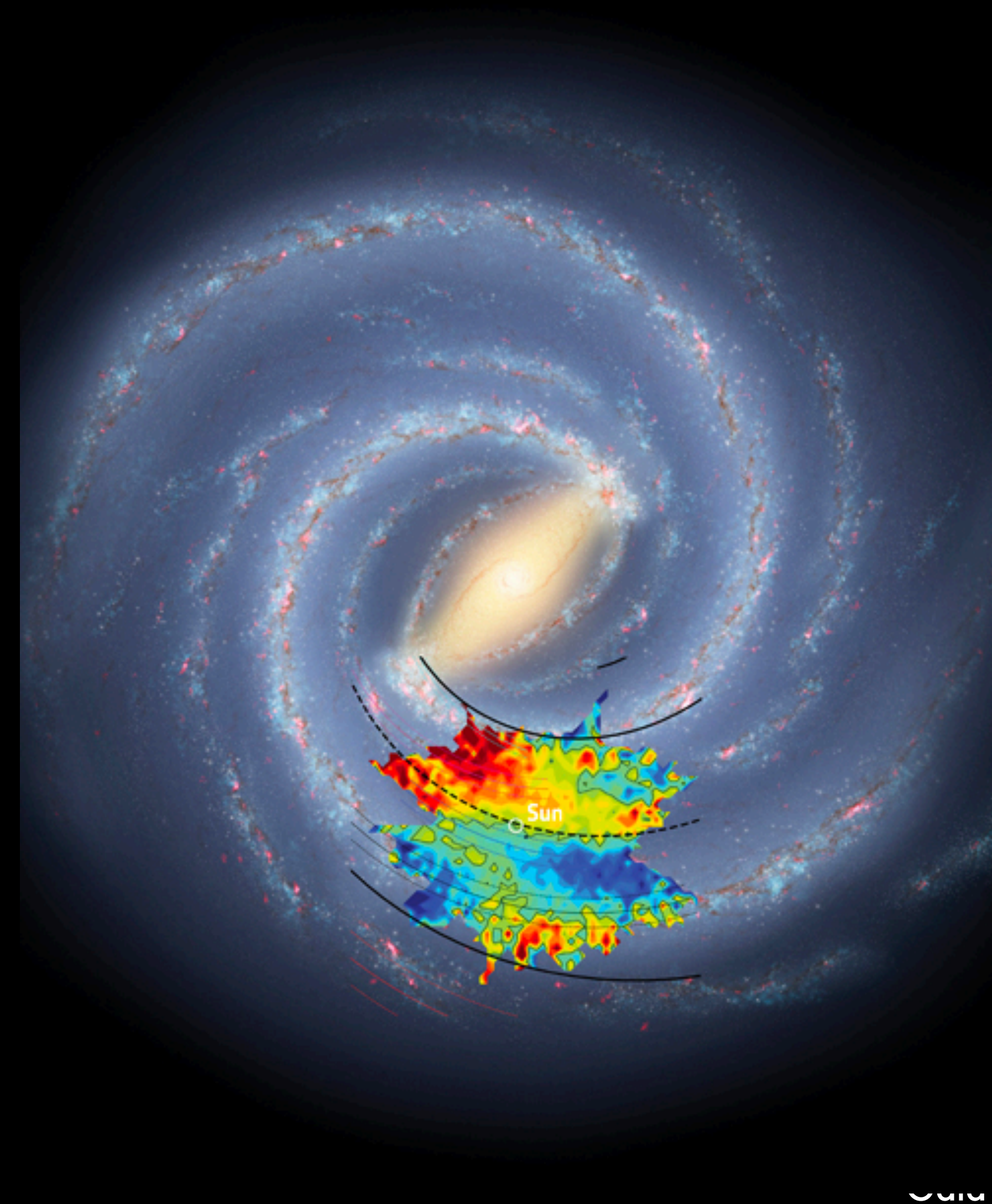
- Electron density fluctuations in ionized plasma give rise to phase fluctuations
- Interference pattern at observer plane with characteristic spatial and spectral scales
- Can lead to 100% intensity modulation on characteristic temporal scales Δt_d , especially towards the Galactic center



Cordes 2002

Next steps: a Galactic Center / Galactic Plane survey

- Target most promising sections of parameter space
- Survey of Galactic plane with interesting targets
- Gaia DR3?



Gaia